



Empowering healthcare 5.0 with deep learning: techniques, trends, and future directions

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Abstract

The rapid advancement of intelligent technologies has driven a transformative shift in healthcare, giving rise to Healthcare 5.0, a new paradigm centered on patient-focused, digitally empowered medical services. This study presents a comprehensive review of cutting-edge Deep Learning (DL) techniques that are redefining Healthcare 5.0 through applications in disease prediction and early diagnosis, medical image analysis and radiology, and multimodal deep learning (MMDL). We analyze a broad spectrum of DL architectures, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTMs), Gated Recurrent Units (GRUs), Generative Adversarial Network (GANs), transformers, autoencoders, transfer learning, and MMDL, highlighting the unique features that make them particularly suited for healthcare tasks. Using bibliometric analysis of 1,342 Scopus-indexed publications (2018-2025) via VOS-viewer, we uncover key research trends, thematic clusters, and emerging focus areas. Furthermore, we assess DL models across diverse real-world healthcare datasets and discuss critical challenges, including data privacy, model interpretability, and system integration. Following the PRISMA methodology, this review also synthesizes recent research on advanced DL architectures for disease diagnosis and Healthcare 5.0, alongside widely used medical imaging and multimodal datasets. The manuscript has been substantially strengthened by incorporating a comparative analysis, quantitative aggregation of reported results, an explicit novelty and contribution statement, and a systematic comparison of DL survey studies across Healthcare 4.0 and Healthcare 5.0, thereby providing a more critical, data-driven, and clearly differentiated perspective beyond prior reviews. Finally, we address significant issues, including ethical considerations, privacy concerns, and model limitations, while outlining promising directions for future research. This work serves as a valuable resource for researchers and practitioners, offering both a broad overview and deep insights into the latest developments in DL for Healthcare 5.0 systems.

Keywords Healthcare · Deep learning · Artificial intelligence · Medical imaging · Disease diagnosis · Multimodality

1 Introduction

Healthcare 4.0 significantly enhances the quality, productivity, flexibility, cost-effectiveness, and dependability of healthcare services while primarily focusing on the business model (Li and Carayon 2021; Haleem et al. 2022; Gupta and Singh 2023; Vladimir V Popov et al. 2022). However, Healthcare 5.0 builds on the innovations of Healthcare 4.0 and focuses on the customer model, marking the next phase of the healthcare revolution. Healthcare 5.0 represents a comprehensive transformation that extends beyond technological innovation (Wazid et al. 2021), shifting from disease-centered to patient-centered care

(Saraswat et al. 2022). Information systems have evolved from localized clinical applications to regional medical informatization (Mbunge et al. 2021), while management practices have advanced from generalized approaches to personalized, preventive healthcare (Tian et al. 2019; Gupta et al. 2022). The practice of healthcare has evolved through the initial evidence-based medicine to industrialized and automated as well as digitally driven medical systems. The next level is Healthcare 5.0, which focuses on personalized care, continuous monitoring, and as well as improvement of well-being. It combines innovative technologies with big data analytics, Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Internet of Things (IoT), cloud solutions and telemedicine (Verma et al. 2022) to facilitate real-time decision making and secure and patient-centered services. Healthcare 4.0 can be defined as a technology-driven paradigm that integrates IoT, big data analytics, cloud computing, and artificial intelligence to enhance automation, connectivity, and efficiency in healthcare delivery systems (Haleem et al. 2022; Gupta and Singh 2023; Popov et al. 2022). In contrast, Healthcare 5.0 extends this paradigm toward a human-centric approach, where advanced technologies such as deep learning are combined with human intelligence to enable personalized, preventive, and ethically grounded healthcare solutions (Tian et al. 2019; Gupta et al. 2022; Li and Carayon 2021; Verma et al. 2022; Mbunge et al. 2021). The key distinction lies in the shift from system efficiency to patient-centric intelligence and collaborative decision-making. To clearly distinguish Healthcare 5.0 from its predecessor, Healthcare 4.0, a concise comparison is presented in Table 1, highlighting differences in technological focus, system design philosophy, and healthcare delivery models.

Table 1 Comparison between Healthcare 4.0 and Healthcare 5.0

Aspect	Healthcare 4.0	Healthcare 5.0
Core paradigm	Technology-driven, automation-focused systems Gupta and Singh (2023); Haleem et al. (2022); Li and Carayon (2021)	Human-centric, personalized healthcare ecosystem Verma et al.(2022); Mbunge et al. (2021)
Focus of care	Efficiency, productivity, and cost reduction Popov et al. (2022); Haleem et al. (2022)	Patient well-being, personalization, and sustainability Saraswat et al.(2022); Verma et al. (2022)
Role of AI & Deep Learning	Automation, prediction, and analytics Li and Carayon (2021); Gupta and Singh (2023)	Collaborative intelligence with human-in-the-loop decision-making Saraswat et al. (2022); Mbunge et al. (2021)
Patient role	Passive data source and service recipient Li and Carayon (2021)	Active participant and decision-maker Verma et al.(2022); Mbunge et al. (2021)
Data utilization	Large-scale but fragmented data processing Popov et al. (2022)	Integrated, context-aware, and personalized data usage Wazid et al. (2021); Gupta et al. (2022)
Ethical considerations	Limited emphasis on explainability and bias Haleem et al. (2022)	Strong focus on explainable, transparent, and trustworthy AI Saraswat et al. (2022)

The implementation of Healthcare 5.0 faces significant challenges arising from the vast scale and intricate nature of healthcare data. Sources such as medical imaging, electronic health records (EHRs), wearable devices, and genomic data produce enormous, heterogeneous, high-dimensional, and mostly unstructured data (Rani et al. 2025; Bertl et al. 2023). Conventional analysis approaches cannot handle such big and dense data. ML techniques are a branch of AI (Haleem et al. 2019), presenting a potential solution to all these limitations, aiming to optimize processes, improve the efficiency of healthcare systems, enhance the quality of patient care, and increase the precision of physicians' work (Li and Carayon 2021; Waring et al. 2020; Javaid et al. 2022; Sendak et al. 2020). They serve as a cornerstone in the advancement of clinical decision support systems, disease diagnosis methodologies, and personalized treatment strategies aimed at optimizing patient outcomes (Ghassemi et al. 2020; Ahmad et al. 2020; Kelly et al. 2019; Siddique and Chow 2021; Cohen et al. 2021).

Regardless of their potential, traditional ML models have serious limitations when involved in contemporary healthcare settings. They usually use handcrafted characteristics, restricting their outcome, scalability, and flexibility to be applied to new data forms Miotto et al. (2018); Martínez-García and Hernández-Lemus (2022). In addition, ML models cannot process high-dimensional and unstructured data that include medical images and free-text clinical notes. Such limitations hamper their capacity to explore the potential of multimodal healthcare data to its fullest extent to pursue the vision of Healthcare 5.0 (Fang et al. 2025). To address these challenges, DL is replacing ML in the healthcare industry to handle bigger, more complex datasets, increase accuracy, and provide better data analysis tools (Chakraborty et al. 2024; Fernandez and Peters 2023; Davenport and Kalakota 2019; Vogt 2018; Schmidhuber 2015). In recent years, the healthcare domain has experienced a quick and striking improvement through DL technology, maintaining the hope for more effective and reasonably priced medical treatment. In low-resource countries with limited healthcare infrastructure, the shortage of trained radiologists and specialized training programs is further compounded by limited awareness among physicians regarding the benefits of diagnostic radiography and interventional radiology (IR) techniques (Jamil et al. 2022; Rosman et al. 2019). These problems help automated reporting and image-based diagnosis to move forward. DL techniques present a promising approach to address the shortage of qualified physicians and experienced radiologists. These techniques facilitate automated report generation and assist radiologists in improving diagnostic accuracy. They are particularly effective in analyzing medical images such as chest X-rays, computed tomography (CT), and magnetic resonance imaging (MRI). Moreover, research over the past several years highlights that EHRs (Wazid et al. 2022) have increasingly incorporated computer-aided detection (CADe) (Pereira et al. 2021; Loizidou et al. 2023) and computer vision (CV) techniques to advance disease prediction from medical images.

Nevertheless, a major concern with DL in healthcare is its tendency to operate as a *black box*. This arises from the non-linear and complex nature of its architectures. As a result, DL models often generate decisions without providing transparent explanations for their outcomes (Chaddad et al. 2023). The absence of transparency gives rise to questions regarding the comprehension, visualization, dependability, impartiality, and ethical consequences of AI-powered healthcare operations (Saraswat et al. 2022; Hulsén 2023). Explainable AI (XAI) presents a hopeful resolution to this obstacle by offering intelligible insights into the decision-making process (Da Silva et al. 2023; Jin et al. 2023). Explainable AI is focused on elucidating the underlying knowledge of DL's black-box model, which uncovers the

decision-making process (Amann et al. 2020; Band et al. 2023; Gohel et al. 2021; Zhang et al. 2022; Yang et al. 2022; Borys et al. 2023).

Recent trend indicates that, MMDL, foundation model, federated learning, and XAI are developing into booming fields because of the applicability in personalized and privacy-conscious Healthcare 5.0 settings. Real-time monitoring, wearable data and cross-modal fusion applications are also becoming popular. Conversely, the trend of single-modality CNN-based image classification and older RNN-based architecture exhibit a downward trend, with newer transformer-based and hybrid models outperforming them and being more scalable. These changes are indicative of a bigger trend towards integrated, context-sensitive systems that can be used to predict illness more comprehensively (Ali et al. 2024; Lin et al. 2025; Mo and Liang 2024).

This survey provides a comprehensive examination of the role of DL within the framework of Healthcare 5.0, structured around the following research questions:

- *RQ1*: In what ways do DL models support the vision of Healthcare 5.0, and how do their advantages compare with those of traditional ML approaches?
- *RQ2*: Which DL algorithms are most commonly applied in healthcare, and what key features make them particularly suitable for clinical applications?
- *RQ3*: What have recent studies done to use DL in disease prediction, imaging, and multimodal healthcare, and what is the role of datasets?
- *RQ4*: What are the primary challenges and limitations of DL in Healthcare 5.0?
- *RQ5*: What open research gaps persist, and which directions appear most promising for advancing efficient, trustworthy, and scalable DL models tailored to Healthcare 5.0?
- *RQ6*: What are the methods of the evaluation and validation of DL models in Healthcare 5.0, and how the related concerns like data leakage, bias, and reproducibility are addressed today?
- *RQ7*: What impact of interpretability, fairness and regulatory compliance on clinical trust and adoption of DL systems in Healthcare 5.0?

This survey aims to offer an in-depth review of the most recent DL algorithms employed in the healthcare field. This paper can be a valuable resource for researchers and professionals who wish to promptly familiarize themselves with the latest advancements in DL for healthcare applications. The main contributions of this study are given below.

- We present a comprehensive state-of-the-art survey of DL techniques, including multimodal architectures such as CNNs, RNNs, LSTMs, GRUs, GANs, diffusion models, and Transformer-based models, applied to Healthcare 5.0, with a particular focus on disease prediction, diagnostic modeling, and medical image analysis.
- In this study, through an extensive literature review, the problems of limited input medical datasets are addressed by the synthesis of generative models in modern healthcare.
- We conduct a bibliometric analysis using VOSviewer on 1,342 Scopus-indexed publications from 2018 to 2025. This analysis reveals dominant keywords, research hotspots, and thematic clusters in the field of DL applications in Healthcare 5.0. The resulting co-occurrence network visualization provides insight into the intellectual structure and evolving focus areas of this interdisciplinary domain.
- We present comparative tables of recent studies employing diverse DL techniques,

along with systematic descriptions of datasets and literature reviews focused on disease prediction and early diagnosis, medical image analysis and radiology, and MMDL in healthcare.

- This study offers an in-depth analysis of fundamental challenges and research gaps in DL-based healthcare systems, highlighting a range of interesting new directions for future research in this field.
- We present novelty and contribution where the proposed review compared with recent related surveys across scope, methodology, data modalities, application focus, and identified gaps of this literature review and also comparative analysis between Healthcare 4.0 and Healthcare 5.0.
- We presents a critical analysis across the three domains of DL in Healthcare 5.0.
- We perform a systematic assessment of methodological quality, validation practices, bias risks, and explainable AI (XAI) adoption across the three domains of DL in Healthcare 5.0.
- We conduct quantitative aggregation of classification accuracies across disease categories using benchmark datasets.

For clarity, the mapping between the stated contributions and their corresponding sections and tables is provided in Table 2.

The paper is organized as follows: Section 2 covers literature selection and bibliometric analysis; Section 3 reviews key DL algorithms in healthcare; Section 4 details the evaluation approach, focusing on disease prediction, medical imaging, and MMDL; Section 5 discusses challenges such as data availability, privacy, ethics, interpretability, and integration; Section 6 outlines future directions, including hybrid models, XAI, and federated learning; and Section 7 concludes the study. The conceptual framework is shown in Figure 1.

Table 2 Mapping of contributions to manuscript sections and tables

Contribution description	Addressed in section(s)	Supporting tables/ figures
Survey of deep learning architectures for Healthcare 5.0	Section 3	Figure 7
Use of generative models to address limited medical datasets	Sections 4.1–4.3	Table 10, 11, Figure 8, 9, 10, 11, 12, 13, 14
Bibliometric analysis of DL-based Healthcare 5.0 research (2018–2025)	Section 2	Table 5, 6, 7, 8 Figure 2, 3, 4, 5, 6
Comparative analysis of studies across three healthcare domains	Sections 4.1–4.3	Table 10, 11, 12, 13, 14, 15, 16 Figure 8, 9, 10, 11, 12, 13, 14
Identification of challenges, research gaps, and future directions	Sections 5–6	Table 17
Comparison with existing surveys and Healthcare 4.0 vs 5.0	Section 1.1	Table 1, 3, 4
Assessment of methodological quality, validation, bias risks, and XAI adoption	Section 5.4	Table 19
Quantitative aggregation of classification performance across disease categories	Section 5.5	Table 18

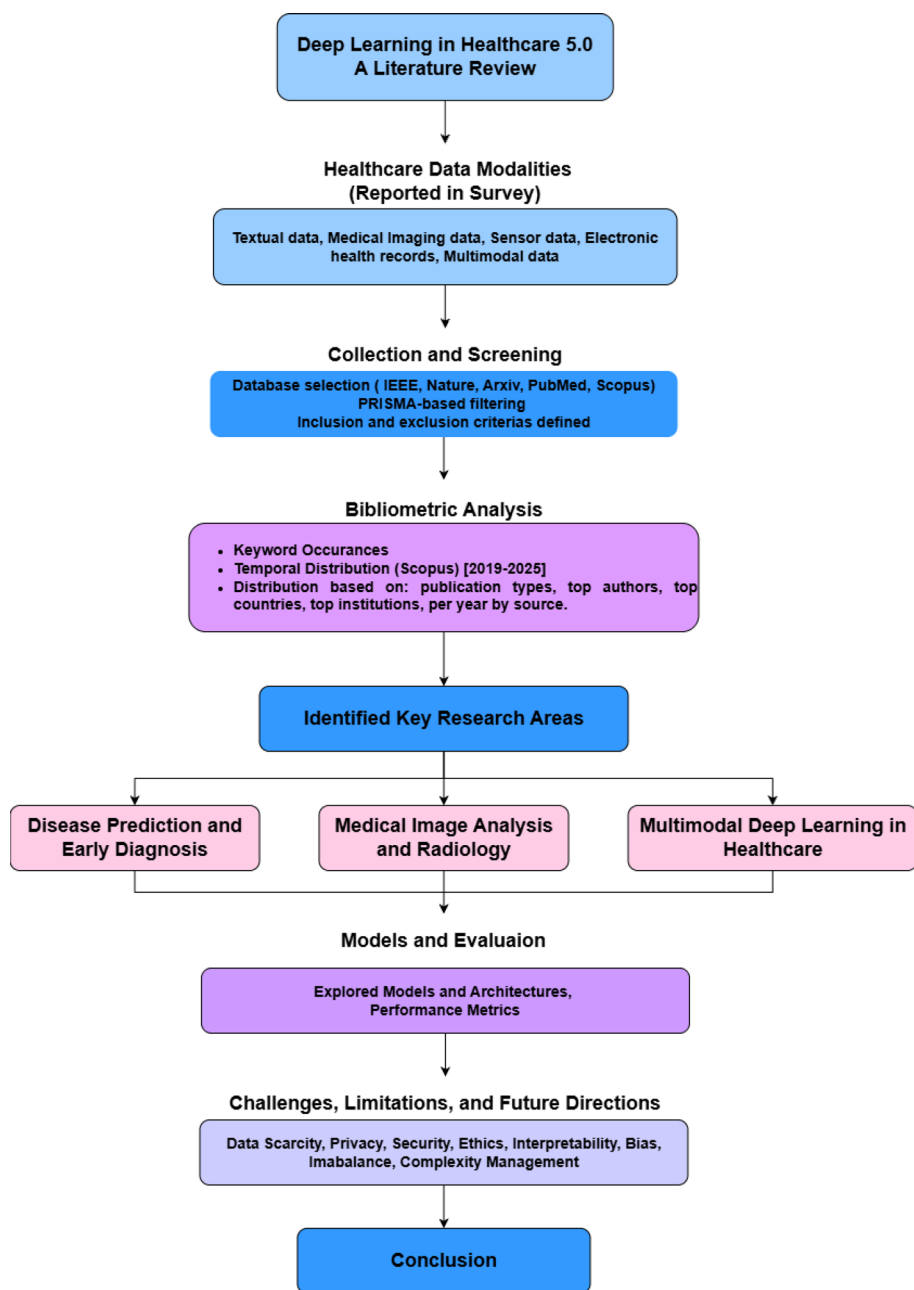


Fig. 1 Conceptual framework of the study

1.1 Novelty and contribution

Table 3 systematically contrasts the proposed review with recent related surveys across scope, methodology, data modalities, application focus, and identified gaps. This table clearly highlights how the present work extends beyond existing reviews by explicitly aligning DL techniques with the Healthcare 5.0 paradigm, integrating human centric, intelligent, and sustainable healthcare perspectives. The comparison clarifies the unique contribution and novelty of the review, thereby positioning it distinctly within the current body of literature.

Table 4 presents a brief comparison between representative prior survey works and our study on deep learning-based disease diagnosis. The reviews available are primarily about Healthcare 4.0, discuss AI/DL usage extensively, and are largely qualitative with limited or partial quantitative analysis. In contrast, our research, falls under the Healthcare 5.0 paradigm, is more comprehensive and structured, by reviewing a larger number of studies, systematizes trends, challenges, and future directions, and is supported by quantitative and critical comparative analysis.

2 Literature selection and bibliometric analysis

The selection of papers for this review on DL techniques in healthcare is conducted using a systematic search strategy on Scopus to ensure the inclusion of high-quality and relevant literature. The search is performed on November, 2025, using the following query:

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(TITLE-ABS-KEY ( ( "deep learning" OR "artificial intelligence" OR "machine learning" ) AND ( "healthcare" OR "health care" OR "medicine" OR "clinical" OR "patient monitoring" OR "clinical decision support" ) ) AND PUBYEAR > 2018 AND PUBYEAR < 2026 AND ( LIMIT-TO ( SUBJAREA , "COMP" ) ) AND ( LIMIT-TO ( DOCTYPE , "cp" ) OR LIMIT-TO ( DOCTYPE , "ar" ) OR LIMIT-TO ( DOCTYPE , "ch" ) OR LIMIT-TO ( DOCTYPE , "re" ) OR LIMIT-TO ( DOCTYPE , "cr" ) OR LIMIT-TO ( DOCTYPE , "bk" ) OR LIMIT-TO ( DOCTYPE , "ed" ) OR LIMIT-TO ( DOCTYPE , "sh" ) OR LIMIT-TO ( DOCTYPE , "no" ) OR LIMIT-TO ( DOCTYPE , "le" ) )
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This query is designed to identify research papers specifically focused on the application of DL techniques in the context of Healthcare 5.0, targeting relevant terms in the title, abstract, and keywords. To maintain a focused yet interdisciplinary scope, publications are limited to key subject areas such as Computer Science, Engineering, Medicine, Health Professions, and Decision Sciences domains that actively contribute to the advancement of Healthcare 5.0. Moreover, standard journal articles, reviews, books, chapters, and conference papers are considered to ensure the inclusion of peer-reviewed, impactful contributions.

VOSviewer (Bukar et al. 2023), a Java-based software tool widely used for constructing and visualizing bibliometric networks, is employed to perform a co-occurrence analysis of keywords extracted from 1,342 Scopus-indexed publications relevant to DL in Healthcare 5.0. Figure 2 presents the resulting keyword co-occurrence network, where each node represents a keyword appearing at least five times in the reviewed literature. The size of each

node reflects its frequency, while the distance and thickness of connecting lines indicate the strength of co-occurrence relationships. This visualization provides meaningful insights into dominant research areas and the conceptual connectivity within the field. The network visualization in Figure 2, generated by VOSviewer, highlights distinct thematic clusters in deep learning research applied to healthcare. Each color represents a major domain and its related concepts.

A. Red Cluster: This prominent cluster centers on "deep learning," "electronic health record," "diagnosis," "diseases," and "machine learning," capturing the intersection of medical informatics and AI. It encompasses topics such as disease detection, diagnosis, cardiology, patient treatment, biomedical engineering, clinical research, and healthcare data analytics. This area showcases how advanced ML and neural models are used for disease prediction, EHR analysis, and personalized medicine.

B. Blue Cluster: The green nodes focus on "machine learning," "human," "prediction," "algorithm," and "article," emphasizing clinical decision-making and AI evaluation practices. It features keywords like risk assessment, support vector machine(SVM), diagnostic accuracy, epidemiology, medical imaging, and procedure optimization. TThe networkis cluster relates to research on algorithmic healthcare solutions, comparative trials, and studies involving healthcare personnel and patient outcomes.

C. Green Cluster: The green nodes focus on "computer vision," "image enhancement," "feature extraction," "object detection," and "semantics," emphasizing image analysis and visual recognition technologies. It features keywords like object recognition, benchmarking, CNN, transformer architectures, feature fusion, extraction methods, semantic segmentation, real-time systems, agriculture applications, defect detection, robotics, and precision agriculture. This cluster relates to research on advanced image processing techniques, automated visual inspection systems, and the application of DL models for feature extraction and pattern recognition across multiple domains including medical imaging and smart agriculture.

D. Yellow Cluster: This cluster focuses on "bioinformatics," "computational biology," "molecular docking," "chemistry," and related biomedical research areas. It includes keywords such as molecular dynamics, brain tumor analysis, lesion segmentation, software tools, genomics, and image-based computational methods. This area represents the intersection of AI with molecular sciences and computational biomedicine, showcasing applications in drug discovery, protein analysis, and precision medicine at the molecular level.

Overall, the diagram illustrates extensive interactions among clusters, representing integrated research efforts in medical data science, clinical AI, diagnostics, and patient-centered innovations.

Table 5 summarizes the distribution of documents used for the bibliometric analysis across different publication types. The majority of the documents are journal articles (928), followed by conference papers (276) and reviews (73). A smaller number of book chapters (22), conference reviews (29), editorials (6), and books (8) are also included. This distribution highlights the predominance of journal articles in the research landscape, reinforcing the academic rigour and depth of inquiry within the domain of Deep Learning powered Healthcare 5.0. Table 6 presents the top 10 keywords by occurrence and link strength in deep learning-based healthcare research. "Deep learning" shows the highest frequency and connectivity, confirming its central role, followed by related AI terms such as "learning systems," "machine learning," and "artificial intelligence." Keywords like "diagnosis," "classification," and "diseases" highlight the dominant clinical applications, while "con-

Table 3 Comparison with existing reviews

Key literature work	Study type	Model types evaluated	Database used	Evaluation metrics	Pretraining analysis	Interpretability analysis	Computational complexity analysis	Comprehensiveness claim justification
Bohmrah and Kaur (2025)	Survey	Deep Neural Networks(DNNs), hybrid DNN+ML classifiers	public medical imaging datasets	Not empirical	Not covered	mentioned briefly	Qualitative remarks on Qualitative remarks on parameter counts and computational cost	The survey covers 117 studies in general in the domain of DNNs, hyperparameter optimization, and MH-based feature selection, but directly not compare it with recent similar reviews
Nia et al. (2023)	Systematic review	Statistical imputation, classical ML, DL, and multimodal fusion	Summarizes multiple disease areas	summarizes metrics used (e.g., imputation error, AUC)	Discusses only representation learning but not dedicated analysis	Not covered systematically	No structured complexity comparison	cover multimodal missing-data methods and applications, but no coverage in comparison to previous related surveys
Wang et al. (2025)	Empirical bibliometric	Bidirectional Encoder Representations from Transformers (BERT)-based text-mining pipeline	Web of Science papers (1,587) and Derwent patent records (1,314) on AI in healthcare (2018–2022)	no clinical performance metrics	first to apply Bidirectional Encoder Representations from Transformers (BERT)-based self-supervised models	No clinical interpretability analysis	qualitative cost comparison between BERT and classical text-mining methods, no de-tailed complexity benchmarks	Novelty by first BERT-based AI-healthcare landscape analysis and combined paper-patient view and not position against disease-specific DL reviews
Xu et al. (2025)	Survey	CNN, DNN, federated	Publicly available dataset	No unified empirical benchmark	Not covered	Not covered	Covered conceptually	Broad focus on edge computing, deep learning, and medical CV, but not provide sufficient comparative support of breadth
Hertel and Ben-lamri (2023)	Survey	CNN, RNN, hybrid DL, transfer learning	Multiple published works	accuracy, sensitivity, specificity, and AUC	Discusses transfer learning qualitatively	Brief discussion	No explicit analysis	Claims complete coverage of COVID-19 DL by modality and task, but does not explicitly compare this to previous coverage of COVID-DL

Table 3 (continued)

Key literature work	Study type	Model types evaluated	Database used	Evaluation metrics	Pretraining analysis	Interpretability analysis	Computational complexity analysis	Comprehensiveness claim justification
Nazir et al. (2025)	Narrative review	CNNs, RNNs, transformers, Variational Autoencoders (VAEs), GANs, Graph Neural Networks (GNNs), attention and holography-based models	Publicly available datasets	Not mentioned	Not covered	Briefly covers	No detailed analysis	Claims comprehensive DL-in-medicine coverage with holography emphasis, but lacks explicit comparison to prior reviews
Our study	Empirical comparative study	DL models (CNNs, RNNs, LSTMs, GRUs, GANs, transformers etc.)	Extensive healthcare datasets	Accuracy, positive predictive value, precision, sensitivity, AUC-ROC, recall rate, F1-score, mean, standard deviation	Comprehensive analysis	Detailed explainability analysis using multiple techniques	In-depth comparative analysis across different model architectures	First study to systematically compare mostly used DL models in healthcare across various datasets, detailed dataset descriptions, evaluation metrics, and interpretability metrics, providing a holistic and empirical assessment of strengths, limitations, and practical applicability

Key literature work	Focus	Health-care era	Novelty/Contribution	Papers reviewed	Quantitative analysis
Kaplanoglu et al. (2023)	AI (ML and DL) in healthcare	Health-care 4.0	Broad overview of AI methods, disease applications, and future challenges	200	Mostly qualitative
Munnangi et al. (2023)	DL in healthcare	Health-care 4.0	Mixed survival review with a proposed MAR-ERNN model	24	Yes
Kumar and Kaur (2024)	ML, DL in healthcare	Health-care 4.0	ML/DL healthcare overview	20	Limited
Our study	DL in healthcare	Health-care 5.0	Detailed review with categorizing trends, challenges, future directions, bibliometric analysis, critical review, comparative analysis	61	Yes

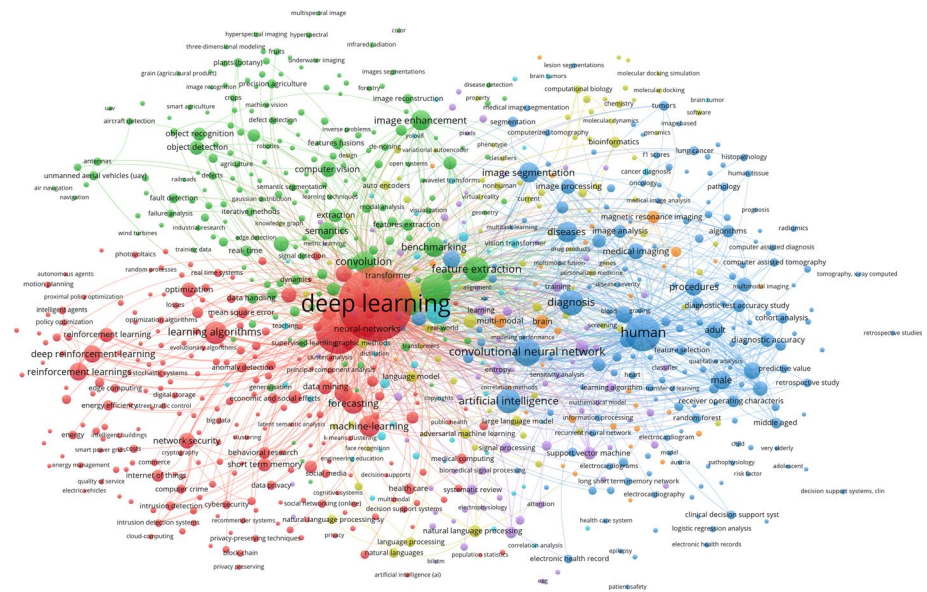
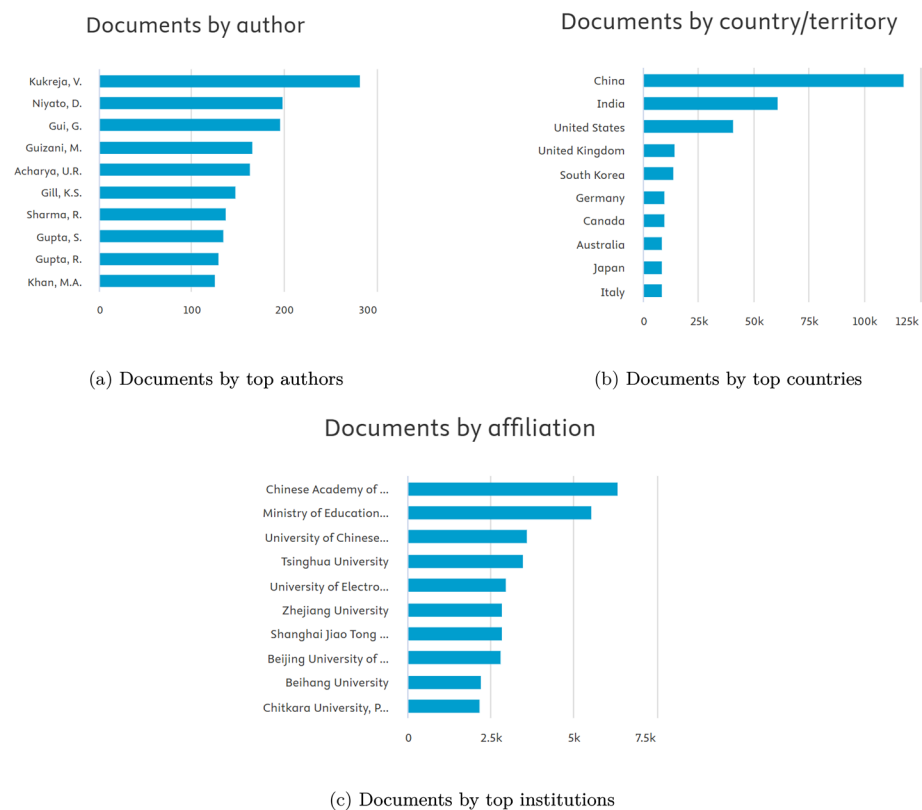


Table 5 Statistics of 1,392 Scopus documents used for the bibliometric analysis 2019-2025

Venue	No. of documents
Article	928
Conference	276
Review	73
Book chapter	22
Conference review	29
Editorial	6
Book	8

Table 6 Top 10 Keywords by Occurrence and Link Strength in Deep Learning in Healthcare

Keyword	Occurrences	Link strength
Deep learning	1299	13889
Learning systems	462	5860
Human	305	5314
Machine learning	221	3115
Convolutional neural networks	174	3052
Feature extraction	145	2499
Diagnosis	146	2311
Classification	148	2193
Artificial intelligence	140	2069
Diseases	101	1747

**Fig. 3** Bibliometric analysis showing the distribution of documents across top authors, countries, and institutions from Scopus

volutional neural networks” and “feature extraction” reflect the strong emphasis on image-driven methodologies. Figure 3 collectively summarize global research productivity across authors, countries, and institutions. The first chart, Documents by author, highlights the most prolific contributors, with Kukreja, V. leading by a significant margin, followed by Niyato, D., and Gui, G., indicating a concentration of publications among a few highly

active researchers. The second chart, Documents by country/territory, shows that China dominates global research output with the highest number of documents, followed by India and the United States, reflecting the strong and rapidly expanding research ecosystems in these regions. The third chart, Documents by affiliation, further emphasizes China's prominence, as leading institutions such as the Chinese Academy of Sciences, Ministry of Education China, and major universities like Tsinghua University and Zhejiang University produce the largest volume of documents. Together, these figures demonstrate that research productivity is heavily concentrated in a small group of top authors and is largely driven by major Chinese institutions and national research programs.

Figure 6 illustrates the systematic selection of studies using the PRISMA framework for DL techniques in Healthcare 5.0. Initially, 502 records were identified through comprehensive database searches in Scopus, with no additional records sourced from other channels. After removing duplicates, 389 unique records were retained and screened based on titles and abstracts. Studies were included if they focused on the application of DL, ML, or AI techniques in healthcare contexts (including disease prediction, medical imaging, diagnostics, and patient-centered care), were published between 2019 and 2025, and were peer-reviewed. Exclusion criteria at this stage included non-English publications, editorials without substantial research content, brief conference abstracts, and studies not directly relevant to Healthcare 5.0 or AI-enhanced medical systems. The screening process excluded 71 records that did not meet the initial inclusion criteria, leaving 288 full-text articles for detailed eligibility assessment. The full-text eligibility assessment involved evaluating these 288 articles for methodological rigor, relevance to Healthcare 5.0 paradigms, clarity of research objectives, quality of evidence, and contribution to the field. All assessments and decisions regarding inclusion/exclusion were conducted systematically by the research team. Criteria considered included research design, dataset characteristics, DL architecture employed, clinical application domain, validation methodology, and reported performance outcomes. Only studies meeting these rigorous standards were included in the final qualitative synthesis, ensuring a focused and high-quality set of sources for the comprehensive analysis of DL applications in Healthcare 5.0. The inclusion and exclusion criteria applied during screening and full-text assessment are summarized in Table 7. Each study was evaluated for relevance to Healthcare 5.0 and DL applications, methodological soundness, clarity of objectives, reproducibility, and quality of evidence. Only studies that satisfied these criteria were included in the final review, resulting in 288 articles for in-depth analysis.

Figure 4 shows temporal analysis of publications reveals a consistent upward trajectory in DL research applied to Healthcare 5.0 from 2019 to 2024, followed by a decline in 2025. Starting from a baseline of 22,316 documents in 2019, the field experienced steady growth, reaching 30,703 publications in 2020 (37.6% increase), 39,397 in 2021 (28.3% increase), and 49,803 in 2022 (26.4% increase). The momentum continued with 62,126 documents in 2023 (24.7% increase) and peaked at 66,319 publications in 2024 (6.7% increase), representing a nearly threefold increase from 2019 levels. However, 2025 shows a decrease to 56,795 documents (14.4% decline), which may be attributed to incomplete data collection as the year has not yet concluded at the time of this analysis. The substantial growth from 2019 to 2024 reflects the rapid advancement and increasing research interest in DL methodologies for healthcare applications, driven by improved computational resources, the availability of large-scale medical datasets, the COVID-19 pandemic's acceleration of digital health initiatives, and the demonstrated success of DL models in clinical diagnostics,

Table 7 Inclusion and Exclusion Criteria for Study Selection in PRISMA

Criteria	Inclusion	Exclusion
Publication year	2019–2025	Prior to 2019 or beyond 2025
Publication type	Peer-reviewed journal articles, conference papers, reviews, and book chapters	Editorials without research content, opinion pieces, short abstracts, preprints, non-peer-reviewed materials
Focus area	Studies focusing on DL, ML, AI applications in healthcare (disease prediction, diagnostics, medical imaging, EHR analysis)	Studies not directly related to healthcare or AI/DL applications in medical contexts
Health-care context	Research applicable to Healthcare 5.0, clinical decision support, patient-centered care, or medical informatics	Non-English publications, studies lacking healthcare relevance
Study content	Studies reporting empirical results, novel architectures, frameworks, datasets, or case studies relevant to Healthcare 5.0 with clear methodologies	Studies lacking methodological detail, clear outcomes, validation, or reproducibility
Subject areas	Computer Science, Engineering, Medicine, Health Professions, Decision Sciences	Studies from unrelated domains without healthcare applications

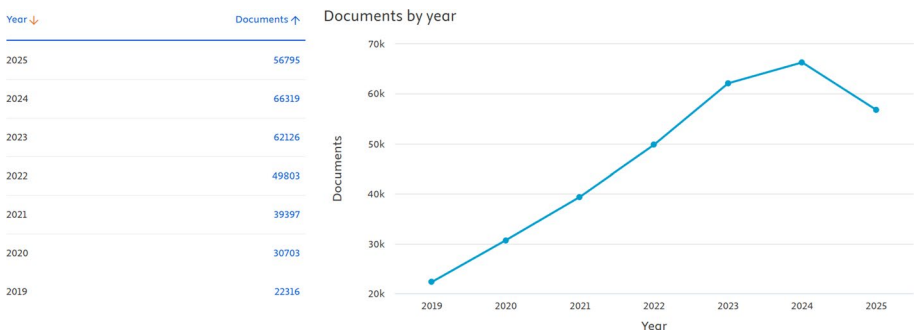


Fig. 4 Temporal distribution of publications on deep learning in Healthcare 5.0 from Scopus (2019-2025)

medical imaging, and personalized medicine. This exponential growth pattern underscores the transformative role of DL in advancing the Healthcare 5.0 paradigm.

Figure 5 illustrates the distribution of publications across major source venues reveals distinct patterns in research dissemination channels and evolving publication preferences within the DL healthcare domain. IEEE Access demonstrates the most dynamic growth trajectory, starting at approximately 1,100 documents in 2019, escalating to a peak of 2,300 publications in 2024, before declining to 2,000 in 2025, establishing itself as the leading venue for this interdisciplinary research. Lecture Notes in Computer Science (including subseries in Artificial Intelligence and Bioinformatics) maintained consistently high output between 1,900-2,050 documents annually from 2019 to 2024, with a notable decrease to approximately 900 in 2025, reflecting its role as a stable conference proceedings venue.

Documents per year by source

Compare the document counts for up to 10 sources.

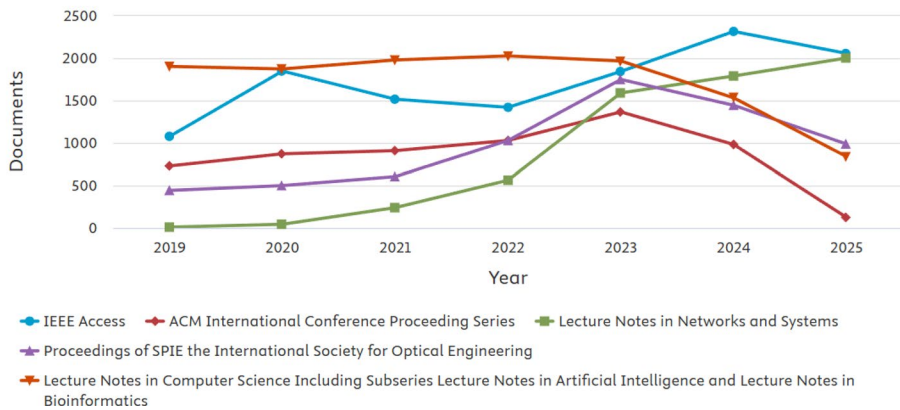
[Compare sources and view CiteScore, SJR, and SNIP data](#)

Fig. 5 Publication trends by major source venues in DL for Healthcare 5.0 from Scopus (2019-2025)

Lecture Notes in Networks and Systems exhibited the most dramatic growth, emerging from near zero publications in 2019 to surpassing 2,000 documents in 2025, indicating its increasing prominence as an outlet for systems-oriented healthcare AI research. Proceedings of SPIE (International Society for Optical Engineering) showed steady growth from 450 documents in 2019 to nearly 2,000 in 2025, highlighting the growing intersection between optical imaging technologies and DL in medical diagnostics. ACM International Conference Proceeding Series maintained moderate levels (750-1,400 documents) from 2019 to 2023, before experiencing a sharp decline to approximately 150 in 2025, suggesting a potential shift in publication venue preferences or changes in conference organization. These venue-specific trends reflect the multidisciplinary nature of DL in healthcare, spanning computer science, biomedical engineering, medical imaging, and clinical informatics communities, while also indicating evolving preferences toward open-access and rapidly published research outlets that facilitate faster knowledge dissemination in this fast-paced field.

Scopus is a widely used and comprehensive academic database, but its coverage carries inherent biases that can influence the results of a literature review. One of the most significant limitations is the strong dominance of English-language publications, which can lead to the underrepresentation of impactful research produced in other languages. This is particularly important for DL applications in Healthcare 5.0, as substantial work from regions such as East Asia, the Middle East, Latin America, and non-English-speaking Europe may not be fully captured. Scopus also tends to prioritize journals with higher citation indices and established publishing practices, creating a bias toward well-resourced institutions and regions with robust research infrastructures, potentially overlooking innovative healthcare AI solutions developed in resource-limited settings or published in regional medical journals. Additionally, the rapid evolution of DL techniques means that cutting-edge developments may first appear in preprints, technical reports, or specialized healthcare AI platforms that fall outside Scopus's indexing scope. Recognizing these limitations allows us to contextualize our bibliometric and PRISMA findings and highlights the need for future reviews to

incorporate multilingual databases, region-specific healthcare repositories, and complementary sources such as PubMed, IEEE Xplore, and arXiv to achieve a more globally inclusive and comprehensive perspective on DL innovations in healthcare. To minimize selection bias, two reviewers independently screened titles, abstracts, and full texts using the pre-defined inclusion and exclusion criteria, with discrepancies resolved through discussion and consensus (Figure 6).

2.1 Summary of included studies across analytical categories

Table 8 presents the distribution of the included studies across major analytical categories, highlighting the relative emphasis of research within the surveyed literature. Among the core domains, Medical Imaging constitutes the largest proportion (27.1%, $n=78$), reflecting the extensive application of deep learning techniques in image-based diagnosis, including modalities such as MRI, CT, X-ray, and histopathology. This is followed by Disease Prediction (22.2%, $n=64$), which encompasses a wide range of clinical applications such as cardiovascular disease, diabetes, Alzheimer's disease, cancer prognosis, and IoT-enabled remote patient monitoring, indicating a strong focus on predictive analytics for early diagnosis and risk assessment. Multimodal Learning accounts for 16.3% ($n=47$), demonstrating a growing trend toward integrating heterogeneous data sources—such as electronic health records, medical imaging, and clinical text to improve diagnostic accuracy and decision-making. Notably, a substantial portion of the literature falls under General/Supporting studies (34.4%, $n=99$), which include foundational topics such as explainable AI, deep learning architectures, ethics, fairness, and federated learning, underscoring the importance of methodological and systemic advancements in healthcare AI. Overall, the table reveals a balanced yet evolving research landscape, with strong dominance in imaging and predictive

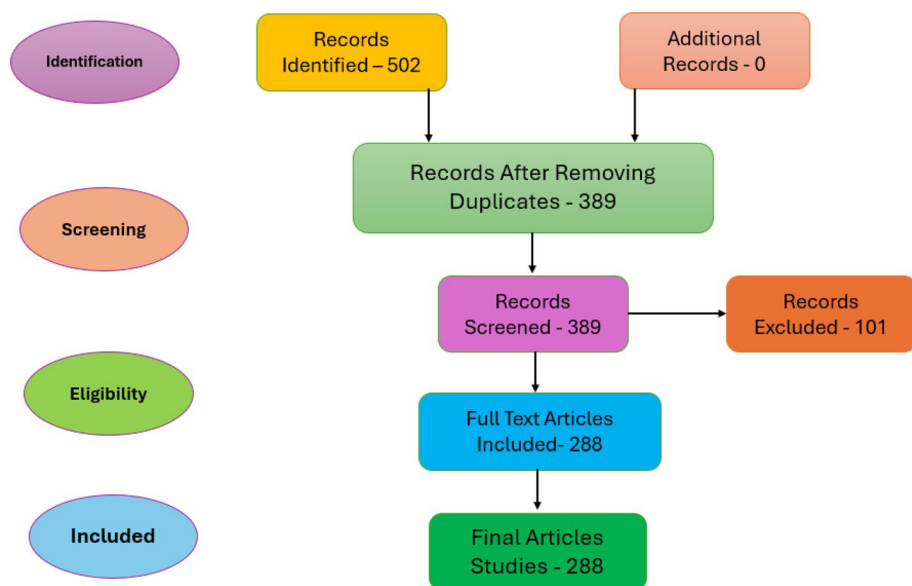


Fig. 6 PRISMA flow diagram for the systematic selection of literature on Deep Learning techniques in Healthcare 5.0

Table 8 Distribution of included studies by major analytical category

Category	Scope/Representative topics	Studies (n)	% of Total
Disease prediction	Cardiovascular, diabetes, Alzheimer's/dementia, kidney disease, arrhythmia detection, chronic disease risk, cancer prognosis, remote patient monitoring, IoT-based prediction	64	22.2%
Medical imaging	Computed Tomography (CT), MRI, X-ray, mammography, fundus/retinal imaging, ultrasound, histopathology; tumour/lesion detection & segmentation, computer-aided diagnosis, COVID-19 imaging	78	27.1%
Multimodal learning	EHR + imaging fusion, text-image models, multi-source clinical data, self-supervised & transformer-based multimodal architectures, multimodal disease diagnosis & prognosis	47	16.3%
General/supporting	XAI methods, DL architectures, ethics, fairness, federated learning, methodology, healthcare systems background	99	34.4%
Total		288	100%

modeling, alongside increasing attention to multimodal integration and supporting frameworks essential for real-world deployment.

3 Background

To provide foundational context for the subsequent discussions, this section outlines the fundamental deep learning architectures and supporting technologies relevant to Healthcare 5.0 applications. DL structures have fundamentally transformed artificial intelligence, enabling models to achieve record-breaking performance across a wide range of applications. In this section, we give an overview of the temporal analysis of the DL studies in healthcare, evolution trend with the initial development of rudimentary statistical and rules-based models, to complex, MMDL models and most important structures that have dominated contemporary AI: Generative Adversarial Networks (GANs), Transfer Learning models, transformers, federated learning, and multimodal large language models. Figure 7 shows various commonly used DL models in the healthcare domain.

3.1 Historical progression and year-by-year research evolution

Prior to 2016, the dominant ML models in healthcare analytics were convincingly standard statistical models and naive ML methods, like logistic regression, decision tree, and SVM. These models worked with structured data but had difficulties with unstructured data with high dimensions such as medical images or EHRs. The major constraints were feature engineering needs and the challenge in capturing non-linear, complex relationships (Schmaltz et al. 2021).

DL (and, in particular, CNNs in medical image and RNNs in sequential data (e.g., EHRs)) was rapidly adopted during the period between 2016 and 2018. Major papers proved that DL is better at image classification, image segmentation, and disease prediction. Indicatively, radiology and pathology have seen the use of CNNs that contributed to the breakthrough in automated detection of a disease such as cancer and diabetic retinopathy. This period was

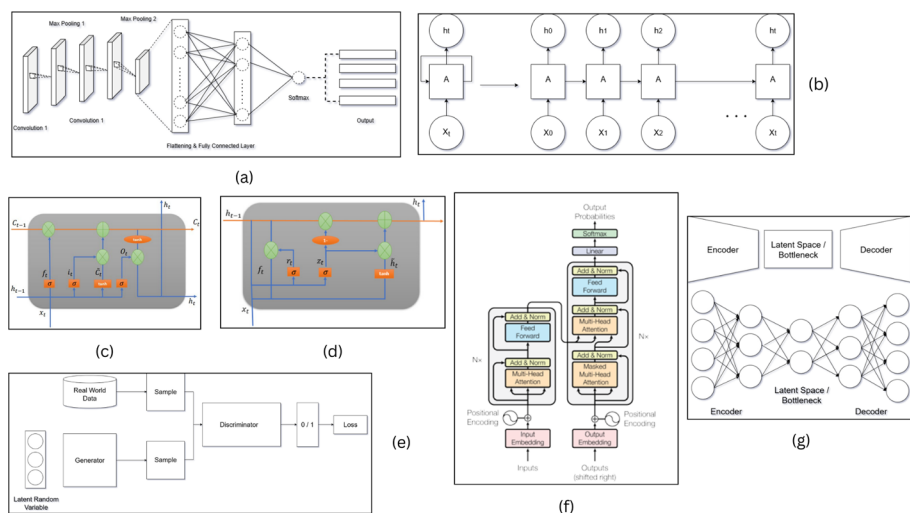


Fig. 7 Commonly used DL models in the healthcare domain, (a) Diagram of a CNN, (b) Diagram of a RNN rolled out to time, (c) Diagram LSTM cell (reproduced from Mateus et al., 2021 licensed under CC BY 4.0), (d) Diagram of GRU cell (reproduced from Mateus et al., 2021 licensed under CC BY 4.0), (e) Diagram of GAN architecture, (f) Transformer architecture (Reproduced from Vaswani et al., 2017 at Google, used under permission for scholarly activities.), (g) Diagram of Autoencoder model

marked by the shift of the hand crafting features towards the end-to-end learning to allow more robust and scalable solutions (Shenavarmasouleh et al. 2023).

From 2019 to 2021, research expanded beyond single-modality data to multimodal fusion, integrating imaging, genomics, and clinical notes. Attention mechanisms and transformers started gaining traction, allowing models to weigh the importance of different data sources and time points. For instance, models like Timeline introduced time decay factors and attention to capture disease progression patterns in EHRs, enabling more interpretable and temporally-aware predictions. This period also saw increased focus on interpretability, fairness, and ethical considerations, as deep learning models began to be deployed in real-world clinical settings (Bai et al. 2018; Kulikowski 2019).

Recent years have witnessed the integration of large language models (LLMs) and foundation models, enabling zero-shot and few-shot learning for healthcare applications. There is a strong emphasis on real-time inference, federated learning for privacy-preserving analytics, and edge computing for deployment in resource-constrained environments. The focus has also shifted toward personalized medicine, with deep learning models tailored to individual patient trajectories and multimodal data streams. Additionally, longitudinal bibliometric studies highlight a surge in publications, collaborations, and interdisciplinary research, indicating a maturing ecosystem (Rahman et al. 2024; Xie et al. 2025).

3.2 Convolutional neural networks (CNNs)

CNNs are applied in healthcare domain due to their ability to automatically generate clinically relevant features, local spatial dependencies, large high-dimensional data, and reliable and interpretable performance across a variety of imaging modalities. For example, CNNs are extensively used in healthcare with applications in oncology (Desai and Shah 2021),

tumor detection (Khan et al. 2022), and disease stage classification (Mahmud et al. 2023), among others.

3.3 Recurrent neural networks (RNNs)

RNNs are especially well applied to clinical problems that deal with sequential or time-dependent or variable data like patient history, physiology, and clinical text. Their memory, dependency modeling, and future health outcome forecasting capabilities render them essential in time-sensitive medical activities such as modeling the course of diseases, real-time monitoring, and predictive analytics. For example, they are used in Alzheimer detection (Nguyen et al. 2018), detecting medical events (Jagannatha and Yu 2016), etc.

3.4 Long short-term memory (LSTM) and gated recurrent units (GRUs)

LSTM and GRU are enhanced variants of RNN, which are designed to overcome the vanishing gradient problem. This makes them more suitable for capturing long term temporal dependencies in data. They find their applications in the healthcare domain, and are often used for ECG signal analysis (Lynn et al. 2019), predicting cardiovascular health risk (Islam et al. 2019), patient monitoring, and prediction of the disease progression, and so on.

3.5 Generative adversarial network (GAN)

A GAN is a neural network architecture, specifically consisting of competing components: a generator (G) and a discriminator (D). The generator creates imitations, and the discriminator predicts if it is real or generated (Creswell et al. 2018). It was first introduced in the groundbreaking study “Generative Adversarial Nets” by Goodfellow et al. in the year 2014 (Goodfellow et al. 2014). The G and D networks play the minimax game. D tries to maximize the capability of correct prediction, 1 for real and 0 for fake, whereas G produces data to minimize D’s ability to distinguish real from fake (Li et al. 2021).

GAN finds applications in various realms of healthcare. This includes generating synthetic data as the imitated ones are very similar to the real ones, as discussed in Aljohani and Alharbe (2022). It is commonly used for data augmentation. Yang et al. used TS-GAN (Time series GAN) to generate data for augmentation, as sensor data is not enough, and the model is able to produce data considering time and space effects (Yang et al. 2023). Vatanparvar et al. in their study proposed a model using GAN, which hides human speech from recorded data by producing sounds similar to humans, thus preserving privacy (Vatanparvar et al. 2019). Equation 1 represents how G tries to minimize losses and produce near results, and D tries to maximize the probability of correctly predicting real from fake.

Although the application of GANs in healthcare is very extensive in terms of data generation and augmentation, it carries with it significant risks that should not be overlooked. Mode collapse is a significant problem, in which the generator synthesizes a restricted range or repetitive patterns of the data distribution as opposed to the entire variety of the data distribution, decreasing the utility of the synthetic samples (Goodfellow et al. 2014; Salimans et al. 2016). The generation of hallucinated or anatomically implausible medical images is another issue, which might visually realistic yet contain errors that can mislead the further diagnostic systems (Yi et al. 2019). These shortcomings make it evident that implementing

GAN-based synthesis in healthcare settings requires strict validation, review by domain experts, and quality-control strategies.

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (1)$$

Where:

- G = Generator network,
- D = Discriminator network,
- $p_{\text{data}}(x)$ = Distribution of true data
- $p_z(z)$ = Prior distribution of input noise,
- x = Sample of real data,
- z = Random noise vector,
- $D(x)$ = Discriminator probability that x is real,
- $D(G(z))$ = Discriminator's probability that the generated sample is real,
- $\mathbb{E}$ = Expectation.

3.6 Transformer architecture

The transformer model was proposed by Vaswani et al. (2017) in their seminal paper titled *Attention Is All You Need*, which brought a great shift in the way researchers think. It is an encoder-decoder model, and mainly focuses on the self-attention mechanism. Equation 2 represents the attention mechanism in transformers, compares the queries with keys to find out the most relevant information, and then takes a weighted sum of values, so the output is focused on the important parts only. It is capable of parallel processing, and multi-headed attention allows understanding different contexts of the same input. It has brought a revolution in natural language understanding and processing. The model components include encoding and embeddings of input, which are passed to the encoder. The computer understands numbers and not words. Therefore, this step is crucial. Positional encoding allows us to generate vectors based on their position. The encoder model has a multi-head self-attention mechanism. This generates attention vectors that are fed to feed-forward Neural Network. Residual connections are added, and layer normalization is performed. The output and positional encoding are fed to the decoder. It has three components: masked multi-head attention, cross multi-head attention, and a position-wise feed-forward layer, two of them similar to the encoder. Masked attention masks the next words in the sequence to prevent the model from peeking, and the results depend on the outputs before the 'i'th stage. The residual connections and layer normalization steps follow. Finally linear and Softmax layers of the model. Softmax produces output in the form of probabilities for the next token.

Dosovitskiy et al., 2020 introduced Vision Transformers in the landmark paper 'An image is worth 16 X 16 words' and compared its performance to CNNs. This is DL model used for computer vision tasks. The model breaks images into patches which are analogous to tokens in Natural Language Processing (NLP) tasks. These are flattened into vectors, positional encodings are added, and passed into self-attention and feed-forward blocks. Finally a class token collects information from all parts of the image, and forms a condensed representation, which is used in the final layer for prediction. This model process the entire image

at once instead of scanning as done in CNN models. Its contribution healthcare includes automatically detecting diseases using X-rays images (Shome et al. 2021), detecting (Pacal 2024) and classifying types of cancer (Ali et al. 2023) among others.

In recent work, it has been demonstrated that transformer based architectures tend to be more effective at disease prediction than CNNs and RNNs due to their ability to treat long-range correlations, heterogeneous data, and scale to large datasets. Although CNNs are still useful in local image pattern, they can be challenging in terms of global context and intermodal association. RNNs can be applied in sequential data, but they have related problems like vanishing gradients and lack of parallelization. Transformers overcome these constraints by using self-attention, which allows enhancing feature extraction on images, text, signals, and EHR data. It renders them highly applicable to multimodal and context-aware Healthcare 5.0 applications.

Transformers are especially well-suited to clinical work since they can capture long-range dependencies through attention, combine multimodal data, transfer learners, scale effectively, and be explainable. This renders them suitable for EHR analysis, clinical text mining, multimodal fusion, drug discovery (Kotkondawar et al. 2024), and medical imaging (Gao et al. 2021) of Healthcare 5.0.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (2)$$

Where:

Q = Query matrix

K = Key matrix

V = Value matrix

d_k = Dimensionality of keys (needed for scaling)

QK^T = Dot product of queries and keys(for measuring the similarities)

$\sqrt{d_k}$ = Scaling factor

3.7 Autoencoders

Autoencoders are an unsupervised neural network model. It is used to learn data representations efficiently, i.e., they try to represent data in as few features as possible. The architecture of an autoencoder consists of three main components: the encoder, the bottleneck (also known as the latent space), and the decoder. The task of the encoder is to compress the input data into a representation that consists only of the essential features. This representation is called the latent vector. This creates a bottleneck as the size is significantly reduced from the input to this vector. This is the smallest layer. Finally, the decoder reconstructs the input data from the latent vector representations.

3.8 Transfer learning

Transfer learning gained prominence following the survey by Pan and Yang published in 2010(Pan and Yang 2009). It refers to a technique in which knowledge acquired from one

domain is applied to a different but related task. This is useful as it reduces the need for having a detailed dataset for each task and also significantly saves computation time by reducing the need for training from scratch. We can either use the whole model and just change the output target, or use only part of the base model and change it based on how similar the task is compared to the base model task. The two important terminologies in Transfer Learning are: Pre-training and Fine-tuning. Pre-training is the process of training a model on a large, generic dataset. This sets the parameters in the model. Fine-tuning, on the other hand, is using these pre-trained models to provide results for a very specific task or problem. They are mainly of three types: Inductive, Transductive, and Unsupervised. Commonly used Transfer Learning models include: Residual Networks (ResNet) (image classification), MobileNet (light-weight model for mobiles) for image data, Generative Pre-trained Transformer (GPT) (text generation), BERT (text classification), BART/T5 (summarization) for NLP, etc. Transfer Learning is most suited in healthcare due to the reduced reliance on large annotated datasets, reduced cost of training, enhanced generalization, and the ability to adapt to domains, which makes it extremely useful in practical healthcare tasks such as medical imaging (Jangam et al. 2022), detecting diseases (Sharma et al. 2021), drug target interactions (Zhang et al. 2024) and multimodal analytics.

3.9 Federated learning

Federated learning was introduced in a paper in 2018, authored by Google researchers Hard et al. (2018), who applied it to next-word generation for mobile keyboards. This is a collaborative method of training ML models, in which model goes to data, instead of collecting data in a central location. This type of learning is usually done to comply with data privacy and sharing norms of the client. Every client learns from data separately, and trains a model. The learning and updates of each model is sent to a central server, which is combined to improve the global model. This cycle is repeated multiple times. However, decentralization does not completely eliminate data security and privacy risks. Common threats to privacy include:

1. **Gradient Leakage Attacks:** An adversary observes and analyses the shared gradients. The attacker initializes a dummy input and iteratively updates the gradients to minimize the difference between the dummy input and to those of the original victim data. This can be curbed by gradient sparsification, which involves sending only the most important gradients from client to the server.
2. **Membership Inference Attacks:** The attacker tries to determine if a specific data record was used in the training set and potentially exposing the identity of the target. Differential privacy can help prevent such attacks by ensuring that no individual data point significantly affects the final model.
3. **Data Reconstruction Attacks:** Attackers can reconstruct data samples from shared gradients using GAN. Differential privacy also helps mitigate such attacks.

In healthcare domain, federated learning is used to collaboratively train models from disease prediction, while keeping patient data private within the premises (Islam et al. 2022).

3.10 Multimodal transformer model

A transformer model consists of an encoder-decoder architecture. The encoder maps the input into a sequence of representations, and the decoder generates an output sequence from those representations. A multimodal transformer architecture takes inputs of multiple modalities and converts the input tokens and embedding. It then performs self and cross attention. Self-attention is the process of finding relations within the same modality, while cross-attention is finding relationships across modalities. Multimodal transformer architecture can be single-stream, multi-stream, and hybrid-stream. In the single-stream model, all modalities are concatenated in one sequence. In a multi-stream architecture, separate transformers are used per modality. A hybrid-stream architecture is like the mix of single and multi-stream architectures (Xu et al. 2023).

In healthcare, it has been used to combine multiple modalities of data and make accurate predictions even if some data is missing. Multimodal Transformers are the future of health care because they combine and harmonize different clinical data (images, text, signals, EHRs) to make more accurate, holistic, and individual decisions in Healthcare 5.0 (Li et al. 2022).

3.11 Multimodal large language models

Instead of relying on text-only inputs, multimodal LLMs can reason and process multiple data types at the same time, thus, providing more context aware responses. They combine multiple types of neural networks into one unified model, and each modality is processed by a specialized encoder. These encoders convert the inputs to numerical representations, which are brought to the shared representation space to interact with each other, and jointly interpret the data.

Healthcare and medicine are also multimodal domains. Google DeepMind and Google Research researchers proposed the Medical Pathways Language Model Multimodal (Med-PaLM M)(Tu et al. 2024), which is capable of interpreting imaging data, clinical text and genomics. Med-PaLM M also outperforms specialist model and is clinically relevant. Large Language and Vision Assistant for BioMedicine (LLaVa-Med), by Microsoft (Li et al. 2023) is a conversational multimodal assistant. This model is cost efficient and is trained to comprehend medical image data, and answer questions about them. These models demonstrate the growing demand of multimodal LLMs, and encourages further exploration in healthcare domains.

3.11.1 Segment anything model

Segment Anything model has been developed by Meta. It is an groundbreaking image segmentation model AI model. It performs zero-shot generalization which means can segment objects from images and videos, which it has not seen before. It has been trained on SA-1B dataset, also introduced by Meta, specifically for this task. This is the largest dataset for image segmentation to-date, with over 1.1 billion segmentation masks and, over 11 million images. It even surpasses supervised learning technique results in many cases. It is "promptable" segmentation model, hence it allows users to generate valid masks based on prompts, and with minimal human input. Its architecture includes an image encoder, a prompt decoder

and a mask decoder, enabling the mask computation. Mazurowski et al. (2023) tested this model against 19 medical imaging datasets and concluded that its performances was variable. On some datasets it performed exceptionally well, while on other, it was moderate or even poor. Therefore, it has to be studied well before applying it in healthcare domain. Dong et al. (2024) adapted SAM for medical imaging and introduced EMedSAM, which outperformed state-of-the-art models in lungs and multiple organ segmentation.

3.12 Evaluation metrics and regulatory considerations

The metrics used to evaluate DL models in healthcare should be able to measure an aspect of predictive accuracy and clinical reliability. Such common measures are accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUROC) which is frequently used to estimate discrimination performance in clinical prediction problems. Provost and Fawcett (2013); Dm (2011). Standard metrics are used to test binary or multi-class predictions in disease diagnosis and early detection.

The accuracy represents the ratio of the samples that have been classified correctly but can be misleading in case of imbalanced samples (Sokolova and Lapalme 2009). Precision is a measure of consistency of the model to detect the presence of positive cases, and is defined as the fraction of times that the model is correct (Powers n.d.). Recall (sensitivity) is used to demonstrate the ability of the model to represent true positive cases, which is essential when a missed cases can be risky for disease diagnosis. The F1-score is a measure that balances between recall and precision and applies to cases where the distribution of the classes is not equal or the costs of errors are uneven. The ROC curve illustrates the trade-off between sensitivity and specificity across thresholds, while AUC indicates how well a model separates classes independent of decision boundaries (Fawcett 2006). Lastly, The confusion matrix generates a total breakdown of prediction results and helps identify systematic model errors. It assists in identifying the weakness of the model and it assists in making specific changes. These measurements jointly provide a holistic view of model performance, and allow more objective and clinically useful analysis of DL systems in Healthcare 5.0. Loss is used as a guide to model training by quantifying prediction errors in optimization, e.g. cross-entropy or mean squared error (Goodfellow et al. 2016).

The analysis of medical imaging is based on out-of-classification segmentation and detection measures. In radiology, tumors or organs Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) measure overlap in radiology segmentations (Hussain et al. 2024). Lesion detection in CT/MRI with sensitivity and specificity and mean Average Precision (mAP) of object detection tasks (Moassefi et al. 2023). These measures point out the compromises in Healthcare 5.0, where the high-resolution 3D imaging requires the effective calculation (Murad et al. 2024).

Multimodal fusion needs to be evaluated in a composite fashion. Concordance Correlation Coefficient (CCC) measures continuous predictions such as times of survival (Shenavarmasouleh et al. 2023). Calibration measures (e.g., Brier score, Expected Calibration Error) are used to make sure that the predicted probabilities are consistent with the results which is crucial when considering clinical decision-thresholds (Abdullah et al. 2022). New metrics encompass fairness-conscious (demographic parity) and uncertainty quantification (predictive entropy) to successful Healthcare 5.0 implementation (Ennab and Mcheick 2024). In addition to the performance indicators, healthcare AI should be in line with the

regulatory requirements. The guidelines on Good Machine Learning Practice (GMLP) by the U.S. FDA focus on the quality, transparency, and sound evaluation of the various phases of the model, whereas the 2021 guidelines by the WHO center on safety, fairness, and responsibility of clinical AI implementation (World Health Organization 2021). Including these considerations provides the foundation for understanding how the models are evaluated before they can proceed to the real world.

3.13 Model auditing frameworks for clinical AI

The auditing frameworks have emerged as one of the significant components of clinical AI since they assist in ensuring that the models act consistently with various groups of patients and in various clinical contexts. Model audits involve systematic checks for bias, robustness, explainability, and data governance to ensure that systems behave consistently within patient subgroups and across different clinical conditions. There are a number of proven structures that aid in this process. The proposed regulatory framework of the FDA of AI/ML-based Software as a Medical Device includes Predetermined Change Control Plan to guarantee the continued monitoring and safety auditing after deploying the system. The regulatory guidance issued by World Health Organization (WHO) on the international level identifies such measures as independent model assessment, documentation requirements, and ongoing audit cycles as the key protections of clinical AI (World Health Organization 2021). Tools that are research-driven also aid in this process. The Model Cards offered by Google are used to standardize its transparency reports, whereas the IBM's AI Fairness 360 toolkit analyze the bias and performance of their models across subgroups in a structured way (Mitchell et al. 2019; Bellamy et al. 2019). These models assist in connecting model creation to reliable clinical practices.

3.14 Explainability in clinical deep learning

Explainability in clinical deep learning can be broadly categorized into different approaches based on their design principles and application contexts.

3.14.1 Categories of explainability approaches

Explainability assists clinicians to recognize DL models arrive at their predictions. It consists of two primary strategies: interpretable-by-design models and post-hoc strategies that study black-box systems in terms of feature attributions, saliency maps, concept scores, counterfactuals, or surrogate rules (Ribeiro et al. 2016; Lundberg et al. 2017). Explainability aids in the auditing of models, emphasis of failure modes, and transparency in clinical contexts in Healthcare 5.0 (Mitchell et al. 2019; Amann et al. 2020).

3.14.2 Model-specific explainability techniques

Various model families depend on different explanation approaches. Grad-CAM, saliency, and LRP are common CNNs tools designed to demonstrate which parts of an image that impacting a diagnosis (Selvaraju et al. 2017; Samek et al. 2021). Temporal attention and relevance curves are explained by sequential models like RNNs, LSTMs, and GRUs that rely

on risk prediction after clarification by clinical events. The attention-map inspection and token-level attribution are used to detect influential text or image patches by transformers (Chefer et al. 2021). Autoencoders have been described using reconstruction-error maps and analysis of latent space, whereas GANs are used to inspect latent space and measure diversity to identify unrealistic generations (Frid-Adar et al. 2018; Yi et al. 2019). The transfer learning models have the advantage of layer-wise attribution in order to distinguish between pretrained and fine-tuned behaviors (Sinno and Yang 2009). Federated learning involves site-level aggregation of explanations in order to identify heterogeneity without information sharing. Cross-modal attention and multimodal counterfactuals are used by multimodal transformers and LLMs in demonstrating the impact of each modality on a decision (Tsai et al. 2019; Azizi et al. 2023). SAM and foundation segmentation models are based on interpretable masks that can be directly checked by clinicians (Kirillov et al. 2023). GNNs are described by subgraph and node-importance attribution to show the relational predictors of predictions (Ying et al. 2019).

3.14.3 General-purpose explainability methods

In all these types of models, general methods like Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP), Integrated Gradients, Testing with Concept Activation Vectors (TCAV), surrogate rules and counterfactual explanations, have remained the focus to generate legible, clinician-friendly interpretations (Ribeiro et al. 2016; Kim et al. 2018; Sundararajan et al. 2017; Wachter et al. 2017). Table 9 summarizes key explainability techniques used in healthcare deep learning, along with their model compatibility, typical applications, and associated limitations.

4 Deep learning in healthcare 5.0

Building upon the foundational concepts presented in the Background section, the following section reviews the major applications of deep learning in key Healthcare 5.0 domains. Deep Learning in Healthcare 5.0 represents the integration of advanced AI techniques with next-generation healthcare systems, emphasizing personalization, real-time decision-making, and human-centric care (Nia et al. 2023; Haleem et al. 2019). Unlike previous phases, Healthcare 5.0 leverages DL models, such as CNNs, RNNs, and transformers (Bhavya et al. 2019; Zengchen et al. 2023) for precise diagnostics, medical image analysis, predictive modeling, and patient monitoring, often powered by big data from IoT devices and EHRs (Chakraborty et al. 2024; Angel Nancy et al. 2022). It promotes smart, connected, and autonomous healthcare ecosystems, enabling early disease detection, robotic-assisted surgery, and intelligent health assistants, all while ensuring patient data privacy and ethical AI deployment (PS and Nataraj et al. 2025).

Table 10 presents an organized (year-wise) summary of the exhibited works for a concise overview; Tables 11 present the details of the datasets utilized. For clarity, the dataset descriptions are divided across three tables. Additionally, Table 17 summarizes the challenges and limitations reported by the authors, along with the identified advantages and disadvantages of the studies. A taxonomy of deep learning architectures commonly used in healthcare, including CNNs, RNNs, Transformers, and multimodal fusion models is repre-

Table 9 Key explainability techniques in healthcare deep learning: methods, applications, and limitations

Method	Type	Model used	Typical application	Key limitation
SHAP	Post-hoc, Model-agnostic	CNN, RNN, Transformers	Feature importance in disease prediction	High computational cost
LIME	Post-hoc, Model-agnostic	CNN, RNN, Multimodal	Local prediction explanation	Sensitive to parameter settings
Gradient-weighted Class Activation Mapping (Grad-CAM)	Post-hoc, Model-specific	CNN	Region visualization in MRI, CT, X-ray	Produces coarse heatmaps
Integrated Gradients	Post-hoc, Model-specific	CNN, RNN, Transformers	Feature attribution in structured data	Requires baseline selection
TCAV	Post-hoc, Concept-based	CNN	Concept-level clinical interpretation	Requires curated concept data
Saliency Maps	Post-hoc, Model-specific	CNN	Pixel-level importance visualization	Noisy outputs
Layer-wise Relevance Propagation (LRP)	Post-hoc, Model-specific	CNN, DNN	Layer relevance explanation	Sensitive to architecture
Deep Learning Important Features (DeepLIFT)	Post-hoc, Model-specific	CNN, RNN	Neuron contribution analysis	Baseline dependency
Attention Mechanisms	Interpretable-by-design	RNN, Transformers, Multimodal	Identify important features/modalities	May not reflect true importance
Counterfactual Explanations	Post-hoc, Instance-based	CNN, RNN, Multimodal	Minimal-change decision explanation	Hard to ensure realistic outputs
Surrogate Models	Post-hoc, Model-agnostic	Any black-box model	Simplified interpretable approximation	Reduced model fidelity
GNN Explainability	Post-hoc, Model-specific	Graph Neural Networks	Node/subgraph importance analysis	High computational complexity

sented. This taxonomy gives an overview of the structure of the reviewed model families in the subsequent subsections and is shown in Figure 8.

Explainable AI (XAI) techniques are incorporated to enhance the interpretability and transparency of DL models in Healthcare 5.0. Likely common examples of XAI tools include LIME, SHAP, Grad-CAM, and DeepLIFT, which motivate interpretability by highlighting which aspects of the features or input area of the model have the most significant influence regarding predictions. For instance, Grad-CAM identifies important areas in medical images that can be used to make a diagnosis, whereas SHAP and LIME measure the importance of features in unstructured and structured data. These tools are used by clinicians to comprehend model reasoning, confirm the prediction with clinical knowledge, and gain some confidence in AI-assisted decision-making.

4.1 Disease prediction and early diagnosis

DL is an emerging method that can be used in healthcare to anticipate and detect diseases before they occur and diagnose them at an early stage (Badawy et al. 2023). It is recognized for its ability to revolutionize the industry by enhancing the precision and timeliness of diagnoses, resulting in enhanced therapy and improved patient care (Mishra et al. 2020;

Table 10 Comparison of recent studies that use various DL techniques

Key literature	Publication year	Disease name	Risk prediction model	Accuracy (%)
1. Disease Prediction and Early Diagnosis				
Ortiz et al. (2016)	2016	Alzheimer's disease	CNNs, VGG16, ResNet, and Inception	acc: 0.90 ± 0.09 , sen: 0.86 ± 0.12 , spec: 0.94 ± 0.10 , AUC: 0.95
Choi et al. (2017)	2016	Heart failure	RNN, GRU	AUCs ranged from 0.777 (12-month window) to 0.883 (18-month window)
Rajendra Acharya et al. (2017)	2017	Heart disease	Deep CNN	Original data: acc: 93.47%, ppv: 97.87%, sen: 96.01%, spec: 91.64% Noise free Data: acc: 94.03%, ppv: 97.86%, sen: 96.71%, spec: 91.54% DDSM database: 97.4%, MIAS database: 96.7%
Jiao et al. (2018)	2018	Breast cancer	Deep CNN (CNNs layers, metric learning layers)	AD vs. NC : 98.74% (detection rate: 100% and false alarm rate: 2.4%)
Bäckström et al. (2018)	2018	AD	3D-CNN	AD vs. NC classification(%): acc: 94.82, sen: 97.7, spec: 92.45, AUC: 96.76 pMCI vs. NC classification(%): acc: 86.36, sen: 83.33, spec: 88.78, AUC: 91.11 MCI vs. NC classification(%): acc: 65.35, sen: 70.59, spec: 59.63, AUC: 69.17
Feng et al. (2019)	2019	AD	3D-CNN, FSBi-LSTM, softmax	AD versus NC (%): acc: 99.10 ± 1.13 , sen: 99.80 ± 0.37 , spec: 98.40 ± 1.17 AD versus MCI (%): acc: 89.40 ± 6.90 , sen: 86.70 ± 9.10 , spec: 84.00 ± 4.80 MCI versus NC (%): acc: 98.90 ± 2.78 , sen: 98.90 ± 3.69 , spec: 98.80 ± 0.63 ternary classification: $95.74 \pm 2.31\%$ acc: 97.01%, sen: 94.7%, spec: 100%, f-Measure: 96.9%
Feng et al. (2020)	2020	AD	3D-CNN-SVM	
Çinar and Yildirim (2020)	2020	Brain tumor	CNN, Alexnet, Resnet50, InceptionV3, GoogleNet and Densenet201	
Alazab et al. (2020)	2020	COVID-19	CNN with VGG16 architecture	Australia region: prediction accuracies for confirmations, recoveries, and deaths 99.94%, 90.29%, and 94.18% Jordan region: obtained prediction accuracies 99.08%, 79.39%, and 86.82%

Table 10 (continued)

Key literature	Publica- tion year	Disease name	Risk prediction model	Accuracy (%)
Angel Nancy et al. (2022)	2022	Heart disease	Fuzzy Inference System (FIS), Bi-LSTM	acc: 98.86%, prec: 98.9%, sen: 98.8%, spec: 98.89%, f-measure: 98.86%
Singh et al. (2022)	2022	CKD	DNN	acc:100% recall:100% prec:100% f-Measure:100%
Aldughayfiq et al. (2023)	2023	Retinoblastoma	Transfer Learning Inceptionv3 with LIME and SHAP	acc: 97%, prec: 98.8%, recall: 99.6%, fl-score: 99.2%
Yashudas et al. (2024)	2024	CVD	BiGRU, IoT network	acc: 99.9%, fl-score: 99.53%
Mandava et al. (2024)	2024	CVD	MDensNet201-IDRSRNet, Relief, LASSO, IDRSNet	spec: 98.85%, sen: 98.90%, prec: 98.93%, recall: 99.01%, fl-score: 98.95%, acc: 99.12%
Mahmud et al. (2024)	2024	AD	VGG16, VGG19, DenseNet169, DenseNet201, Ensemble 1 (VGG16, VGG19), Ensemble 2 (DenseNet169, DenseNet20), proposed model (Efficient-NetB3, CNN), XAI method (saliency maps, Grad-CAM)	acc:96%, prec: 89%, recall: 93%, fl score: 91%
Wani et al. (2024)	2024	Lung cancer	CNN, XGBoost with SHAP	acc: 97.43%, sen:98.71%, and fl-score: 98.08%
Tanim et al. (2025)	2025	Type-2 diabetes	DeepNetX2	without feature selection: PIMA dataset: acc: 92.21%, prec: 94%, recall: 92%, fl-score: 92% Local Private dataset: acc: 95.74%, prec: 96%, recall: 96%, fl-score: 96% Type-2 diabetes dataset: acc: 99.50%, prec: 100%, recall: 99%, fl-score: 100%
Ramu et al. (2025)	2025	CKD	Hybrid CNN-SVM model	CKD: precision: 86%, recall: 100%, fl-score: 92% Non-CKD: precision: 100%, Recall: 75%, fl-score: 86% overall accuracy: 96%

Table 10 (continued)

Key literature	Publication year	Disease name	Risk prediction model	Accuracy (%)
Shehta et al. (2025)	2025	Blood cancer	ResNetRS50, RegNetX016, AlexNet, Convnext, EfficientNet, Inception V3, Xception, VGG19	ResNetRS50: accuracy: 97%, recall: 99%, f1 score: 98% RegNetX016: accuracy: 96.4%, recall: 99%, f1-score: 98% AlexNetacc: 87.9%, recall: 90%, f1-score: 88% ConvNext: accuracy: 94.2%, recall: 94.6%, f1-score: 95% EfficientNet: accuracy: 93.0%, recall: 88%, f1-score: 91% Inception_v3: accuracy: 92.7%, recall: 88%, f1-score: 91% Xception: accuracy: 91%, recall: 88%, f1-score: 92% MOD-ID-CNN: accuracy: 70%, precision: 68%, recall: 69%, f1-score: 68% MOD-2D-CNN: accuracy: 95%, precision: 93%, recall: 93%, f1-score: 93%
Maji et al. (2025)	2025	Dementia	MOD-ID-CNN and MOD-2D-CNN	
2. Medical Imaging Analysis and Radiology				
Hua et al. (2015)	2015	Lung cancer	Deep Learning frameworks: DBN, CNN	Sensitivity 1. CNN: 73.4% 2. DBN: 73.3%
Gulshan et al. (2016)	2016	Diabetic retinopathy	Deep Learning algorithm	EyePACS-1: sensitivity: 97.5%, specificity: 98.1% Messidor-2: sensitivity: 96.1%, specificity: 98.5%
Cheng et al. (2016)	2016	Breast and lung cancer	Stacked Denoising Autoencoder (SDAE)	Lung CT: i. Single strategy: 87.4% $\pm$ 3.3 accuracy ii. All strategy: 94.4% $\pm$ 3.2 accuracy Breast US: 82.4% $\pm$ 4.5 accuracy
Zilly et al. (2017)	2017	Glaucoma	CNN	94.1%
Horie et al. (2019)	2018	Esophageal cancer	Single shot multi-box detector CNN	Superficial cancer: 99% advanced cancer: 92% ESCC diagnosis: 99% EAC diagnosis: 90%
Li et al. (2018)	2018	Glaucomatous optic neuropathy	Deep CNN with Inception-v3 model	AUC of 0.986 sensitivity of 95.6%, specificity of 92.0%
Jeyaraj and Samuel Nadar (2019)	2018	Oral cancer	Regression-based partitioned deep CNN	91.4% (small dataset), 94.5% (large dataset); sensitivity: 0.94, Specificity: 0.98

Table 10 (continued)

Key literature	Publication year	Disease name	Risk prediction model	Accuracy (%)
Song et al. (2019)	2019	Thyroid nodules (identifying benign and cancerous)	Deep Learning algorithm (DLA) using Inception-V3	Internal dataset: sensitivity: 95.2%, Negative Predictive Value (NPV): 95.5%. External dataset: Sensitivity: 94.0%, NPV: 90.3% 99.60%
Khamparia et al. (2021)	2020	Skin cancer	Pretrained CNN models (VGG19, ResNet50, Inception V3, and SqueezeNet) and Transfer learning	99.60%
Pandit and Bandy (2020)	2020	COVID-19	VGG-16 CNN	Binary classification: Accuracy: 96%, sensitivity: 92.64%, specificity: 97.27% Multi-class classification: Accuracy: 92.53%, sensitivity: 86.7%, specificity: 95.1% F1-score: 97.31%
Kuchana et al. (2021)	2020	COVID-19	2D CNN with U-Net	
Elsharkawy et al. (2021)	2021	COVID-19	MGRF model and a Neural Network-based fusion system for analysis	The CAD system achieved: sensitivity: 100% specificity: 97% $\pm$ 3% accuracy: 98% $\pm$ 2% DSC: 98% $\pm$ 2% 99.74%
Chattopadhyay and Maitra (2022)	2022	Brain tumor	CNN with RMSProp optimizer and Softmax activation in final layer	
Naz et al. (2023)	2023	Pulmonary diseases (Nodules, tuberculosis, pneumonia, COVID-19, and pulmonary edema)	ResNet50 CNN Model, LIME	COVID-CT dataset: 93% COVID Net dataset: 97%
Huy and Lin (2023)	2023	Tuberculosis	CBAMWDnet	Accuracy: 98.80% sensitivity: 94.28% specificity: 95.7% F1-score: 96.35% precision: 98.50%

Table 10 (continued)

Key literature	Publica- tion year	Disease name	Risk prediction model	Accuracy (%)
Sharma and Guleria (2023)	2023	Pneumonia	Neural Networks with VGG-16	1. Dataset 1: 92.15% accuracy 2. Dataset 2: 95.4% accuracy
Mahmud et al. (2023)	2023	Brain tumor	CNN	Accuracy: 93.3% AUC: 98.43% Recall: 91.1% Loss: 0.25
Odusami et al. (2023)	2023	Alzheimer's disease	In-3-channel ResNet18	Accuracy: 73.90% specificity: 94.32% sensitivity: 66.74%
Hroub et al. (2024)	2024	Lung disease (COVID-19 and pneumonia)	Inception-V3, CROP, ROT, HF	Pneumonia dataset: 94.55% Covid19- & Pneumonia dataset: 97.44%
Kar et al. (2024)	2024	Intracranial hemorrhage (ICH)	CNN	Accuracy: 99.76% precision: 99.53% recall: 99.68%
Tawfeek et al. (2025)	2025	Lung cancer	CNN	Accuracy: 98.5% specificity: 88.7 % sensitivity: 89 % precision: 89.2% recall: 99.68%
3. Multimodal Deep Learning in Healthcare				
Liu et al. (2014)	2015	AD	Stacked Autoencoder (SAE) with a softmax logistic regressor	Binary classification (MR-only): Accuracy :82.59%, sensitivity: 86.83% Multiclass classification (MR-only): Accuracy: 46.30%, specificity: 77.78% Multiclass classification (MR and PET): Accuracy: 53.79%, specificity: 86.98%
Hsu et al. (2018)	2018	Thoracic disease (Retrieval relevance according to ICD-9, this information does not apply at the training time however)	DenseNet-121: Image features, Deep Averaging Network (DAN) encoder: Textual, Embedding Alignment (EA), Adv(Adversarial alignment), EA and Adv (Semi-supervised), Principal Component Analysis (PCA): dimensionality reduction	Retrieval tasks: Text to image, image to text Measure: MRR, cosine similarity, nDCG@1/10/100 Best unsupervised bi-gram EA obtained nDCG@100

Table 10 (continued)

Key literature	Publication year	Disease name	Risk prediction model	Accuracy (%)
Hao et al. (2019)	2019	Cerebral infarction	MD-RCNN (Multimodal Recurrent Convolutional Neural Network) : [Deep Belief Network (DBN), RCNN (Recurrent Convolutional Neural Network)] Multimodal RNN (multi-GRU + logistic regression)	Accuracy: 96%, Recall: 98.08%
Lee et al. (2019)	2019	AD		Accuracy: 81% Sensitivity: 84 % Specificity: 80 %
Li et al. (2020b)	2020	Retinal disease diagnosis [Age-related macular degeneration (AMD), Pathologic Myopia (PM), Diabetic Retinopathy (DR)]	CycleGAN, ResNet18 Backbone	Unsupervised: 1. Ichallenge-AMD dataset: AUC: 74.58%, Accuracy: 86.58%, Precision: 83.20%, Recall : 74.58%, F1-score: 77.33% 2. Ichallenge-PM dataset: AUC: 98.55%, accuracy: 98.65% , precision: 98.60%, recall : 98.55%, F1-score: 98.57%
Venugopalan et al. (2021)	2021	AD, mild cognitive disorders (MCI), controls	3D-CNN, Auto-encoders	Supervised: 1. Ichallenge-AMD dataset: AUC:77.19% 2. Ichallenge-PM dataset: AUC: 98.04
Sheu et al. (2022)	2022	Pneumonia	Deep Learning Classification (MDA-PSP) system [Hybrid DNN with Batch Normalization (Dense-BN)]	(MRI, EHR, and SNP)- internal cross-validation accuracy-79% and external test set accuracy-78% Dense-BN accuracy: 0.77 (vitals), 0.92 (CXR), 0.75 (combined with weights)
Lee et al. (2023)	2023	CVD	Multimodal Deep Neural Network (DNN + CNN: DenseNet-169)	AUROC: SMC: 0.781 (95% CI 0.766–0.798) UK Biobank: 0.872 (95% CI 0.857–0.886) for multimodal model

Table 10 (continued)

Key literature	Publication year	Disease name	Risk prediction model	Accuracy (%)
Zhang et al. (2023)	2023	Parkinson's disease	TL-TAGCN: Time-contrastive learning with Topology Adaptive Graph Convolutional Network	1. PPMI data set: Accuracy: 94.6 % Precision: 0.963±0.011 Recall rate: 0.969±0.017 F1 score: 0.964±0.006 2. PS data set: Accuracy: 92.9 % Precision: 0.969±0.062 Recall rate: 0.945±0.045 F1 score: 0.955±0.027
Li et al. (2023)	2023	Colorectal cancer AD	DeAF: A two-stage Deep Learning framework: MA-SimSiam (unsupervised multimodal alignment) and SAF (Self-Attention Fusion) module	1. Colorectal cancer: Accuracy: 55.15 % 2. ADNI: Accuracy: 69.9 %
Intriago et al. (2023)	2023	AD	Self-Supervised Multimodal Representation Learning (SSMRL) framework: Balanced Random Forest (BRF) + CyCLIP cost function + 3D CNN (sMRI) + BiLSTM (fMRI)	AUC scores: 1.sMRI: 98.30 % ± 1.53 % (multi) 2. fMRI: 97.70 % ± 1.47 % (multi)
Thiruvenkadam et al. (2024)	2024	Brain tumor	Five-layer CNN incorporating dropout and max pooling, assessed using XAI techniques	98.9% validation accuracy after 40 training epochs
Mahalakshmi et al. (2024)	2024	Breast cancer	CNN, RNN	Metrics shown as 1.0; exact values unspecified
Krishnaveni et al. (2024)	2024	Chronic obstructive pulmonary disease (COPD)	MultimodNet; FCN (for PFTs) + DenseNet121 (for CXR)	Accuracy: Initial stage: 95.231% Progressive stage: 95.008% Complicated stage: 94.989% Critical stage: 94.256%

Table 10 (continued)

Key literature	Publica- tion year	Disease name	Risk prediction model	Accuracy (%)
Ding et al. (2024)	2024	Type 2 Diabetes Mellitus (T2DM)	LLMs (LLMs) including BiomedBERT, ClinicalBERT, and GPT-2, DNNs, and SHAP	Multimodal : (A) textual laboratory values, (B) clinical notes, and (C) laboratory values BiomedBERT-93, ClinicalBERT-90, SciFive-93, RoBERTa-92, Flan-T5-base-220 M-93, Flan-T5-large-770 M-92, BERT-93, GPT-2-92
Ranjith and Sakthivanitha (2025)	2025	Cataracts and glaucoma	Hybrid MMDL framework: ResNet/EfficientNet, attention mechanisms, CNN, RCNN, GNN and explainability tools: SHAP and Grad-CAM	Accuracy: 95 % (cataract), 92 % (glaucoma) AUC-ROC: 0.97 (cataract), 0.94 (glaucoma)
Kumar et al. (2025)	2025	AD	DenseNet (primary); CNN, VGG-16, MobileNet (comparative baseline models), Adam optimizer	DenseNet accuracy: 81.5% MobileNet accuracy: 57.4% CNN accuracy: 50.8% VGG-16 accuracy: 26.7 %
Gezimati and Singh (2025)	2025	Breast cancer	Custom 23-layer CNN; decision-level (late) fusion architecture	Validation Accuracy:96.6% Sensitivity: 96.3% AUC: 0.99
Shi et al. (2025)	2025	Glioma (brain tumor)	SAM [ViT-based image encoder, mask decoder, and prompt encoder]	FAHZU private dataset: Dice score: 88.81% and HD95: 9.83 BraTS2020 public dataset: Dice score 89.25% and HD95: 10.03 Best AUC: 0.9346 (MAND-LSTM for stroke)
Tsai et al. (2025)	2025	Diabetes mellitus, heart disease, stroke, hypertension	Multimodal Multitask Learning (MTL) [Multimodal Attention Network for Dementia (MAND) with logistic regression (LR), multi-layer perceptron (MLP), LSTM, and Multi-Head Self-Attention (MHSA), CTR models: Fac-torization Machines (FM) and Deep Convolutional Network (DCN)]	

acc = accuracy, ppv = positive predictive value, prec = precision, sen = sensitivity, spec = specificity, AD = Alzheimer’s disease NC = Normal control, pMCI = progressive state, MCI = Mild cognitive impairment

Table 11 Dataset description

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
1. Disease Prediction and Early Diagnosis						
Ortiz et al. (2016)	ISIC archive dataset (International Skin Imaging Collaboration)	Contains dermoscopic images of skin lesions	900 dermoscopic images	Images are labeled as benign or malignant skin lesions. The dataset is specifically curated for skin cancer classification	https://challenge.isic-archive.com/landing/2016/41/?utm_source=chatgpt.com	Python, TensorFlow, Keras, OpenCV, and NumPy
Choi et al. (2017)	Sutter Palo Alto Medical Foundation (Sutter-PAMF) primary care patients	EHRs	32,787 patient records (3,884 HF cases and 28,903 controls)	Time-stamped clinical data such as: Diagnoses, medication orders, procedure orders	-	Theano, Skip-gram
Rajendra Acharya et al. (2017)	PhysioBank MIT-BIH arrhythmia database	Imbalanced ECG data	104.3 MB	-	https://physionet.org/content/mitdb/1.0.0/	MATLAB (Mathlab, Natick, MA, USA) software
Jiao et al. (2018)	DDSM and MIAS database	Mammography (X-ray imaging)	SM database- 600 images MIAS database- 1.51 GB	-	DDSM database- http://www.eng.usf.edu/cvprg/mammography/database.html MIAS database https://www.repository.cam.ac.uk/handle/1810/250394	Deep Learning toolbox, MatConvNet and CUDA
Bäckström et al. (2018)	ADNI	Brain MRI scans	-	subjects- 340 MRI brain scans-1198 (AD- 600, NC-598)	https://adni.loni.usc.edu/data-samples/adni-data/	Python, Keras Deep Learning library built on TensorFlow backend
Feng et al. (2019)	ADNI	MRI, PET	-	AD-93, pMCI-76, sMCI-128, NC-100	https://adni.loni.usc.edu/	Keras library, TensorFlow as backend

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Feng et al. (2020)	ADNI	Brain MRI images	-	subjects-469 MRI samples-3127	https://adni.loni.usc.edu/	Python version 2.7.12, Keras, TensorFlow
Çinar and Yildirim (2020)	Kaggle Repository	Brain MRI images	-	without tumor-98 with tumor-155	https://www.kaggle.com/datasets	Matlab
Alazab et al. (2020)	Kaggle	Chest X-ray images	Original dataset: 97 MB 128 images Augmented dataset: 1000 images	Two types of data: A healthy dataset (chest X-ray images of healthy persons) and A COVID-19 dataset (chest X-ray images of COVID-19 patients)	https://www.kaggle.com/datasets/nabeelsajid917/covid-19-x-ray-10000-images	Python, ARIMA, Fpprophet, ImageDataGenerator, Keras, LSTM, Matplotlib, NumPy, Pandas, Scikit, SciPy, TensorFlow
Angel Nancy et al. (2022)	Cleveland and Hungarian dataset UCI machine learning repository	Physiological data, electronic clinical data (ECD)	658 Kb	-	https://archive.ics.uci.edu/dataset/45/heart+disease	Wireless body sensor networks, TensorFlow ML package, Apache Spark, and Cassandra
Singh et al. (2022)	UCI Machine Learning repository	Clinical data (patient health records)	400 patients	24 clinical features. 250 instances of CKD and 150 instances of non-CKD	https://archive.ics.uci.edu/	Python Scikit-learn

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Aklughayfiq et al. (2023)	Retinoblastoma fundus images-MathWorks Retinoblastoma dataset, Google images non-retinoblastoma fundus images-Messidor dataset	Fundus images	-	800 fundus images (400 retinoblastoma and 400 normal images)	-	Python with Keras, Lime, and SHAP libraries for model training and explainable AI methods, and Google Colab with a Tesla T4 GPU for computation
Yashudas et al. (2024)	1. Framingham's heart dataset: US National Institutes of Health's (NIH) Biospecimen Information Coordination Center and Data Warehouse 2. Statlog heart dataset: Kaggle	Epidemiological data, physiological data	1. Framingham's heart dataset: 2. Statlog heart dataset: 141 Kb	4240 records, 15 attributes	1. Statlog heart dataset: https://archive.ics.uci.edu/dataset/et/98/statlog+project	biosensors, Cloud Simulator (CloudSim)
Mandava et al. (2024)	Hungarian, Cleveland, Long Beach, VA, Switzerland	Statlog dataset: UCI	1. Hungarian, Cleveland, Long Beach, VA, Switzerland dataset: 125.9 KB 2. Statlog dataset: 4.3 KB	303 records, 14 attributes	Hungarian, Cleveland, Long Beach, VA, Switzerland dataset: http://www.eng.usf.edu/cvprg/mammography/dataset.html Statlog dataset: https://www.repository.ca.m.ac.uk/handle/1810/250394	Python

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Mahmud and Barua (2024)	Kaggle Oasis	MRI scans	36 MB	Mild dementia-896 images moderate dementia- 64 images Non-dementia-3200 images Very mild dementia- 2240 images	-	Python, TensorFlow, Keras, OpenCV, Scikit-learn, Google Colab
Wani et al. (2024)	Survey Lung Cancer (SLC)	Numerical data and categorical data	-	309 samples with 16 features	-	TensorFlow, Keras
Tanim et al. (2025)	1. Type-2 diabetic dataset (IEEDDataPort) from Frankfurt hospital, Germany 2. Local private dataset from the Pabna diabetes hospital 3. PIMA Indian diabetes dataset	Numerical data and categorical data	1.Type-2 Diabetes dataset: 60.60 KB 2. Local private dataset 3. PIMA Indian diabetes dataset: 45.8 KB	1. Type-2 diabetes data-set: 2000 individuals (684 diabetic, 1316 non-diabetic) 2. Local private dataset: 465 participants (373 diabetic, 92 non-diabetic) 3. PIMA Indian diabetes dataset: 768 records (268 diabetic, 500 non-diabetic)	1. Type-2 diabetes dataset: https://ieee-dataport.org/documents/type-2-diabetes-dataset 2. Local private dataset: - 3. PIMA Indian diabetes dataset: https://data.mendeley.com/datasets/7zcc8v6hvp/1	Python, TensorFlow, Keras, and Scikit-learn, SHAP, LIME for XAI techniques, assembled on Kaggle's free tier

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Ramu et al. (2025)	Clinical database	Numerical data and categorical data	-	8 essential clinical features including age, blood pressure, specific gravity, albumin, sugar levels, red and white blood cell counts, hemoglobin, blood urea, serum creatinine, and other relevant factors	-	-
Shehta et al. (2025)	Chinese National Medical centre (C-NMC) dataset	Medical images of blood cancer (leukemia, lymphoma, and multiple myeloma)	15,135 images total, 10.44 GB	Open-access dataset of blood cancer images, Labeled images showing specific forms of blood cancer	https://www.cancerimagingarchive.net/collection/c-2019/	Google Colab, Python, TensorFlow
Maji et al. (2025)	1. Health metric data -Apollo diagnostic center and hospitals. 2. Facial image dataset collected by the research team	-	Health metric data: 1000 samples. Facial image dataset: 1800 images; 900 demented, 900 non-demented	Quantitative health metrics (diabetic status, age, blood oxygen level, heart rate, body temperature, weight) and Facial images	-	Python programming within Jupyter notebook environment, Javascript, HTML, and CSS

2. Medical Imaging Analysis and Radiology

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Hua et al. (2015)	Lung Image Database Consortium (LIDC) dataset	CT images	Data of 1010 patients	Nodes with size larger than 3 mm selected. Total 2,545 selected	1. https://pmc.ncbi.nlm.nih.gov/articles/PMC3041807/ 2. https://pubmed.ncbi.nlm.nih.gov/15333795/	-
Gulshan et al. (2016)	EyePACS-1, Messidor-2 datasets	Retinal fundus images	1. EyePACS-1: 9963 images from 4997 patients 2. Messidor-2: 1748 images from 847 patients	EyePACS and 3 hospitals in India for development and Messidor-2 for validation	Messidor: https://www.ias-iss.org/ojs/IAS/article/view/1155	StatsModels, SciPy, Python
Cheng et al. (2016)	Breast US from Taipei Veterans general Hospital, LIDC: Lung image database consortium	Breast ultrasound and CT scans	1. 520 breast US images from 520 patients 2. 1400 nodules from LIDC	1. Breast US dataset: 520 images (245 malignant, 275 benign) 2. Lung CT: 1400 (700 benign, 700 malignant)	1. https://pmc.ncbi.nlm.nih.gov/articles/PMC3041807/ 2. https://pubmed.ncbi.nlm.nih.gov/15333795/	
Zilly et al. (2017)	DRISHTI-GS dataset and RIM-ONE v3 dataset	Retinal fundus images	DRISTI GS-1,016 MB	DRISHTI-GS dataset includes 50 images, RIM-ONE v3 has 159 stereo fundus images	DRISTI-GS: https://evit.iit.ac.in/projects/mip/drishti-gs/mip-dataset2/Home.php RIM-ONE v3: http://medimrg.webs.ull.es/research/retinal-imaging/rim-one/	MATLAB

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Horie et al. (2019)	Cancer Institute Hospital, Japan	Endoscopic images with white-light imaging (WLI) and narrow-band imaging (NBI) at high resolution	Training: 8428 images; testing: 735 images	Training: 8428 images of esophageal cancer lesions (397 ESCC and 32 EAC) Test: 1118 images (162 cancerous and 376 non-cancerous) 48 116 fundus photographs	-	Framework: Caffe Deep Learning framework algorithm: Single shot multiBox detector (SSD)
Li et al. (2018)	LabelMe (Healgoo Ltd. LabelMe dataset)	Retinal fundus photographs			-	Stata software version 14.0 (for statistical analyses)
Jeyaraj and Samuel Nadar (2019)	BioGPS data portal: UCI Machine Learning repository, TCIA archive: Standard GDC data set: Standard	Hyperspectral imaging data encompassing malignant, benign, and normal tissue areas	1300 image patches	BioGPS: 100 image patches TCIA: 500 image patches GDC: 700 image patches	TCIA Archive: https://www.cancerimagingarchive.net/ GDC data set: https://portal.gdc.cancer.gov/	GoogLeNet Inception V3 architecture, executed on hardware with an NVIDIA GeForce GPU
Song et al. (2019)	Internal dataset: Seoul Metropolitan Government Seoul National University (SMG-SNU) Boramae Medical Center, Korea External dataset: Kuma Hospital, Japan	Ultrasonographic (US) thyroid nodule images in TIFF or DICOM format	1513 images	Training data: 1358 US images (670 benign, 688 malignant) Internal test set: 55 US images (34 benign, 21 malignant) External Test Set: 100 US images (50 benign, 50 malignant)	https://cdn-links.lww.com/permalink/md/c/md_2019_04_02_chai_md-d-18-06935_sdc1.pdf	Python, NumPy

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Khamparia et al. (2021)	International Skin Imaging Collaboration (ISIC) image archive	Skin lesion images	Augmented dataset: 5,000 images (3,800 training, 1,200 testing)	Benign: 1,900 (training), 600 (testing) Malignant: 1,900 (training), 600 (testing)	https://www.isic-archive.com/	Python, MongoDB, TensorFlow, .NET framework (C#)
Pandit and Bhanday (2020)	1. Cohen's COVID-19 image data collection, 2. Kaggle competition data set of chest x-rays	Chest X-ray images		A total of 1,428 X-ray images: 224 positive for COVID-19, 700 instances of bacterial pneumonia, 504 healthy cases	1. https://github.com/ieee8023/covid-chestxray-dataset 2. https://www.kaggle.com/datasets/andrewmvd/covid19-x-rays	
Kuchana et al. (2021)	1. Kaggle dataset 2. Github dataset	CT scan images	1. Kaggle dataset: 1 GB 2. Github dataset: 309.3 MB	1. Kaggle dataset: 20 complete chest CT scans (301 images each) 2. Github dataset: 929 CT slices from over 50 patients	1. https://www.kaggle.com/datasets/andrewmvd/covid19-ct-scans 2. http://medicalsegmentation.com/covid19/	MedSeg annotation tool, python, TensorFlow

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Elsharkawy et al. (2021)	1. Publicly available archives of COVID-19 positive cases 2. COVID-19 Open Research Dataset Challenge (CORD-19) 3. Data from the University of Louisville and Mansoura University	Chest X-ray images	1. Publicly available archives of COVID-19 positive cases: 525 MB 2. COVID-19 Open Research Dataset Challenge (CORD-19): 20 GB	X-rays of 200 COVID-19-positive individuals, 100 of whom died and 100 recovered	1. https://github.com/ieee8023/covid-chestxray-dataset	
Chattopadhyay and Maitra (2022)	2020 BraTS dataset	2D MRI images	7 GB	2892 images (2473 training and 273 testing, 9:1)	https://www.med.upenn.edu/cbica/brats/brats2020/data.html	Python, TensorFlow, Keras
Naz et al. (2023)	COVID-CT, COVID-NET	Computed Tomography (CT) scans and chest radiographs (X-rays)		COVID-CT dataset: 800 images, 349 CT scans with COVID-positive results (216 persons), and 397 CT scans with COVID-negative (397 persons). COVID-NET Dataset: 19,000 chest radiographs (CXRs)	COVID-CT: https://arxiv.org/abs/2003.13865	Python, Google Colaboratory (COLAB), TensorFlow, LIME

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Huy and Lin (2023)	1. Montgomery dataset 2. Shenzhen dataset 3. Tuberculosis (TB) chest X-ray dataset	Chest X-Ray images	5000 CXR images	3906 in normal category, 1094 in TB category from 3 datasets	Datasets 1 and 2. https://pmc.ncbi.nlm.nih.gov/articles/PMC4256233/ Dataset 3. https://arxiv.org/abs/2007.14895	
Sharma and Guleria (2023)	1. Dataset 1: Kaggle 2. Dataset 2: Kaggle and GitHub	CXR radiographs	1. 5856 images from dataset 1 2. 6436 images from dataset 2	1. Pediatric CXR, having normal vs. pneumonia classes 2. Multi-class images having normal, pneumonia and COVID-19 images	1. https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia	Orange 3.31.1 simulator used
Mahmud et al. (2023)	Brain tumor dataset: Kaggle	MRI images	3264 images	The dataset contains images in 4 classes: 937 images of Meningioma, 500 of no tumor, 900 of pituitary tumor, and 926 of glioma tumor	https://www.kaggle.com/code/sayedgoma/brain-tumor/notebook	Google Colab Pro+

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Odusami et al. (2023)	ADNI	MRI, PET images	824 images (412 MRI, 412 PET)	The dataset contains spatially normalized MRI and PET images, which consist of early and late MCI (Mild Cognitive Impairment) groups	https://adni.loni.usc.edu/	Pytorch, Nvidia GTX1660
Hroub et al. (2024)	Pneumonia dataset: Kaggle Covid19- &-Pneumonia dataset Kaggle	Chest X-Ray image data	1. Pneumonia dataset: 2.31 GB 2. Covid19- &-Pneumonia dataset: 2.14 GB	Pneumonia dataset: 5856 images, a training set (5232 images) and a testing set (624 images). Covid-19 and pneumonia dataset: 6432 images in a training set (5144 images) and a testing set (1288 images)	1. https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia?resource=download	Python, PyTorch, vit_pytorch API, pytorch-grad-cam
Kar et al. (2024)	Kaggle	-	-	82 patients, around 30 image segments each	-	Python, TensorFlow, Keras, OpenCV, NumPy

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Tawfeek et al. (2025)	Kaggle	Chest CT images	613 images	It has normal and affected groups of images. The affected group has squamous, large cell, and adenocarcinoma categories	https://www.kaggle.com/datasets/mohamedhanyyy/chest-ctscan-images	Python, TensorFlow, ResNet-50, Modified AlexNet, VGGNet, Keras, Pandas, Matplotlib, NumPy, Seaborn, Sklearn
3. Multimodal Deep Learning in Healthcare						
Liu et al. (2014)	Alzheimer's Disease Neuroimaging Initiative (ADNI)	T1-weighted MRI, FDG-PET scans	758 MR-only patients (180 AD, 204 NC, 214 neMC, 160 cMCI) multimodal patients (MRI and PET): 331	Total brain ROIs utilized 83, PET aligns with MRIs	https://adni.loni.usc.edu/	MATLAB 2013a, Image Registration Toolkit (IRTK)
Hsu et al. (2018)	Medical Information Mart for Intensive Care (MIMIC) dataset	X-ray images, Radiology reports (Text: Findings and impressions)	473,057 chest X-rays images and 206,563 reports of 63,478 patients	473,057 chest X-rays images and 206,563 reports of 63,478 patients	-	-
Hao et al. (2019)	Grade-A, second-class hospital in Wuhan, China	-	20,320,848 data from 31,919 patients (2013–2015)	multimodal clinical data from 706 patients, 606 for training and 100 for testing	-	Python framework

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Lee et al. (2019)	ADNI	Neuroimaging data, cerebrospinal fluid biomarkers, cognitive performance metrics, and participant demographic characteristics	1,618 ADNI participants; 307 MCI-converter, 558 MCI non-converter, 415 cognitively normal older adult controls (CN), 338 AD	Longitudinal and cross-sectional data	http://adni.loni.usc.edu	TensorFlow, PyTorch
Li et al. (2020b)	1. Ichallenge-AMD dataset 2. Ichallenge-PM dataset 3. EyePACS dataset; Kaggle's Diabetic Retinopathy Detection Challenge (EyePACS) dataset 4. Fundus-FFA	Color fundus images and Fundus Fluorescein Angiography (FFA) images	1. Ichallenge-AMD dataset: 1200 images 2. Ichallenge-PM dataset: 1200 images 3. EyePACS dataset: 88.29 GB (88,702 images) 4. Fundus-FFA dataset: 59 subjects (30 healthy, 29 DR patients)	1. Ichallenge-AMD dataset: AMD patients (23%) and non-AMD patients (77%) 2. Ichallenge-PM dataset: PM cases (50%), Non-PM (50%), and PM cases (50%) 3. EyePACS dataset: DR grading ranges from 0–4	1. EyePACS dataset: https://www.kaggle.com/c/diabetic-retinopathy-detection/data	PyTorch
Venugopalan et al. (2021)	ADNI (ADNI1, ADNI2, ADNI GO, and ADNI3	Clinical data, imaging (MRI, PET), Genetic	Genetic data-3 million	SNP-808 patient, MRI imaging-503 patients, clinical and neurological test data -2004 patients	https://adni.loni.usc.edu/	-

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Sheu et al. (2022)	Taichung Veterans General Hospital (TCVGH), Taiwan	Vital signs, Chest X-ray images	3972 patient records for pneumonia	Adults (≥ 18); days 1 to 3; CXR and vitals (2014 to 2018) ICU, short stays, deaths excluded	-	Python (Keras, OpenCV, Pandas, Matplotlib, Seaborn, NumPy)
Lee et al. (2023)	1. Samsung Medical Center (SMC) dataset, Korea 2. UK Biobank dataset	-	SMC: 517 patients (2954 images) for internal validation; 2026 patients (3518 images) for development UK Biobank: 11,091 patients (11,298 images) for external validation	Retrospective EMR and fundus data (2010–2016); CVD defined by ICD-10; externally validated with UK Biobank	UK Biobank: https://www.ukbiobank.ac.uk/use-our-data/apply-for-access/	TensorFlow (version 2.9.0), R (version 3.6.3)
Zhang et al. (2023)	The Parkinson's Progression Markers Initiative (PPMI) dataset Parkinson Speech dataset (PS): Department of Neurology in Cerrahpasa Faculty of Medicine, Istanbul University	Multivariate time series (from sensor data, voice recordings, clinical attributes)	1. PPMI dataset: 683 subjects, 15,798 records, 212 features 2. PS dataset: 68 subjects, 1208 records, 26 features	1. PPMI dataset: Clinical assessments 2. PS dataset: Acoustic voice features (e.g., jitter, shimmer, pitch)	PPMI dataset: http://www.ppmi-info.org	TAGCN architecture using a PyTorch-based framework

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Li et al. (2023)	1. Colorectal cancer dataset (Sun Yat-sen University, Fudan University, Cancer Hospital, Zhejiang Hospital of Southern Medical University, Fujian Cancer Hospital, and Fujian Medical University Cancer) ADNI dataset	Multimodal (medical images and clinical data)	1. Colorectal cancer dataset: 185 patients 2. ADNI dataset: 1675 samples	1. Colorectal cancer dataset: Includes CT images and clinical data 2. ADNI dataset: MRI images, volume of anatomical brain structures (138-dimension), phenotypic information (2-dimension), and genetic data	ADNI dataset: http://adni.loni.usc.edu	PyTorch
Intriago et al. (2023)	Dataset from different pathologies	Brain imaging data (labeled data and unlabeled data)	1. Labeled dataset: 487 participants (31 AD, 456 NC) 2. Unlabeled dataset: 468 participants (177 SCC, others with psychiatric conditions)	Multimodal MRI (Structural MRI (sMRI), Functional MRI (fMRI)) data from aged 18–98, encompassing SCC, AD, and psychiatric groups	-	FastSurfer, fMRIPrep

Table 11 (continued)

Key literature work	Database used	Data types	Database size	Details of the dataset	Link	Software/ tools used
Thiruvengadam et al. (2024)	BraTS (Brain Tumor Segmentation Challenge dataset)	Multimodal MRI brain images (T1c, T2, FLAIR)	3300 images	Labeled MRI slices fused at the pixel level and categorized into tumor and non-tumor classes. 10 volumes of each MRI modality image	-	Google Colaboratory
Mahalakshmi et al. (2024)	-	Mammograms (image), prescription data, blood reports (textual/sequential)	-	Contains benign/malignant data; augmented mammograms; sequential clinical records	-	TensorFlow, Keras
Krishnaveni et al. (2024)	1. Chest X-ray14 dataset 2. COPDGene dataset	-	Chest X-ray14 dataset: Over 100,000 images	CXR resized to 224A—224; PFTs normalized; data aligned and balanced using augmentation and SMOTE	1. Chest X-ray14 dataset: https://mhcc.app.box.com/v/ChestXray-NIHCC	TensorFlow, Keras

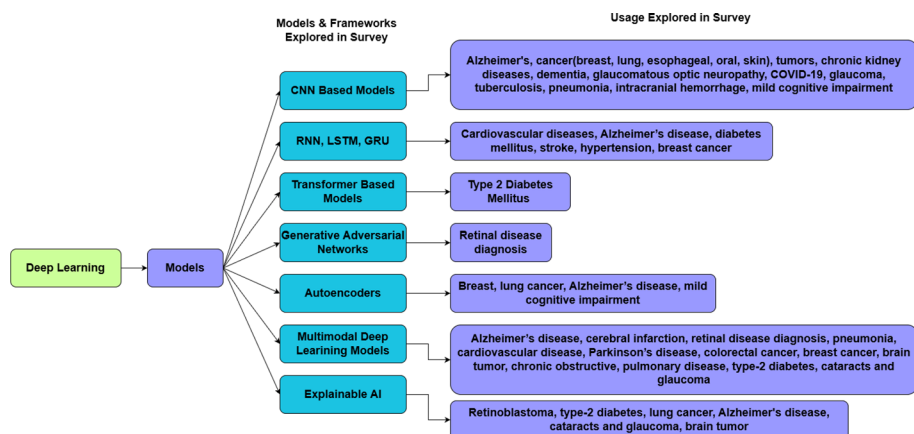


Fig. 8 Diagram showing the taxonomy of deep learning architectures in healthcare 5.0, illustrating the core deep learning models and frameworks and their specific usage

Nazir et al. 2025). DL can improve the precise prediction of diseases in patients, leading to enhanced quality of care. Computer-aided diagnosis (CADx) is becoming increasingly vital for disease diagnosis, particularly in regions with restricted healthcare access (Chan et al. 2020; He et al. 2018; Tao et al. 2025). Neural networks in DL evolve and enhance with new data, rendering them more effective than conventional algorithms in identifying illnesses such as cancer (Sompawong et al. 2019), Alzheimer's disease (AD), heart disease (Izci et al. 2019; Oh et al. 2018; Chen et al. 2020), brain tumors, etc. (Malakar et al. 2022). These AI-driven systems provide enhanced precision, less human error, and economical, accessible healthcare solutions (Nishat et al. 2025; Srivastava et al. 2025). The continuous progress in CADx significantly influences preventive healthcare and early disease diagnosis, enhancing patient outcomes worldwide (Jiménez-Gaona et al. 2020; Tao et al. 2025; Al-Antari et al. 2020). We have showcased recent research aimed at predicting and diagnosing various diseases via diverse deep-learning techniques.

This study presents a review of recent advancements in the field of disease prediction and early diagnosis. Rajendra Acharya et al. (2017) developed a deep CNN model capable of autonomously detecting and classifying electrocardiogram (ECG) signals into five distinct heartbeat categories. Jiao et al. (2018) employed a deep CNN to obtain deep representations of breast mass images. Employing a 3D CNN, Bäckström et al. (2018) demonstrated excellent performance in detecting AD on a large dataset. Feng et al. (2019) tackled the difficulties associated with leveraging 3D-CNN and fully stacked bidirectional LSTM (FSBi-LSTM) for AD diagnosis in the context of limited imaging data. Feng et al. (2020) proposed a novel brain imaging framework that integrates 3D-CNN with SVM to improve the diagnostic accuracy of AD. Çınar and Yildirim (2020) employed a CNN model to detect brain tumors from MRI scans. Angel Nancy et al. (2022) employed bidirectional Long Short-Term Memory (Bi-LSTM) models to effectively predict the risk of heart disease. Similarly, Yashudas et al. (2024) utilized Bidirectional Gated Recurrent Units (BiGRU) to enable timely identification, medical intervention, and nutritional guidance for cardiovascular conditions. Mandava and Vinta improved the precision of cardiovascular disease prediction through a DL-based modified DenseNet201 (MDenseNet201) model (Mandava et al. 2024).

Mahmud and Barua (2024) proposed an interpretable AI framework for diagnosing AD by analyzing MRI data with a modified CNN. Similarly, Bader Aldughayfiq et al. (2023) enhanced retinoblastoma diagnosis using InceptionV3 with transfer learning, integrating XAI to improve model transparency. Wani et al. (2024) introduced an interpretable hybrid DL approach, ConvXGB, combining CNN and XGBoost for explainable lung cancer detection. In the domain of infectious diseases, Alazab et al. (2020) developed an automated CNN-based system to accurately detect COVID-19 from chest X-ray images and forecast future cases, recoveries, and deaths using historical data. Singh et al. (2022) designed a DNN for early detection of chronic kidney disease (CKD), identifying key predictive features via Recursive Feature Elimination (RFE). Janghel and Rathore (2021) proposed a CNN-based diagnostic model for early Alzheimer's detection using Functional Magnetic Resonance Imaging (fMRI) and Positron Emission Tomography (PET) images, applying a 3D-to-2D preprocessing conversion before feature extraction. Tanim et al. (2025) employed XAI methodologies through a Deep Neural Network (DeepNetX2) to improve interpretability and accuracy in diabetes diagnosis. Ramu et al. (2025) enhanced CKD detection accuracy by adopting a hybrid CNN-SVM framework designed to mitigate overfitting, computational inefficiency, and class imbalance. Shehta et al. (2025) developed a classification model for blood cancer, achieving high diagnostic accuracy and supporting early intervention to improve survival outcomes. Ortiz et al. (2016) advanced skin cancer diagnosis with an Ensemble DL model that combined multiple CNN architectures to boost classification accuracy on dermoscopic images. Choi et al. (2017) improved early heart failure detection by leveraging temporal relationships in EHR. Furthermore, Maji et al. introduced an interactive cognitive assessment platform that integrates modified CNNs (MOD-1D-CNN for health metrics and MOD-2D-CNN for facial image analysis) with gamified evaluation strategies to support the early diagnosis of dementia. Figure 9 provides a consolidated visual summary of how different DL architectures perform across multiple disease prediction and early diagnosis tasks, addressing cross-task performance variation.

4.2 Medical imaging analysis and radiology

The discovery of medical image analysis came from the invention of the microscope by Leeuwenhoek in the 17th century (Meijer et al. 1997). Then X-rays and CT scans set the foundation for more advanced systems. With the advancement of technology, AI was integrated into the medical image analysis process. It has become a highly studied and researched field, and large amounts of data known as big data contribute greatly to research (Savadjiev et al. 2019). The boom in ML and DL research during the 2010's revolutionized this field; hence, the demand for automated and accurate systems grew. These systems provided interpretations of complex images in the medical domain and were then used in diagnostics to provide better treatment and detect serious issues early on Shen et al. (2017); From machine learning to deep learning (2021). Figure 10 was adapted from the work of Anwar et al. (2018), which shows the various imaging techniques used in medical imaging.

Several studies have used different methodologies in the domain of medical image analysis; for example, Hua et al. (2015) used DL frameworks to eliminate explicit feature extraction in classifying pulmonary nodules in CT scans as benign or malignant. Gulshan et al. (2016) developed and validated a DL algorithm for the automated detection of referable diabetic retinopathy (RDR) and diabetic macular edema (DME) using retinal fundus

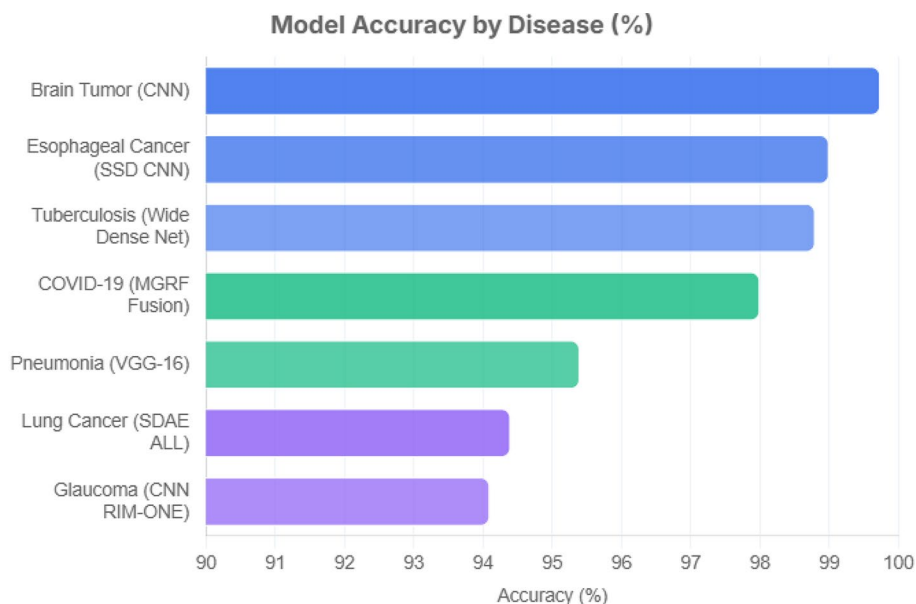


Fig. 9 Comparative accuracy of DL based models for disease prediction and early diagnosis across multiple disease conditions

images. Cheng et al. (2016) explored a CADx to simultaneously identify benign and malignant lesions and nodules. Zilly et al. (2017) provide a computationally efficient and precise technique for the automatic segmentation of the optic disc and the optic cup of the fundus images to facilitate early detection of glaucoma. Horie et al. (2019) evaluated the diagnostic performance of a CNN in detecting esophageal cancer from endoscopic images. Li et al. (2018) evaluated the efficacy of a DL algorithm in identifying referable glaucomatous optic neuropathy (GON) using color fundus images. Jeyaraj and Nadar (2019) created an automated, computer-assisted diagnostic method for the early identification of oral cancer. Song et al. (2019) categorize thyroid nodules as benign or malignant to avoid unnecessary fine needle aspiration (FNA). Khamparia et al. (2021) created a DL Internet of Health Things (IoHT)-driven system for automated skin cancer identification and classification from dermoscopic images to improve accuracy and reduce diagnostic errors. Pandit and Banday (2020) created a cost-effective, automatic chest X-ray COVID-19 detection model. Kuchana et al. (2021) aimed to improve COVID-19 diagnosis and tracking by DL-based lung CT scan segmentation. Elsharkawy et al. (2021) developed a computer-assisted diagnostic (CAD) system to assess pulmonary function and predict mortality risk in COVID-19 patients through chest X-ray analysis. Chattopadhyay and Maitra (2022) created a tumor detection model for speedier diagnosis and treatment. Naz et al. (2023) classify pulmonary conditions and explain classification results to help radiologists discover early disease. Huy and Lin (2023) developed a model to detect tuberculosis with high accuracy, specificity, and sensitivity. Sharma and Guleria (2023) developed a system for early detection and classification of pneumonia using DL and transfer learning using chest X-ray images. Mahmud et al. (2023) developed a CNN-based model for detecting brain tumors accurately in the early stages using MRI images. Odusami et al. (2023) developed a model for early detec-

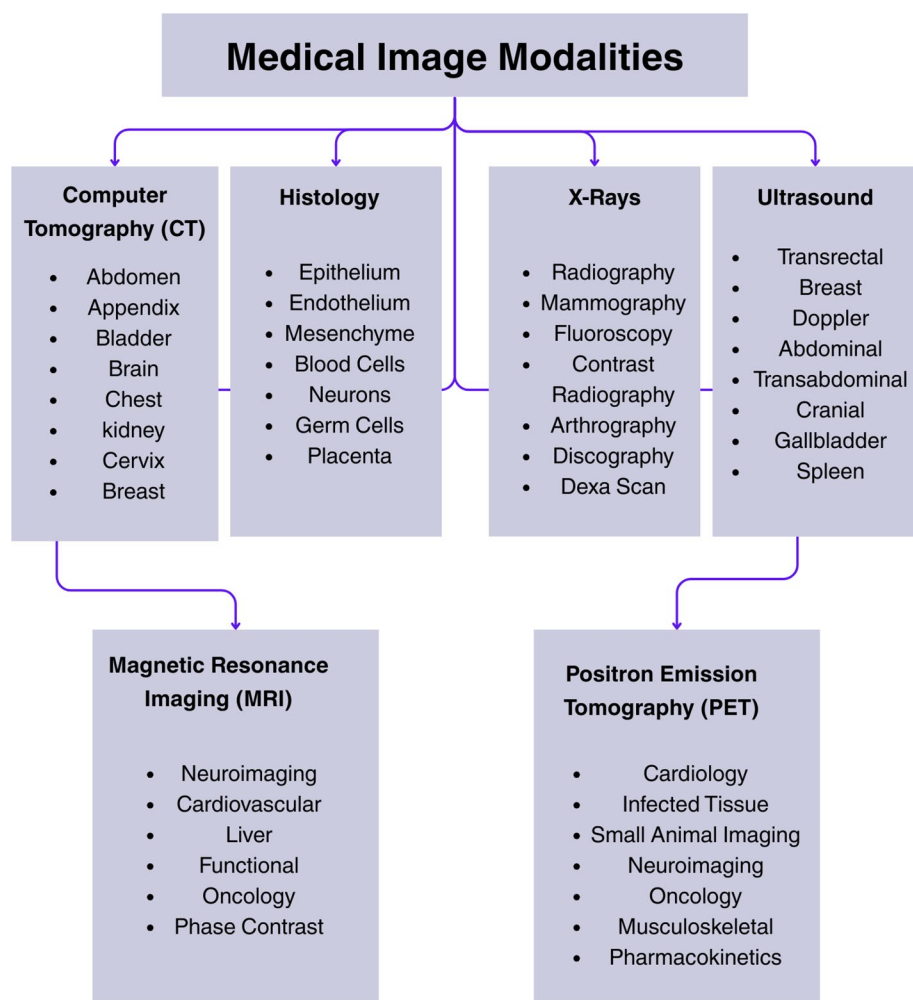


Fig. 10 Common modalities used in medical image analysis

tion of Alzheimer's disease, combining MRI and PET images for a better analysis. Nussair et al. (2024) developed an explainable DL lung disease prediction tool from chest X-rays. Kar et al. (2024) employed cutting-edge DL techniques to automatically detect intracranial hemorrhage (ICH). Tawfeek et al. (2025) developed a DL model to detect lung cancer.

Huy and Lin (2023) in their paper, proposed a CBAMWDNET hybrid DL model with two components: Convolutional Block Attention Model and Wide DenseNet(CBAMWDNet). CBAM is an attention mechanism used with CNN to improve model performance by focusing only on important parts and ignoring irrelevant details. This is made possible by channel and spatial attention in CBAM, which highlights which features are important and where in the input data the features are located, respectively. This makes the model interpretable and also reduces overfitting. WDN enables the extraction of detailed features as it widens the layers instead of deepening them.

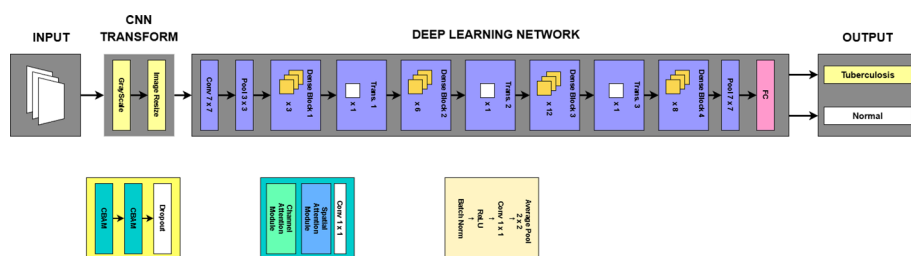


Fig. 11 CBAMWDNet Architecture diagram for "An Improved Densenet Deep Neural Network Model for Tuberculosis Detection Using Chest X-Ray Images", Vo Trong Quang Huy; Chih-Min Lin (Huy and Lin 2023)

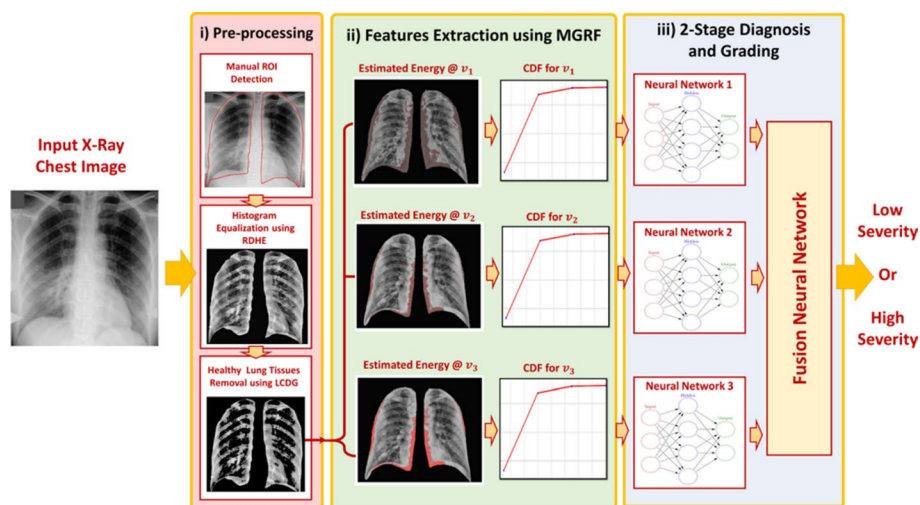


Fig. 12 Pipeline of the CAD System of the MGRFNN model (adapted from Elsharkawy et al. 2021 licensed under CC BY 4.0.)

In short, this approach combines the focusing feature of CBAM and the representation power of Wide DenseNet into a model that produces accuracy as high as 98.8% on different datasets. The architecture of the model has been shown in Figure 11.

Elsharkawy et al. (2021), in their paper built a computer-aided diagnostic system combining the Markov-Gibbs Random Field (MGRF) model with neural networks to predict the severity of lung infection of patients due to COVID-19. The MGRF model calculates the Gibbs energy at varied distances in infected areas to detect how uneven or abnormal the tissues are. These values are then converted to graphs called the CDF—cumulative distribution functions—which are then input into the neural networks. The neural network analyzes the inputs and classifies the infection as mild or severe. The model achieved an accuracy of 98% and could classify all severe cases correctly during testing. The model's architecture has been represented in Figure 12.

4.3 Multimodal deep learning in healthcare

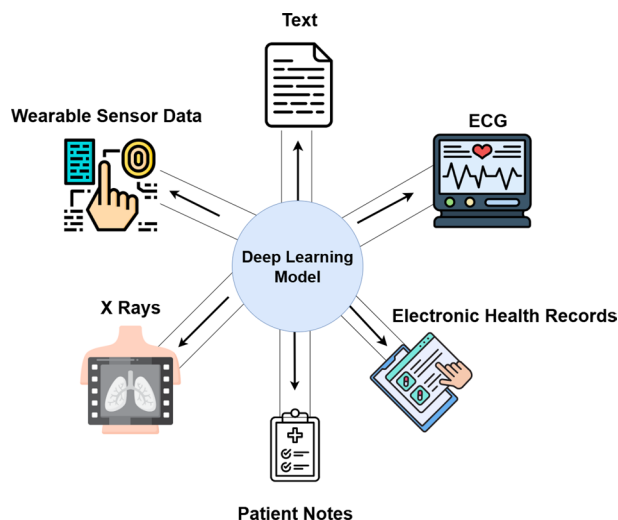
In the era of smart healthcare, medical systems increasingly rely on diverse data modalities ranging from imaging (e.g., MRI, CT, and histopathology) to EHR, wearable sensor signals, and clinical notes (Bianchi et al. 2017).

There are usually three types of multimodal fusion strategies adopted by MMDL. The input stage in early fusion takes raw or low-level data across several modalities and thus, the model learns joint representations early on. Intermediate fusion takes a combination of features that are independently learned across both modalities and provides a balance between common learning and representations of the modalities. Late fusion is an algorithm that combines the end predictions or decision scores of individual models, and is applicable when modalities are strongly dissimilar in structure or quality. Such fusion approaches are useful in generating more situational and multi-faceted models that can be adapted easily to the Healthcare 5.0 setting.

The data types offer a complete picture of the patient, with imaging data recording the spatial and structural patterns (Bianchi et al. 2017), EHR giving the comprehensive clinical history, wearable devices providing real-time physiological measurements, and clinical notes offer contextual insights. DL models use comparatively heterogeneous sources to combine them, which can be broadly divided into feature-level and decision-level fusion. During feature-level fusion, the features extracted in separate modalities (i.e. CNN features on images and LSTM embeddings on EHR sequences, transformer embeddings from text) are inputted together into a common latent space where they are jointly learned. On the other hand, decision-level fusion takes the result of the unimodal models and combines them to obtain a final consensus prediction. The preprocessing operations (alignment, normalization and synchronization of data) are necessary to allow compatibility of modalities and to improve the overall predictive capabilities of the system.

MMDL enables the fusion of these heterogeneous sources to create richer, context-aware representations that drive improvements in diagnostics, prognostic modeling, and personalized treatment planning. For example, joint image-text models have been explored for tasks like visual question answering and automated clinical report generation, as surveyed in a scoping review of biomedical image-text integration (Sun et al. 2023). In oncology, combining imaging modalities such as CT/MRI with structured clinical biomarkers has shown promise in improving liver cancer detection and prognosis (Siam et al. 2023). More recently, comprehensive reviews highlight the rise of transformer-based fusion architectures (including intermediate fusion strategies), which consistently outperform unimodal baselines across tasks such as computer-assisted diagnosis and survival prediction (Guarrasi et al. 2025), and emphasize improved AUC through multimodal integration in clinical applications (Schouten et al. 2025). These works underline how MMDL lies at the core of intelligent and adaptive healthcare systems, capable of enabling real-time decision support, automated reporting, and patient-centric care pathways.

In our study, we have reviewed some of the recent works in the MMDL domain. In healthcare and related fields, there is a growing demand for such models. Figure 13 represents different kinds of data that are frequently used in MMDL models. Hsu et al. (2018) trained a shared space between image and text using unsupervised (and semi-supervised) learning, using a condition of cross-modal retrieval and representation learning in medical imaging. Hao et al. (2019) performed chronic disease prediction using multimodal hospital

Fig. 13 Commonly used modalities in MMDL

data (Fine-grained text features and structured data). Lee et al. (2019) forecasted the progression from Mild cognitive impairment to AD by integrating longitudinal and multimodal biomarkers. Li et al. (2020b) created a self-supervised feature learning algorithm utilizing multimodal retinal images that will be robust to defeating disease diagnosis without imposing manual labeling. Venugopalan et al. (2021) proposed a MMDL framework for precise staging of AD by integrating imaging data, genetic data (single-nucleotide polymorphisms (SNPs)), and clinical test records. Sheu et al. (2022) developed a pneumonia status prediction system to assess whether early-stage patients can be discharged within or after 7 days of hospitalization. Lee et al. estimated current Cardiovascular Disease (CVD) risk by integrating fundus images and clinical data; they evaluated model generalizability across datasets (Lee et al. 2023). Zhang et al. (2023) detected Parkinson's disease by using unsupervised, uncertainty-aware learning to extract informative representations from multivariate time series. Intriago et al. (2023) proposed early identification of SCC individuals at elevated risk for developing Alzheimer's disease by multimodal MRI representation learning. Li et al. (2023) enhanced disease prediction by independently optimizing feature alignment and fusion processes within an MMDL framework. Thiruvankadam et al. (2024) improved brain tumor detection using CNN on fused multi-modal MRI with XAI support. Mahalakshmi et al. (2024) utilized multi-modal fusion to enhance diagnostic accuracy and facilitate personalized healthcare; breast cancer detection can be made early and precise. Krishnaveni et al. (2024) achieved a more thorough classification of Chronic Obstructive Pulmonary Disease (COPD) stages through the integration of chest X-ray imaging and pulmonary function test data. Ding et al. (2024) employed Large Language Multimodal Models (LLMMs) for the prediction of chronic diseases, specifically diabetes. Ranjith and Sakthivanitha (2025) designed a multi-modal framework that facilitates accurate, interpretable, and scalable early detection of ocular diseases. Kumar et al. (2025) developed a robust, scalable end-to-end system for early-stage Alzheimer's diagnosis, enabling timely intervention and improved patient outcomes. Gezimati and Singh (2025) established an MMDL framework for breast cancer diagnosis utilizing infrared and (theoretically) terahertz imaging. Shi et al. (2025) enhanced glioma MRI segmentation via SAM fine-tuning with modality-specific fusion

adapters. Tsai et al. (2025) designed an effective multitask learning model for simultaneous prediction of multiple chronic diseases using multimodal patient information. Figure 14 shows how DL models deal with multiple modalities and how the data are processed, fused, and learned.

4.4 Comparative analysis of DL models in healthcare tasks

Over the past decade, the sphere of deep learning (DL) models has achieved profound improvements in the healthcare sector with regard to predicting diseases, especially as a part of Healthcare 5.0. Comparisons of various diseases, including AD, kidney disease, lung disease, heart disease and breast cancer, indicate that DL models like CNN, LSTM, hybrid and ensemble models are always more effective than the traditional machine learning models in medical imaging and clinical data. A brief comparison of model based on the prominent studies is provided in the Tables 12, 13, 14, 15, 16 below and the main distinctions in terms of Model/Method, accuracy, disease domain are pointed out.

5 Challenges and limitations

Despite the remarkable progress achieved in deep learning-based healthcare systems, several methodological, ethical, and deployment-related challenges remain that must be addressed for reliable real-world implementation. DL methods have changed the scenario of many healthcare sectors, dealing with high percentages of improvement in medical image analysis, disease prediction, and remote patient monitoring. However, use of such techniques and implementation in the clinical setting has several obstacles and hindrances (Shamshirband et al. 2021; Amin et al. 2021; Karri et al. 2025). These problems include issues relating to the quality of data and the capacity to interpret results, computationally intensive demands, and the necessity of large amounts of data (Nisar et al. 2021). It is important to recognize these problems for effective use of DL in healthcare environments.

Fig. 14 Deep learning architecture for multimodal fusion

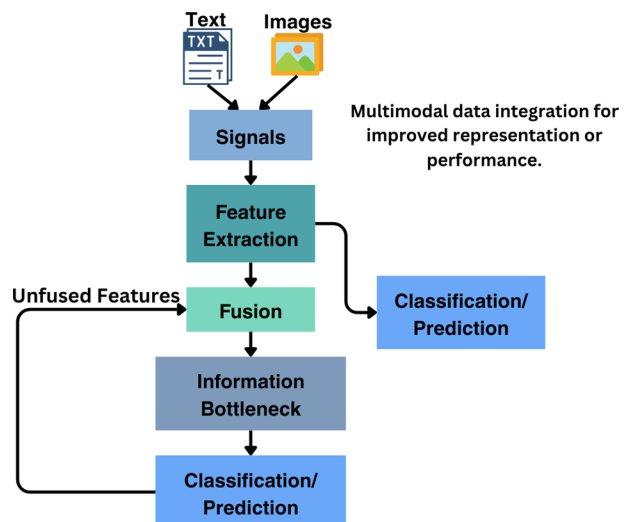


Table 12 Summary of comparative table for neurodegenerative disease diagnosis (alzheimer's disease, dementia, and parkinson disease)

Key literature work	Model	Accuracy (%)
Ortiz et al. (2016)	CNNs, VGG16, ResNet, and Inception	90%
Bäckström et al. (2018)	3D-CNN	98.74%
Feng et al. (2019)	3D-CNN, FSBi-LSTM, softmax	94.82%
Feng et al. (2020)	3D-CNN-SVM	99.10%
Mahmud and Barua (2024)	VGG16, VGG19, DenseNet169, DenseNet201, Ensemble 1 (VGG16, VGG19), Ensemble 2 (DenseNet169, DenseNet20), proposed model (Efficient-NetB3, CNN), XAI method (saliency maps, Grad-CAM)	96%
Oduami et al. (2023)	In-3-channel ResNet18	Accuracy: 73.90%, specificity: 94.32%, sensitivity: 66.74%
Liu et al. (2014)	SAE with a soft- max logistic regressor	Binary classification (MR-only): Accuracy :82.59%, sensitivity:86.83% Multiclass classification (MR-only): Accuracy: 46.30%, specificity: 77.78% Multiclass classification (MR and PET): Accuracy: 53.79%, specificity: 86.98%
Oduami et al. (2023)	Multimodal RNN (multi-GRU + logistic regression)	Accuracy: 81% Sensitivity: 84 % Specificity: 80 %
Intriago et al. (2023)	Self-Supervised Multimodal Representation Learning (SSMRL) framework: Balanced Random Forest (BRF) + Cy-CLIP cost function + 3D CNN (sMRI) + BiLSTM (fMRI)	AUC scores: 1. sMRI: 98.30 % $\pm$ 1.53 % (multi) 2. fMRI: 97.70 % $\pm$ 1.47 % (multi)
Kumar et al. (2025)	DenseNet (primary); CNN, VGG-16, MobileNet (comparative baseline models), Adam optimizer	DenseNet accuracy: 81.5% MobileNet accuracy: 57.4% CNN accuracy: 50.8% VGG-16 accuracy: 26.7 %
Janghel and Rathore (2021)	VGG-16 CNN + SVM/KNN/Tree	99.9%(fMRI)
Maji et al. (2025)	MOD-CNN [MOD-1D-CNN and MOD-2D-CNN]	MOD-1D-CNN : 70.50%, MOD-2D-CNN : 95.72%
Venugopalan et al. (2021)	3D-CNN, Auto-encoders	(MRI, EHR, and SNP)- internal cross-validation accuracy-79% and external test set accuracy-78%
Zhang et al. (2023)	TL-TAGCN	1. PPMI data set: Accuracy: 94.6 % Precision: 0.963 $\pm$ 0.011 Recall rate: 0.969 $\pm$ 0.017 F1 score: 0.964 $\pm$ 0.006 2. PS data set: Accuracy: 92.9 % Precision: 0.969 $\pm$ 0.062 Recall rate: 0.945 $\pm$ 0.045 F1 score: 0.955 $\pm$ 0.027

Table 13 Summary of comparative table for heart disease diagnosis

Key literature work	Model	Accuracy (%)
Choi et al. (2017)	RNN, GRU	UCs ranged from 0.777 (12-month window) to 0.883 (18-month window)
Rajendra Acharya et al. (2017)	Deep CNN	Noise free Data: 94.03%, Original data: 93.47%
Angel Nancy et al. (2022)	Fuzzy Inference System (FIS), Bi-LSTM	98.86%
Yashudas et al. (2024)	Bi-GRU, IoT network	99.90%
Mandava et al. (2024)	MDensNet201-IDRSRNet, Relief, LASSO, IDRSNet	99.12%
Lee et al. (2023)	DNN + CNN: DenseNet-169	AUROC: SMC: 0.781 (95% CI 0.766–0.798) UK Biobank: 0.872 (95% CI 0.857–0.886) for multimodal model
Rao et al. (2024)	AttGRU-Hybrid Deep Learning	95.42%
Bilal et al. (2024)	Hybrid ETCXGB (CNN-XGBoost)	99.26%
Baashar et al. (2022)	Pure DL (MLP, CNN, RNN)	84.3% (AUC)

The literature review presented in Table 17 covers all three areas: disease prediction and early diagnosis, medical imaging and radiology, and MMDL, summarizing the advantages, disadvantages, and research gaps of existing Deep Learning approaches. This section outlines the key challenges and unresolved issues identified across the reviewed studies.

5.1 Challenges and limitations for disease prediction and early diagnosis

Early detection of illnesses and anticipating illnesses are critical elements of health care, emphasizing the improvement of patient outcomes by specifying the illnesses during the first stages (Nia et al. 2023). However, several challenges and limitations hinder the efficacy of the predictive models.

5.1.1 Challenges in data handling

- *Preprocessing of data:* Real-world healthcare data often contains some inconsistency, incompleteness, and noise, which make the DLA very challenging to implement. Therefore, feature extraction needs additional processing that can result in enhanced diagnosis skills. This indicates that there has been an inadequate and ineffective way of learning the model efficacy in detecting the disease at its early stages (Choi et al. 2017; Jiao et al. 2018; Angel Nancy et al. 2022).
- *Data quality and data complexity:* Various studies point to the problems of data quality and complexity, especially when it comes to EHRs. The types of data covered include but are not limited to numerical, categorical, and textual information; they often have missing values and noise, making it challenging to extract the data and impeding the training of DL models (Ortiz et al. 2016; Rajendra Acharya et al. 2017; Choi et al. 2017; Mandava et al. 2024; Tanim et al. 2025). In most instances, the reliance on health variables entered by the user can result in the inaccuracy of the data, which harms the level of diagnostic accuracy (Maji et al. 2025).

Table 14 Summary of comparative table for lung disease (pneumonia, COVID-19, cancer) diagnosis

Key literature work	Model	Accuracy (%)
Wani et al. (2024)	CNN, XGBoost with SHAP	97.43%
Hua et al. (2015)	DBN, CNN	Sensitivity 1. CNN: 73.4% 2. DBN: 73.3%
Alazab et al. (2020)	CNN with VGG16 architecture	Australia region: 94.80% Jordan region: 88.43%
Cheng et al. (2016)	Stacked Denoising Autoencoder (SDAE)	Lung CT: i. Single strategy: $87.4\% \pm 3.3$ accuracy ii. All strategy: $94.4\% \pm 3.2$ accuracy Breast US: $82.4\% \pm 4.5$ accuracy
Pandit and Banday (2020)	VGG-16 CNN	Binary classification: Accuracy: 96%, sensitivity: 92.64%, specificity: 97.27% Multi-class classification: Accuracy: 92.53%, sensitivity: 86.7%, specificity: 95.1%
Kuchana et al. (2021)	2D CNN with U-Net	F1-score: 97.31%
Elsharkawy et al. (2021)	MGRF model and a Neural Network-based fusion system	sensitivity: 100% specificity: $97\% \pm 3\%$ accuracy: $98\% \pm 2\%$ Dice Similarity Coefficient (DSC): $98\% \pm 2\%$
Naz et al. (2023)	ResNet50 CNN Model, LIME	COVID-CT dataset: 93% COVID Net dataset: 97%
Sharma and Guleria (2023)	Neural Networks with VGG-16	1. Dataset 1: 92.15% accuracy 2. Dataset 2: 95.4% accuracy
Hroub et al. (2024)	Inception-V3, CROP, ROT, HF	Pneumonia dataset: 94.55% Covid19- &-Pneumonia dataset: 97.44%
Tawfeek et al. (2025)	CNN	Accuracy: 98.5% specificity: 88.7 % sensitivity: 89 % precision: 89.2% recall: 99.68%
Sheu et al. (2022)	Dense-BN	Dense-BN accuracy: 77%(vitals), 92%(CXR), 75% (combined with weights)
Krishnaveni et al. (2024)	MultimodNet: FCN + DenseNet121	Accuracy: Initial stage: 95.231% Progressive stage: 95.008% Complicated stage: 94.989% Critical stage: 94.256%

Table 15 Summary of comparative table for kidney disease diagnosis

Key literature work	Model	Accuracy (%)
Singh et al. (2022)	DNN, SVM, KNN, Logistic Regression, Random Forest, Naive Bayes	100%
Ramu et al. (2025)	Hybrid CNN-SVM model	96.8%

- *Data availability:* Availability of large, high-quality datasets is a big challenge since DL models need vast amounts of data to learn correctly. In the case of healthcare, the condition is highly problematic, as the considerations of data privacy limit access to the full dataset (Rajendra Acharya et al. 2017; Jiao et al. 2018; Feng et al. 2020; Çinar and

Table 16 Summary of comparative table for breast cancer diagnosis

Key literature work	Model	Accuracy (%)
Jiao et al. (2018)	Deep CNN (CNNs layers, metric learning layers)	DDSM database: 97.4%, MIAS database: 96.7%
Mahalakshmi et al. (2024)	CNN, RNN	Metrics shown as 1.0; exact values unspecified
Gezimati and Singh (2025)	CNN, decision-level (late) fusion architecture	Validation Accuracy: 96.6% Sensitivity: 96.3% AUC: 0.99
Al-Antari et al. (2020)	YOLO detector with CNN, ResNet-50	DDSM dataset: YOLO 99.17% INbreast dataset: 97.27%

Yildirim 2020; Alazab et al. 2020; Yashudas et al. 2024; Maji et al. 2025).

- *Data diversity and data size:* The data diversity and data size are also vital aspects in the realization of the high performance of the model. The shortage of a range of datasets that possess a wide variety of demographics and disease progression levels is a factor limiting the accuracy of the predictive models (Ortiz et al. 2016; Choi et al. 2017; Jiao et al. 2018; Bäckström et al. 2018; Singh et al. 2022; Aldughayfiq et al. 2023; Mandava et al. 2024; Mahmud and Barua 2024; Tanim et al. 2025; Shehta et al. 2025; Ramu et al. 2025). Indicatively, the health metric and facial dataset has low representation of different ethnicities, age groups and socio-economic backgrounds thus it limits the capability of the model to generalize within large populations (Maji et al. 2025).
- *Data imbalance:* Imbalanced datasets, where individual classes are drastically under-represented as compared to others, pose one of the most serious issues when it comes to the development of models that represent an effective generalization across different patient groups. This has been the case especially when the datasets are used in predicting chronic illnesses and other health aspects that are associated with them (Rajendra Acharya et al. 2017; Wani et al. 2024; Tanim et al. 2025). As an illustration, the Deep Learning approach that has been suggested to predict chronic kidney disease was tested on fairly small data volumes, hence limiting its generalizability (Singh et al. 2022) or the limited access to labeled health data hampers the ability of the MOD-1D-CNN model to succeed in being generalized across various clinical settings (Maji et al. 2025).
- *Reproducibility crisis:* Reproducibility is an increasing concern in clinical AI because many studies are not publicly available with code, model checkpoints, and clinical datasets to reproduce their findings. Experiments with AI are fragile to implementation and training parameters and by empirical audits, the literature results are usually difficult to reproduce without complete provenance (Hutson 2018; Gundersen et al. 2018). Cross-site evaluation, in particular within the healthcare context, has shown significant amounts of generalization errors as models that are trained using data of one hospital are tested to the data of another hospital, highlighting how the heterogeneity of the data set and missing data can conceal fragile performance (Zech et al. 2018). The field is shifting to more rigorous reporting and release practices (e.g., CONSORT-AI, TRIPOD+AI), standard outputs, and routine sharing of code, seeds, and environment descriptions to allow results to be separately validated and compared (Liu et al. 2020).

Table 17 Literature review

Key literature work	Pros	Cons	Research gap
Ortiz et al. (2016)	1. High accuracy 2. Automated diagnosis 3. Uses pre-trained models for feature extraction	1. Requires significant computational power 2. Limited dataset size, only 900 images 3. Interpretability of models remains a challenge	1. Limited dataset size 2. Need insights into feature importance or decision-making of the model 3. Not validated in real clinical settings or diverse demographic groups
Choi et al. (2017)	1. Improved predictive performance 2. Temporal modeling 3. Scalability	1. Require significant computational resources and expertise 2. Data limitations that may limit generalizability 3. Interpretability	1. Dataset limitation 2. Further exploration into more feature representations might enhance performance 3. Do not address how predictions could be integrated into decision-making processes 4. Lacks clinical implications
Rajendra Acharya et al. (2017)	1. Fully automatic 2. Unaffected by the ECG signal quality 3. Incorporates a ten-fold cross-validation technique	1. Extensive hours of training, sophisticated technology such as GPUs, computationally demanding	1. Class imbalance in datasets 2. Lack of comprehensive validation in clinical settings 3. Not a fully automated system; additional feature extraction or selection required 4. More extensive datasets are needed
Jiao et al. (2018)	1. Improved performance 2. Effective deep feature extraction capabilities 3. A reliable baseline for breast mass categorization	1. Mammography breast masses differ in appearance 2. Challenging to catalog in the original data space 3. Time-consuming	1. Restricted dataset size 2. Dependence on pretraining 3. Lack of cross-dataset validation 4. High Computational Requirements 5. Omission of Interpretability Analysis 6. Limited Integration of Clinical Parameters
Bäckström et al. (2018)	1. Efficient and simple 2. High performance 3. Automatic feature extraction 4. Large dataset	1. Performance drop on large datasets 2. Preprocessing dependent 3. Higher computation cost 4. Absence of multi-modality 5. Restricted Generalization	1. Integration of Multiple Modalities 2. Enhanced diversity in datasets 3. Enhanced architectural efficiency 4. Interpretability of predictions 5. Validation in clinical practice
Feng et al. (2019)	1. No prior knowledge required for manual feature extraction 2. Eliminate subjectivity 3. Faster convergence 4. High accuracy	1. Limited diagnostic performance for sMCI 2. Lack of direct identification of cerebral lesion structures 3. Underutilization of longitudinal MRI data	1. Lack of use of longitudinal data 2. Direct localization of brain lesions is not feasible 3. Lacks the exploration of shared feature representations across different imaging modalities

Table 17 (continued)

Key literature work	Pros	Cons	Research gap
Feng et al. (2020)	1. Higher performance 2. Efficiency 3. Non-invasive and non-irradiating 4. Automated classification 5. 3D Feature extraction	1. Insufficient data, 2. Inadequate comparison with radiologist-based diagnoses, 3. Inability to handle unusual presentations 4. Uncertain long-term effectiveness	1. Insufficient data 2. Requires validation with follow-up imaging and external datasets 3. Limited generalizability without cross-platform evaluation 4. Performance in early or atypical AD remains unclear
Çinar and Yildirim (2020)	1. Elevated precision 2. Improved feature extraction 3. Robust validation metrics	1. Computationally demanding 2. Limited dataset 3. Risk of overfitting	1. Necessity for an expanded dataset 2. Comparative analysis with alternative hybrid models 3. Empirical clinical validation
Alazab et al. (2020)	1. High detection accuracy 2. Increased generalization and resilience 3. Quick, economical X-ray diagnosis	1. Limited CT scan exploration 2. Small dataset size 3. Although infected, some patients test negative	1. Limited diagnostic methods 2. Need for rapid detection 3. Insufficient data availability
Angel Nancy et al. (2022)	1. Precision and efficiency 2. Real-time monitoring 3. Remote patient surveillance 4. Effectual data handling	1. Security and privacy of data; 2. Reliance on IoT devices 3. Flexibility and cooperation 4. Latency and bandwidth usage 5. Price and availability	1. Lack of clinical validation 2. Evaluation based on synthetically augmented datasets
Singh et al. (2022)	1. Achieved high accuracy 2. Uses RFE for feature selection, improving efficiency 3. Outperforms traditional ML classifiers in accuracy	1. Limited dataset 2. Limited feature used 3. Overfitting risk 4. Lack of real-world Testing	1. Need larger datasets 2. Need for advanced features
Aldughayfiq et al. (2023)	1. Good categorization accuracy 2. Improved readability using LIME and SHAP	1. Bias in dataset composition 2. Insufficient clinical evaluation 3. Restricted availability of imaging modalities	1. Restricted dataset heterogeneity 2. Clinical validation 3. Relevance to additional ocular disorders 4. Enhancement of explainability 5. Accessibility and economic viability
Yashudas et al. (2024)	1. Remarkable precision 2. Utilization of IoT network 3. Implementation of the Bi-GRU Attention Model 4. Recommendation framework 5. Validation of cloud simulation	1. data integrity 2. Adaptive medical conditions 3. Requirement for additional testing	1. Limited integration of recommendation systems 2. Higher computational overhead 3. Lack of adaptability to dynamic real-time patient data

Table 17 (continued)

Key literature work	Pros	Cons	Research gap
Mandava et al. (2024)	1. High accuracy 2. Efficient feature selection 3. Advanced pre-processing techniques 4. Combines multiple methods 5. Comprehensive testing 6. Practical application 7. Handles large datasets	1. High computational resources required 2. Potential over-fitting 3. Complexity 4. Clinical validation needed 5. Real-time application challenges 6. Dependence on quality of data 7. Limited generalizability	1. Insufficient clinical validation 2. Larger dataset needed 3. Management of missing data 4. Techniques for feature selection 5. Flexibility of interpretation 6. Level of resource consumption
Mahmud and Barua (2024)	1. Exceptional precision 2. Elucidation 3. Resilient Ensemble models 4. Thorough assessment	1. Complication 2. Generalizability 3. Incorporation into clinical workflows	1. Need for generalization 2. Advancement of XAI methods 3. Assessment of varied datasets 4. Integration of multimodal data 5. Lack of real-world clinical validation 6. Performance enhancement through optimization
Wani et al. (2024)	1. Elevated predictive precision 2. Interpretability 3. Effective computation 4. Hybrid model architecture 5. Augmented trust	1. Hardware-intensive 2. Dataset imbalance 3. Accessibility challenges	1. Explainability in AI 2. Generalizability issue 3. Hardware dependence 5. Model scalability
Tanim et al. (2025)	1. Higher accuracy 2. Enhanced transparency using Explainable AI 3. Improved interpretability and efficiency due to effective feature selection 4. Reduced computational complexity 5. Faster inference time	1. High computational complexity. 2. Extensive preprocessing is required 3. Dependent on higher-quality data 4. Difficult to generalize due to data set restriction	1. Limited dataset 2. Need for optimization to minimize model complexity 3. Limited model transparency
Ramu et al. (2025)	1. improved feature extraction 2. Better accuracy 3. Manages imbalanced data through SMOTE 4. Minimized overfitting	1. High computational complexity 2. Dataset size affects SVM's performance 3. Real-time optimization is needed	1. More optimization is required to lower computational overhead 2. Different classifiers could increase productivity 3. Need varied datasets 4. Improved generalization required
Shehta et al. (2025)	1. High accuracy 2. Comparative analysis 3. Comprehensive metrics 4. Clinical relevance 5. Automation potential	1. Ethical considerations 2. Data limitations 3. Limited clinical validation 4. Need for more explainable AI approaches 5. Need more hyperparameter optimization	1. More thorough ethical consideration is needed 2. Need larger and more diverse datasets covering all demographics 3. Clinical validation is necessary

Table 17 (continued)

Key literature work	Pros	Cons	Research gap
Maji et al. (2025)	1. Innovative design 2. User-friendly installation 3. High accuracy 4. Portability 5. Reliability and security 6. Rapid responsiveness 7. User-friendliness 8. Cost-effectiveness	1. Requires enhancements in the 2D gaming application's user interface layout. 2. Necessitates improvements in robustness and scalability of health metrics and facial imaging.	1. Restricted demographic representation 2. Inconsistencies due to variable lighting conditions 3. Insufficient availability of health-related data 4. Variability in input accuracy based on user interaction 5. Inconsistent behavioral patterns during gameplay 6. Limitations imposed by technological infrastructure
2. Medical Imaging Analysis and Radiology			
Hua et al. (2015)	1. Handcrafted features not required. 2. Good results in classification without using texture or morphology features	1. 3D features not incorporated 2. Resizing input images leads to loss of information related to size, which is an important indicator	1. Incorporating diverse datasets 2. Utilizing advanced Deep Learning models 3. Difficulty in feature extraction
Gulshan et al. (2016)	1. High performance accuracy 2. Consistent results 3. Scalable 4. Evaluation based on Specificity and sensitivity-2 operating points	1. Limited scope as only suitable for RDR and DME 2. Algorithm may miss minute features as ophthalmologists missed them 3. High dependency on training data 4. Cannot replace traditional systems	1. Testing algorithm on more datasets 2. Integrator variability 3. Assessing feasibility in real-world situations
Cheng et al. (2016)	1. Handcrafted features not required 2. Works well with varied modalities 3. Noise-resistant	1. Limited breast US data size 2. Data quality dependency 3. Not validated across modalities	1. Requirement of a larger dataset for reliable results 2. Validation across modalities required
Zilly et al. (2017)	1. Elevated segmentation precision 2. efficient, robust, and scalable 3. Novelty	1. Dataset dependency 2. Stochastic element 3. Limited training 4. Computational limits	1. Depends on varied datasets for validation 2. Imbalanced sampling impacts 3. Entropy sampling needs improvement 4. Underperforms deep CNNs on huge datasets
Horie et al. (2019)	1. High sensitivity (98%) for detecting esophageal cancer 2. Identification of Small Lesions 3. Rapid analysis time (27 s for 1118 images) 4. Capability to distinguish between superficial and advanced cancers	1. Positive predictive value relatively low (40%) 2. Misdiagnosis due to shadows and normal structures resulted in a negative predictive value of 95% 3. Dependent on image quality	1. Need for a diverse dataset 2. Real-time validation needed 3. Poor-quality image handling 4. Need for further training

Table 17 (continued)

Key literature work	Pros	Cons	Research gap
Li et al. (2018)	1. Attained exceptional performance 2. Potentially advantageous for large-scale, cost-efficient glaucoma screening 3. Effective model construction utilizing an extensive, annotated dataset	1. Absence of interpretability 2. Constrained to photos from a certain place (China) 3. Elevated misclassification rates for instances with coexisting ocular problems (e.g., high myopia)	1. Generalizability is needed 2. Interpretability issue 3. Need for clinical validation 4. Managing confounding variables 5. Limitation in data collection and population diversity 6. Constraints related to image quality and evaluation accuracy
Jeyaraj and Samuel Nadar (2019)	1. Elevated precision, sensitivity, and specificity 2. Efficient cancer segmentation via hyperspectral imaging 3. Decreases the need for specialized oncologists	1. Small data limits generalization 2. Computationally intensive 3. Not clinical workflow-integrated	1. Needs more diversified data 2. Requires real-time clinical implementation 3. Only limited comparison to advanced Deep Learning algorithms
Song et al. (2019)	1. Elevated sensitivity 2. Real-time image predictions (0.07 s) 3. A promising FNA-reduction tool 4. Training photos need little pre-processing	1. Reduced specificity 2. Comparatively limited training dataset 3. Poor augmentation or segmentation may reduce model robustness	1. Relatively small training datasets 2. Insufficient data augmentation to simulate variability 3. Clipped nodules may overlook anatomical characteristics 4. Validation on larger, more diverse datasets is needed
Khamparia et al. (2021)	1. High categorization accuracy 2. Integration with IoHT allows remote diagnosis 3. Automated detection cuts errors 4. Effective data augmentation boosts model performance	1. Intensive computer instruction 2. Lack of dataset diversity hinders generalization 3. Highly dependent on IoHT infrastructure and resources	1. Dependence on the quality of input data 2. Validation needed on varied and larger datasets 3. Consider sophisticated IoHT integration for scaling 4. Human Error in diagnosis 5. Optimize computation
Pandit and Bandy (2020)	1. High classification accuracy 2. Quick, affordable diagnosis 3. Non-contact diagnosis 4. Depends less on RT-PCR	1. Small dataset 2. Possible over-fitting despite data augmentation 3. Image quality and variety affect model performance	1. Limited training data limits generalization 2. More diverse, larger datasets improve robustness 3. Clinical validation needed
Kuchana et al. (2021)	1. High performance 2. Potential integration into healthcare systems 3. Real-time monitoring and recovery tracking	1. Limited by the quality and variety of the training datasets 2. Need for significant computing resources 3. Different patient demographics and conditions may affect the model's efficacy 4. Human assistance may be needed for preprocessing	1. Need for a larger dataset 2. Requirement for enhanced 3D-models 3. Integration of multimodal data 4. Real-world evaluation

Table 17 (continued)

Key literature work	Pros	Cons	Research gap
Elsharkawy et al. (2021)	1. High accuracy and sensitivity in predicting mortality 2. Provides objective metrics for assessing lung infection severity	1. Limited by the variability of X-ray quality across different machines 2. Relies on a relatively small dataset for training and validation	1. Variability in the quality of chest X-ray images 2. Integration with other modalities
Chattopadhyay and Maitra (2022)	1. Achieved high accuracy 2. Tumor detection is automated, saving time 3. Well-manages tumor size, form, and intensity	1. Needs huge, diversified datasets to train 2. Computationally costly training	1. Limited clinical and dataset generalization 2. Improve intensity biases and MRI fluctuations
Naz et al. (2023)	1. Very accurate pulmonary disease categorization 2. LIME makes clinical decisions more explainable 3. Effective Transfer Learning small datasets	1. Small datasets hinder generalization 2. Reliance on ResNet50-trained models 3. Explainability evaluation needs expert acceptance	1. Insufficiently diversified, huge datasets for generalization 2. Limited demographic and clinical tests 3. Additional features needed to strengthen the model
Huy and Lin (2023)	1. High accuracy without the use of pre-trained processes 2. Model generalizes well 3. Good results in few(only 50) epochs	1. Computationally intensive due to a large number of parameters 2. Limited labelled data 3. High resource demanding during training	1. Using CBAMWDNet for other diseases 2. Incremental learning and continuous training with new data 3. Optimizing performance with ensemble models. 4. Improving weights with multi-objective optimization
Sharma and Guleria (2023)	1. Good performance on both binary and multi-class datasets 2. High accuracy 3. Robust feature extraction using VGG-16	1. Computationally expensive, the model has a high number of parameters 2. Validation on real-world scenarios is required	1. Validation on a larger dataset is required 2. Optimize architecture 3. Use of other architecture
Mahmud et al. (2023)	1. High accuracy and AUC score 2. Better performance than InceptionV3, ResNet-50, VGG16	1. Many CNN layers 2. Requires a good GPU 3. Post hoc visualization lacking on tumor areas	1. Validation with diverse datasets is required 2. Do tumor localization and post hoc visualization
Odusami et al. (2023)	1. Handles heterogeneous MRI and PET data effectively 2. XAI provides interpretability 3. Better classification performance due to the fusion of features	1. Low sensitivity 2. Requires a high GPU	1. Validation with diverse datasets required 2. Optimization required
Hroub et al. (2024)	1. High accuracy through data augmentation 2. Class activation mapping (CAM)-based explaining predictions 3. Easy, affordable implementation	1. Computer-intensive models 2. Optimized ViT (vision transformers) training requires larger datasets	1. Performance of ViTs is limited by dataset size 2. More robust augmentations and lightweight models are possible

Table 17 (continued)

Key literature work	Pros	Cons	Research gap
Kar et al. (2024)	1. High accuracy 2. Efficiency 3. Process huge datasets fast 4. Improved interpretable mechanism 5. Robustness across classifications 6. Handling large data sets accurately	1. Limited data size 2. Highly resource-intensive	1. Limited availability of external dataset 2. Use of diverse clinical scenarios 3. Investigating methods to enhance the interpretability of the model 4. Advanced interface required
Tawfeek et al. (2025)	1. High accuracy 2. Better segmentation accuracy 3. End-to-end pipeline	1. Limited data size 2. Highly resource-intensive 3. Low transparency	1. Need for interpretability 2. Diverse dataset usage requirement
3. Multimodal Deep Learning in Healthcare			
Liu et al. (2014)	1. Improved accuracy 2. Effective multimodal fusion 3. Well performance on low label data 4. High-level feature learning 5. Unsupervised capability	1. Dataset limitation 2. Computational-demanding training needed 3. Limited performances in small classes	1. Validation on a more diverse dataset is needed 2. The unlabeled data performance is not widely studied 3. Test needed for real-time applicability and scalability
Hsu et al. (2018)	1. Achieve closer to supervised performance with very few labels 2. The mixture of annotated and unannotated data is even more advantageous 3. Best performance when applying the report sections of the clinic description	1. Text embeddings trained without domain awareness are poor 2. Pre-trained Models offer limited generalizability 3. Unsupervised rocustes refinement minimally effective 4. Low Medical Record Review (MRR) demonstrates report similarity	1. Domain-specific text encoders needed 2. Generation tasks exploration 3. Broader modality fusion inability 4. Sparse evaluation of downstream clinical activities
Hao et al. (2019)	1. Integration of multimodal data 2. Detailed feature extraction 3. Efficient feature integration 4. Exceptional precision 5. Outstanding efficacy	1. Cost of computational 2. Dependency on data	1. Ineffective fusion strategy for multimodal information
Lee et al. (2019)	1. Incorporates heterogeneous data modalities 2. Effectively processes irregular temporal sequences 3. Utilizes both overlapping and non-overlapping datasets 4. Strong predictive accuracy 5. Improved sensitivity and balanced accuracy	1. GRU parameters remain static during the final integration phase 2. Initial learning occurs independently across modalities 3. Risk of missing cross-modal feature interactions 4. Predictive accuracy declines over extended time horizons	1. Lacks unified optimization across data modalities 2. Cross-modal feature integration remains suboptimal 3. Future direction includes incorporating genomic and imaging data 4. Intends to design a feedback-enabled, unified GRU framework

Table 17 (continued)

Key literature work	Pros	Cons	Research gap
Li et al. (2020b)	1. Efficacy of multimodal data 2. Advantages compared to other self-supervised methods 3. Resistance to computer-generated images 4. Generalizable and transferable features	1. Limited FFA fundus images 2. Initially unsupervised, but final classification relies on labeled data	1. Fundus-FFA images are limited 2. Not fully unsupervised 3. Lacks multi-modal mutual information modeling 4. Needs to extend to the other imaging modalities 5. Synthesis required to generate another modality via adversarial learning
Venugopalan et al. (2021)	1. Better performance than shallow learning models 2. DL-assisted multimodality data analysis performs better than single-modality DL models 3. Can extract high-performing features	1. Limited sample size of the ADNI dataset 2. Combined SNP and imaging modality exhibited reduced performance 3. Model is not fully trained for all modalities	1. Limited dataset 2. No evaluation on early/atypical AD 3. Limited generalizability 4. No validation on external datasets
Sheu et al. (2022)	1. Enhances prediction accuracy 2. Supports clinical decisions 3. Enables early discharge 4. Integrates multi-modal data with interpretability	1. Requires physician training 2. Limited model flexibility 3. Opaque deep layers 4. Potential data privacy issues	1. Incomplete hospital-standard analysis 2. Limited multi-modal studies 3. Absence of post-treatment recovery prediction 4. Lack of clinical-CXR integration for decision simulation
Lee et al. (2023)	1. Enhanced accuracy 2. Non-invasive risk prediction 3. Future event association 4. Model interpretability	1. Retrospective design 2. Restricted SMC data access 3. Geographic bias 4. Need for prospective validation	1. No prospective validation 2. Geographic limitation 3. Absent real-time use 4. Need for broader multi-modal integration
Zhang et al. (2023)	1. Unsupervised representation learning 2. Models complex temporal relationships 3. Incorporates uncertainty estimation 4. Learns discriminative features 5. Achieves high accuracy 6. Demonstrates broad applicability	1. Prior methods emphasize linear correlations 2. Assumes fully supervised settings 3. Ignores intra-variable temporal patterns 4. Susceptible to overconfidence and low reliability 5. Simplistic graph construction 6. Risk of representation over-smoothing	1. Fails to capture complex temporal correlations 2. Limited use of unlabeled clinical data 3. Lacks uncertainty quantification 4. Ignores temporal intra-correlations 5. Overconfident models lack reliable uncertainty scoring
Li et al. (2023)	1. Enhanced clinical-image feature synergy 2. Spatial information retention 3. Superior to unimodal and prior fusion models 4. Robust unsupervised feature learning	1. Limited sample size 2. Incomplete clinical data 3. Persistently low mean accuracy	1. Small dataset impacts model stability 2. Improved clinical interpretability required 3. Future work to include larger, more diverse datasets

Table 17 (continued)

Key literature work	Pros	Cons	Research gap
Intiago et al. (2023)	1. Multimodal approach improves AUC by 2.45 percent 2. Enables early risk identification in SCC cases 3. Leverages unlabeled data effectively	1. Dataset access is restricted 2. Relies solely on MRI data 3. SCC outcome prediction remains uncertain	1. Cross-site validity untested 2. Lacks PET/clinical data 3. No real-time use shown
Thiruvengadam et al. (2024)	1. Achieves high accuracy 2. Improved tumor visibility via fusion 3. Enhanced interpretability with XAI	1. Restricted to pixel-level fusion 2. Lacks segmentation methods 3. Uses limited architectural diversity	1. Current models provide limited interpretability 2. Advanced fusion techniques are required 3. Limited transparency in CNN architectures
Mahalakshmi et al. (2024)	1. Improved diagnostic accuracy 2. Integrative methodology 3. Efficient processing 4. Early-stage detection 5. Resilient architecture 6. Adaptable design	1. Limited data reliability 2. Reduced model transparency 3. High computational demand	1. Unclear dataset scope 2. Lack of clinical validation 3. Absence of real-time testing 4. Incomplete multimodal data integration
Krishnaveni et al. (2024)	1. High diagnostic accuracy 2. Holistic analysis 3. Handling Imbalanced Data efficiently 4. Better outcomes 5. Strong data fusion	1. Limited generalization 2. Dataset dependence 3. Untested real-time use	1. No external validation 2. Lacks demographic data 3. Needs multimodal model improvement
Ding et al. (2024)	1. Enhanced precision 2. Efficient utilization of electronic health records (EHRs) 3. Integration of multimodal data 4. comprehensibility	1. Computational intensity 2. Restricted generalizability 3. Reliance on superior EHR quality	1. Requirement for expanded, varied datasets 2. Enhanced fusion methodologies 3. Capabilities for real-time prediction
Ranjith and Sakthivanitha (2025)	1. Enhanced predictive accuracy 2. improved model interpretability 3. Effective integration of multimodal data 4. Strong generalizability across diverse populations 5. Increased robustness 6. High AUC-ROC performance metrics 7. Positive clinician feedback	1. Dataset is restricted 2. Clinical validation incomplete 3. computationally intensive 4. Lacks evaluation in varied clinical settings	1. Deficiency in Deep Learning advancements 2. Inadequate multi-modal data fusion 3. Restricted model interpretability 4. Challenges in early-stage disease recognition 5. Obsolete data fusion techniques 6. Limited use of explainable AI methods
Kumar et al. (2025)	1. Superior DenseNet performance 2. Enhanced feature propagation and reduced gradient issues 3. Strong model robustness 4. Multimodal framework potential 5. Effective early-stage detection	1. Limited accuracy of MobileNet 2. Suboptimal performance of VGG-16 3. Moderate results from CNN 4. High computational resource demands 5. Restricted dataset scale	1. Constraints of the current diagnostic approach 2. Limited sensitivity to early-stage detection 3. Insufficient generalizability across diverse populations

Table 17 (continued)

Key literature work	Pros	Cons	Research gap
Gezimati and Singh (2025)	1. Safe, non-invasive imaging 2. Enables early cancer detection 3. High sensitivity and tumor specificity 4. Captures spatial-temporal data 5. Robust multimodal integration 6. Enhanced prediction accuracy 7. Benefits of decision-level fusion 8. Achieves high classification accuracy 9. No overfitting or underfitting observed	1. Scarcity of THz datasets 2. Expensive THz imaging systems 3. Difficulty in tissue sample acquisition 4. Challenges in manual diagnosis 5. THz is still under development 6. Restricted spatial resolution	1. Limited Deep Learning use in THz/IR imaging 2. THz datasets are scarce 3. Lacks THz post-processing frameworks 4. Data management and interoperability require improvement 5. Multimodal validation needed
Shi et al. (2025)	1. High accuracy 2. Efficient tuning 3. Strong fusion 4. Good generalization 5. Minimal overfitting	1. Large model size 2. Limited multi-scale capability 3. 2D-only segmentation	1. No native 3D support 2. Weak multi-scale modeling 3. High storage demand
Tsai et al. (2025)	1. Concurrent prediction 2. Efficient computation 3. Correlation modeling 4. Interpretability 5. Robustness 6. Rich feature representation	1. Imbalanced data 2. Dependence on clinical records 3. Fixed prediction time-frame	1. Exploration of chronic disease interrelations 2. Temporal comorbidities, early risk factors in youth 3. Long-term prediction from shorter histories 4. Data imbalance mitigation 5. Annual disease incidence forecasting

5.1.2 Methodological constraints

- Model interpretability and complexity:** Although Deep Learning models have high predictive power, they are often criticized because they lack interpretability and can be described as black boxes. Such a lack of transparency is an obstacle to acceptance in general clinical practice because confidence in the model predictions is fundamental in creating trust between the professionals in the medical field (Jiao et al. 2018; Wani et al. 2024; Mahmud and Barua 2024; Tanim et al. 2025). Techniques like LIME and SHAP are used to provide more interpretable predictions for individual predictions. However, there are still major issues in achieving transparent and comprehensive interpretability of complex models (Jiao et al. 2018; Bäckström et al. 2018; Aldughayfiq et al. 2023). Moreover, the high computational demands of such models may limit their applicability in resource-constrained settings (Rajendra Acharya et al. 2017; Mandava et al. 2024; Tanim et al. 2025).
- Feature extraction and selection:** Traditional Machine Learning models mainly rely on manual feature extraction, which is resource-intensive, subjective, and error-prone (Rajendra Acharya et al. 2017; Mandava et al. 2024). In many cases, Deep Learning

systems do not produce the optimal results. For example, the CNN-LSTM architecture can never accurately localize the lesions of the brain (Feng et al. 2019). Whereas dimensionality reduction option in feature selection algorithms (e.g., Recursive Feature Elimination(RFE)) may lead to improved model performance, it does not always succeed in identifying the most influential features in disease predictions and as such, could lower accuracy and stability of derived models (Singh et al. 2022).

- *Combining multiple data modalities:* The integration of data from different modalities—such as medical imaging, EHR, and genomic data, clinical notes—has the potential to enhance the accuracy of predictive models. Nonetheless, there is also a lot of complexity in this integration to the design and interpretability of such models. Bäckström et al. (2018). Implementing multimodal data integration requires the use of advanced and complex model designs, the development and optimization of which can be complicated in the circumstance of insufficient access to the dataset, in particular (Feng et al. 2020; Mahmud and Barua 2024).

5.1.3 Computational and technological limitations

- *Computational demands:* DL models need substantial computational resources, which can present a considerable barrier in resource-limited settings (Choi et al. 2017; Rajendra Acharya et al. 2017; Jiao et al. 2018). This challenge is particularly dominant in the requirement for high-performance hardware and specialized software to manage and process huge volumes of data effectively (Yashudas et al. 2024; Mandava et al. 2024; Wani et al. 2024; Maji et al. 2025; Tanim et al. 2025; Ramu et al. 2025). The complex architectures like DeepNetX2 or three-dimensional convolutional network (3D ConvNet) have extensive computation requirements that might not always be feasible and sustainable, particularly with limited technology infrastructure in a health facility (Tanim et al. 2025; Bäckström et al. 2018).
- *Real-time functionality and scalability concerns:* A significant percentage of models cannot be extended or scaled and would not easily apply in real-time, therefore, decreasing their applicability in dynamic and time-sensitive cases, such as in clinical practice. This limitation is especially profound in models that require a lot of extensive computational resources, making them less feasible for routine clinical practice (Choi et al. 2017; Yashudas et al. 2024; Wani et al. 2024).
- *Generalizability across different demographics:* Models trained on small-scaled, defined datasets often exhibit poor performance when deployed in diverse clinical settings, mainly due to data quality disparities, data collection procedures, and patient demographics. (Feng et al. 2020; Mahmud and Barua 2024; Wani et al. 2024; Tanim et al. 2025; Ramu et al. 2025). This difficulty is particularly evident in studies focused on retinoblastoma diagnosis, where the practical validity of AI models remains questionable (Aldughayfiq et al. 2023). Moreover, the widespread adoption of AI technologies in healthcare is hindered by the absence of affordable and convenient methods to access high-quality clinical data, such as retinal imaging, which is necessary to achieve reliable and accurate performance of models (Aldughayfiq et al. 2023).
- *Energy efficiency and carbon footprint of DL training:* Large-scale AI systems have been questioned in their impact on energy consumption and the environment because

of the computational requirements of training and deploying DL models. The resource intensive nature of modern architectures, particularly transformers, multimodal models and large foundation models necessitate the heavy use of both GPUs and TPUs, which in turn leads to high electricity usage and lots of carbon emission (Strubell et al. 2019; Patterson et al. 2021). This is a major problem in the healthcare environment due to the possibility of resource-intensive training pipelines, which can be unfeasible in hospitals with a small computational infrastructure. Recent research points to ways of reducing energy and carbon expenses, including model pruning, quantization, knowledge distillation, energy-conscious neural architecture search (NAS) and the progression of lightweight clinical models optimized for edge deployment (Tan and Le 2019). The federated and distributed learning can also decrease the necessity of centralized high-powered training because they allow updating locally at a low cost (Li et al. 2020a). The use of energy metrics in model evaluation, including FLOPs, inference latency, and training-energy reporting, can be useful to promote sustainable, scalable, and environmentally friendly development of clinical AI.

5.1.4 Clinical validation and trustworthiness

Clinical validation of Deep Learning models is hindered by the complexity and heterogeneity of EHR data, demographic and setting-related variability, and the challenge of modeling temporal dynamics (Choi et al. 2017). Overfitting and limited generalizability due to constrained datasets further impact performance (Aldughayfiq et al. 2023; Tanim et al. 2025). Clinical validation is imperative to ascertain that predictive models demonstrate consistent performance and reliability within real-world healthcare settings. Meaningful collaboration with healthcare professionals is essential to support the design and execution of clinical studies that assess the safety, efficacy, and practical utility of these models in routine medical practice (Ortiz et al. 2016; Rajendra Acharya et al. 2017; Bäckström et al. 2018; Feng et al. 2020; Çinar and Yildirim 2020; Angel Nancy et al. 2022; Aldughayfiq et al. 2023; Mandava et al. 2024; Mahmud and Barua 2024; Shehta et al. 2025).

5.1.5 Ethical considerations

Ethical issues, including the imperative to ensure that models do not produce discriminatory outcomes, are of critical importance in the development of healthcare AI systems. The involvement of ethics experts throughout the development and evaluation process can aid in identifying and mitigating potential biases, thereby promoting fairness and equitable treatment across diverse patient populations (Shehta et al. 2025).

5.1.6 Fairness, equity, and bias

The problem of equity issues in clinical AI arise because DL models usually reproduce demographic and socioeconomic imbalances present in real-world healthcare data (Obermeyer et al. 2019; Rajkomar et al. 2018). The underrepresentation of the minority will result in the lack of the model generalizability and the unbalanced diagnostic performance within the race, gender, and socioeconomic levels. It is also demonstrated that models can have implicitly encoded demographic characteristics that can amplify risk of biased predictions

and unequal clinical outcome (Gichoya et al. 2022). In addition, differences in quality of data, sensor stability and digital access may also further skew model behavior and increase already existing health inequities (Buolamwini and Gebru 2018). To overcome these challenges, it is necessary to have more representative data, subgroup-consciousness in evaluation, and algorithmic design oriented towards fairness, to achieve equitable deployment.

5.1.7 Fairness metrics

- *Demographic parity*: Demographic Parity (DP) is a measure to determine whether a model gives positive results equally on different demographic groups irrespective of the ground-truth labels. In clinical prediction problems, the value can be used to determine whether an algorithm disproportionately flags some groups as either high-risk or low-risk (Hardt et al. 2016; Sufian et al. 2024; Feng et al. 2024).
- *Equalized odds*: Equalized Odds (EO) demands that both the true positive rate (TPR) and false positive rate (FPR) of a model are equal between groups. This is especially relevant in disease prediction systems due to unequal error rates, which may result in either overdiagnosis or underdiagnosis of particular groups of patients (Hardt et al. 2016; Rajkomar et al. 2018; Sufian et al. 2024; Liang et al. 2024).
- *Equal opportunity*: A relaxation of equalized odds which only demands similar true positive rates across groups (e.g., sensitivity is similar across all), and is concerned with missed detections in any group. This is important in healthcare where all the patients with a condition stand equal opportunities of being rightly diagnosed and reducing disparities in access to timely treatment (Hardt et al. 2016; Mitchell et al. 2018; Bateni et al. 2022).
- *Calibration within groups*: Calibration across groups examines whether predicted risk scores correspond to the actual outcome probabilities are similar across various demographics. Even with high accuracy, poor calibration leads to making harmful clinical decision (Mitchell et al. 2018; Van Der Meijden et al. 2025).

5.1.8 Real-time constraints

Healthcare 5.0 applications which include continuous monitoring and emergency response and wearable-assisted diagnostics require real-time performance (Kumar et al. 2023; Hayyolalam et al. 2021). Deep learning models require high availability, and can run at resource-limited IoT/edge devices and should provide low-latency predictions with clinical reliability, but most of the state-of-the-art architectures are computationally costly and cannot be implemented on-device or at the bedside. According to previous surveys on IoMT, H-IoT, and smart healthcare, the real-time problems of latency, bandwidth, and computational capabilities and power consumption are still significant unresolved problems, particularly with edge-based and safety-critical deployments (Saifuzzaman and Ananna 2024; Kumar et al. 2023).

5.2 Challenges and limitations in medical imaging analysis and radiology

Medical imaging analysis and radiology encounter a range of challenges and limitations that can impede accurate diagnosis and optimal treatment outcomes. These issues arise from technological constraints and human-related factors, collectively affecting the effectiveness and reliability of medical imaging within clinical practice. This section presents a comprehensive overview of the common challenges and limitations encountered in medical imaging analysis and radiology.

5.2.1 Data constraints

- *Limited availability of training data:* DL models, particularly CNNs, often require large and diverse datasets to perform effectively. Limited data, such as a few images or the absence of rare or abnormal cases, can restrict the model's ability to learn comprehensive features and reduce its generalizability to real-world applications (Hua et al. 2015; Cheng et al. 2016; Zilly et al. 2017; Jeyaraj and Samuel Nadar 2019; Song et al. 2019; Tawfeek et al. 2025). Like the small sample size may hinder the CNN's generalization and diagnostic accuracy for esophageal cancer (Horie et al. 2019).
- *Heterogeneity and quality of datasets:* The effectiveness of a model is influenced not only by the volume of data but also by its quality. Poor data quality can compromise model performance, while insufficiently diverse datasets may hinder the model's ability to generalize across varying populations or imaging environments (Hua et al. 2015; Gulshan et al. 2016; Li et al. 2018; Horie et al. 2019; Song et al. 2019; Jeyaraj and Samuel Nadar 2019; Khamparia et al. 2021; Chattopadhyay and Maitra 2022; Huy and Lin 2023; Medhat et al. 2025).
- *Artifacts in medical imaging:* Medical images often contain noise that impacts diagnostic accuracy. While the SDAE can address this, the persistent noisy nature of medical data remains a challenge (Cheng et al. 2016).

5.2.2 Model complexity and algorithmic challenges

- *Probabilistic behavior of algorithms:* Certain algorithms demonstrate stochastic behavior as a result of point sampling within images, which can cause variability in performance across different executions (Zilly et al. 2017).
- *Complexity in feature extraction:* Traditional methods require complex feature extraction and selection, which can be time-consuming and may not offer significant advantages over simpler morphological features used in clinical practice (Cheng et al. 2016). Sometimes, deep networks implicitly learn features that may be unknown or overlooked by humans, limiting the clarity and interpretability of model decisions (Hua et al. 2015; Gulshan et al. 2016; Zilly et al. 2017; Khamparia et al. 2021).
- *Algorithmic inefficiencies:* The performance of image processing methods like segmentation and classification is often limited by current algorithms, which struggle to manage the complexity and variability of medical images. For example, automated glioma segmentation is hindered by the complexity of MRI data, driven by variability in tumor

size, shape, and location, making effective algorithm development difficult (Hua et al. 2015; Zilly et al. 2017; Horie et al. 2019; Khamparia et al. 2021; Chattopadhyay and Maitra 2022).

- *Integration of diverse imaging techniques*: Analyzing images from diverse modalities poses challenges due to variations in image characteristics and the necessity for modality-specific algorithms (Cheng et al. 2016; Song et al. 2019; Jeyaraj and Samuel Nadar 2019).
- *Restricted generalization capability*: DL models trained on specific datasets may lack generalizability, potentially reducing performance across diverse populations and settings, highlighting the need for further research in broader demographic contexts (Li et al. 2018; Song et al. 2019).

5.2.3 Explainability and interpretability challenges

- *Explainability and accountability*: Similar interpretability concerns arise in medical imaging models, particularly when complex feature hierarchies are involved (Li et al. 2018; Song et al. 2019; Tawfeek et al. 2025).
- *Erroneous classification outcomes*: Automated systems can yield false positives, prompting unwarranted interventions, or false negatives, leading to missed diagnoses (Li et al. 2018); in many cases, these errors are primarily linked to overlapping or co-existing eye conditions (Horie et al. 2019).

5.2.4 Ethical and compliance considerations

The application of AI in medical imaging presents regulatory and ethical challenges, such as ensuring transparency and mitigating algorithmic bias, which may impede its integration into clinical practice (Cheng et al. 2016; Li et al. 2018; Song et al. 2019; Jeyaraj and Samuel Nadar 2019).

5.2.5 Legal accountability and responsibility

One of the essential dilemmas in implementing the DL-based systems to Healthcare 5.0 is the establishment of clear responsibility and liability frameworks in the case of the AI wrong diagnosis of patients. Contrary to the conventional methods of clinical decision-making, AI-based diagnosis contains multifaceted interactions between software developers, medical professionals, and organizations, and it is difficult to determine accountability, which is legally ambiguous. The current medical malpractice legislation is poorly suited to deal with the errors caused by algorithmic bias, software bugs, and model drift. The new regulatory environment of the European Union, with new AI Act and amended Product Liability Directive, makes the providers of AI systems very demanding in terms of safety, transparency, and risk management, but there still are challenges of implementation and enforcement. Moreover, the medical workers will still be responsible in the circumstances when they do not correctly incorporate AI outputs into clinical decision-making or overruling inappropriate recommendations. These unresolved legal issues create barriers to adop-

tion, restrict innovation, and obscure pathways for error reporting and patient compensation (Vardas et al. 2025; Van Leeuwen et al. 2025).

5.2.6 Real-world and clinical integration

- *Overburdened radiologists:* Radiologists handle large workloads, leading to fatigue and the risk of overlooking critical findings. While automated systems can ease this burden, their integration presents significant challenges (Gulshan et al. 2016; Medhat et al. 2025).
- *Difficulties in manual interpretation:* Current diagnostic processes rely on manual image interpretation, which is time-consuming and prone to human error (Jeyaraj and Samuel Nadar 2019). This method also depends on skilled radiologists, who may be scarce, particularly in underserved regions (Gulshan et al. 2016; Li et al. 2018; Horie et al. 2019; Khamparia et al. 2021; Tawfeek et al. 2025).

5.2.7 Contextual limitations

- *Computer-aided diagnostic systems:* Deep Learning shows potential in CADx but remains less explored. Conventional CADx systems' reliance on explicit feature design presents a challenge that Deep Learning seeks to overcome (Hua et al. 2015; Cheng et al. 2016).
- *Necessity for advanced methods:* There is a critical need for precise automated disease detection, as traditional methods are often inadequate. Advanced approaches, particularly those employing CNNs, are essential to enhance the accuracy and efficiency of tumor identification (Hua et al. 2015).
- *Limited healthcare access:* In numerous regions, particularly in underserved or rural areas, access to specialized healthcare services is limited, impeding early disease detection and intervention (Tawfeek et al. 2025).

5.3 Challenges and limitations of multimodal deep learning in healthcare

There are several limitations and challenges related to healthcare MMDL and their application that occur during data collection and pre-processing, as well as model deployment and understanding. Important challenges are data imbalance in the classes, insufficient amounts of data, prevalence of pre-processing, interpretability of models, and the challenges associated with choosing the best fusion methods. It is essential to overcome these current challenges in the deployment of advanced models in clinical practice (Raza et al. 2021; Behrad and Abadeh 2022; Cui et al. 2023; Yan et al. 2023; Zong 2024).

5.3.1 Data-related challenges

- *Data source complexity:* A healthcare system has diverse entrants of data, some of which may be an image (X-rays, CT scan), clinical measurements (PFTs), or EHR, and

even at a more fundamental level, genetic information. It is a great challenge to integrate these divergent types of data that are of different formats, scales, and internal structures (Venugopalan et al. 2021; Zhang et al. 2023; Ding et al. 2024; Krishnaveni et al. 2024).

- *Lack of adequate integration strategies:* Numerous current multimodal methods suffer due to inferior integration strategies, making the methods have a low diagnostic capability. Concatenation of features may fail to find the complex relationship among different modalities (Lee et al. 2019; Li et al. 2020b; Sheu et al. 2022; Lee et al. 2023; Thiruvankadam et al. 2024; Krishnaveni et al. 2024; Ranjith and Sakthivanitha 2025).
- *Use of simulated patient data:* Another major limitation of the past research with linguistic models is that they have used simulated patient data, and they lack authenticity in a large-scale clinical environment. This leaves doubts about the autonomous applications and practical deployment (Hao et al. 2019; Mahalakshmi et al. 2024).
- *Contextual information deficiency:* Most of the available schemes fail to consider the context of a piece of information when dealing with text data of medical texts or some missing values. It has the potential to result in less detailed feature extraction, thereby overlooking potentially important diagnostic or prognostic information (Hao et al. 2019). In particular, the status of gene mutation may be characterized by a high rate of missingness (e.g., 10.56% in total, 7.41% in gene data, and 3.15% in the other clinical data), making model training and reliability complicated (Li et al. 2023).
- *Data quality and volume dependency:* The quality and quantity of the multimodal data are very important factors in the accuracy and effectiveness of the multimodal models. Moreover, a deep integration process itself is so crucial that all the shortcomings in the quality of data or a lack of its quantity may influence the work of the model to a great extent (Hao et al. 2019; Li et al. 2020b; Venugopalan et al. 2021; Zhang et al. 2023; Li et al. 2023; Ding et al. 2024; Ranjith and Sakthivanitha 2025).
- *Single disease oriented:* Most past studies on AI imaging and multimodal imaging are single-disease oriented and do not easily integrate with others that are single-disease oriented. This discontinuity among disciplines reduces the general functionality and poses a threat of the generalizability of the models because of spectrum bias.
- *Small public dataset and scarcity of multimodal dataset:* In particular, in healthcare, the lack of a huge sample size in the datasets represents a major bottleneck for the data integration of DL-based models (Li et al. 2020b, 2023; Zhang et al. 2023; Ding et al. 2024; Ranjith and Sakthivanitha 2025). Indicatively, the ADNI dataset contains only a few thousand samples, and even fewer have the three modalities (imaging, EHR, and genetic data) (Venugopalan et al. 2021; Kumar et al. 2025), and publicly shared small THz imaging datasets hinder the development of robust AI-enabled THz applications (Gezimati and Singh 2025). This shortage may cause suboptimal performance because of insufficient training data for the dense networked DL. This field is also lacking in multimodal datasets that would offer one-to-one mappings between other representations (e.g., text and images, or images that belong to different sources). This compels existing research to often use labels from image datasets as text input, which in turn results in inferior model performance (Venugopalan et al. 2021).
- *Dependence on sensitive medical records:* Multimodal models tend to rely heavily on the information provided in the medical records of the patients, which include highly confidential medical information. Such dependency is not very practical when implementing in real-life operations because of privacy and data governance regulations. Any

need to access and use such data demands strict ethical clearances and compliance with the data protection legislation, restricting its usage to the general population and wide-scale research cooperation (Sheu et al. 2022).

5.3.2 Interpretability challenges

These interpretability challenges are further intensified in multimodal architectures due to the complexity of integrating heterogeneous data sources (Li et al. 2023; Zhang et al. 2023; Intriago et al. 2023; Thiruvankadam et al. 2024; Krishnaveni et al. 2024; Ding et al. 2024; Gezimati and Singh 2025; Ranjith and Sakthivanitha 2025).

5.3.3 Generalization and validation challenges

A model trained with a particular dataset or on a set of population may fail to apply to other previously unseen data in other hospitals, areas, or even demographics. Clinical deployment Real-world deployment of the model depends crucially on ensuring its robustness and its generalizability (Venugopalan et al. 2021; Lee et al. 2023; Zhang et al. 2023; Krishnaveni et al. 2024). Although there are good internal results of cross-validation, certain multimodal combinations, especially those with extremely weak overlapping data and small, perform poorly in the external validation sets. This demonstrates that combining or clustering on several datasets is a challenge in making sure that these models work well on the unseen set of data (Venugopalan et al. 2021; Lee et al. 2023; Mahalakshmi et al. 2024; Krishnaveni et al. 2024).

5.3.4 Clinical adoption and workflow integration

The current research suggests that XAI does not intrinsically increase the performance of clinicians and can often lead to automation bias or over-reliance when clinicians follow inaccurate AI results because it is backed by a plausible-looking explanation (e.g., a heat-map emphasizing a relevant organ when the diagnosis is incorrect). Contrarily, complicated or unclear explanations may reduce trust and raise cognitive load, and interfering optimal clinical practices instead of simplify them. It needs a more human-centered evaluation framework since technical metrics (such as SHAP fidelity) are unsuitable to be correlated with the real-world requirements of bedside decision making (Kayser et al. 2024; Badhon et al. 2025; Gambetti et al. 2025; Bienefeld et al. 2023).

5.3.5 Deployment and practical problems

- *Computational resource:* The time and costs of training and deploying MMDL models can be prohibitively expensive, as they often demand high-powered computing resources (e.g., access to a high-performance Graphics Processing Unit (GPU)), which may make it infeasible to deploy MMDL models at most research institutions or by healthcare providers (Lee et al. 2019; Intriago et al. 2023; Mahalakshmi et al. 2024; Ranjith and Sakthivanitha 2025; Kumar et al. 2025).
- *Real-world deployment needs:* Problems that could arise with real-world deployment,

including how to fit these systems into a real-world clinical workflow and how to support scale and robustness in a variety of clinical environments, have yet to be fully addressed (Intriago et al. 2023; Mahalakshmi et al. 2024; Ranjith and Sakthivanitha 2025).

- *Clinician feedback integration*: Although the improvements in areas of feedback loops occur to capture clinical realities, ongoing adjustments and confirmation through clinician input are required in the implementation of any real-life situation incorporation (Sheu et al. 2022; Intriago et al. 2023; Ranjith and Sakthivanitha 2025).
- *Training optimization conflicts*: End-to-end multimodal models may encounter conflicts during backpropagation, as the model finds it challenging to concurrently train both the sub-model (e.g., for medical images) and the fusion module. This competing learning pressure can cause the model to under-train the sub-module, limiting its ability to capture critical features (Li et al. 2023)
- *Loss of information*: In numerous multimodal fusion methodologies, medical images characteristics must be transformed into one-dimensional (1D) tensors to align with the format of clinical data. This procedure may lead to a substantial loss of spatial information, essential for precise diagnosis (Li et al. 2023).

5.3.6 Cross-cutting ethical, regulatory, and fairness challenges

The deployment of deep learning models in Healthcare 5.0 requires careful consideration of cross-cutting ethical, regulatory, and fairness-related challenges. Regulatory approval and compliance demand transparent model behavior, robust validation, and clear accountability to ensure patient safety and system reliability. Ensuring fairness across diverse patient populations and minimizing bias are essential to prevent unequal healthcare outcomes. Addressing these interconnected issues is critical for achieving trustworthy and responsible real-world implementation of AI-driven healthcare systems.

5.4 Methodological quality, validation practices, and bias assessment

In the three areas of DL in Healthcare 5.0 Disease Prediction and Early Diagnosis, Medical Imaging Analysis and Radiology and MMDL validation practices display significant heterogeneity and heterogeneity that affects the methodological soundness, with the majority of studies using internal schemes e.g. k-fold cross-validation or hold-out splits of individual institutions or datasets e.g. ADNI of neuroimaging and small Kaggle-style COVID-19 X-ray collections, with very few studies employing external multi-center cohorts or temporal separations (Krishnaveni et al. 2024). The nature of datasets differs wildly: small datasets (thousands of patients with rare outcomes) are used in early-diagnosis EHR models, tiny (<200 originals) sets are used in imaging studies, medium-sized multi-center MRI/PET datasets have class imbalances (fewer MCI converters), and multimodal works are large EHR-centric datasets to moderate single-center fusions (such as COPD X-ray+Pulmonary Function Test (PFTs)). There are widespread data leakage risks, such as augmentation or patch extraction prior to splitting in imaging and multimodal pipelines, per-visit (as opposed to per-patient) splits in EHR prediction, and slice-level CV in neuroimaging, which interfere with clinical belief in Healthcare 5.0 (Thiruvankadam et al. 2024; Sheu et al. 2022). These dimensions are specifically recorded in the accompanying representative studies summary table, which in turn allows organization and cross-domain comparison of the studies, but

also reveals that patient-level external validation, leakage audits, balanced reporting, and standardized XAI are required to promote methodological rigor (Liu et al. 2014; Li et al. 2023). Table 19 presents a critical analysis across the three domains of Deep Learning in Healthcare 5.0, detailing validation type, Metrics used, Data balance, Data leakage risks, XAI used.

5.5 Quantitative aggregation of results

Tables 18, 19 presents a quantitative aggregation of reported classification accuracies for different disease categories using deep learning based models across multiple benchmark datasets. For each disease, accuracy values reported in the literature were statistically aggregated to compute the *mean accuracy* and the *sample variance*, providing a consolidated measure of overall model performance as well as performance consistency across heterogeneous datasets.

Let $x_1, x_2, \dots, x_n$ denote the reported accuracy values (in %) for a given disease across n studies or datasets. The mean accuracy (μ) is calculated as the arithmetic average of the reported accuracies:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i \quad (3)$$

The sample variance (s^2) is computed to quantify the dispersion of accuracy values around the mean, reflecting the robustness and stability of model performance across different datasets:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2 \quad (4)$$

6 Future research directions

Addressing the identified limitations requires continued research efforts and technological advancements, which are discussed in the following future research directions. Future research in healthcare plays a vital role in advancing medical knowledge and enhancing patient care. With the increasing incorporation of technology into healthcare systems, there is a pressing need to investigate novel methodologies and tools aimed at improving diagnostic precision, therapeutic effectiveness, and patient involvement. Emphasizing these areas can help overcome existing challenges and contribute to the development of more efficient and accessible healthcare solutions.

6.1 Future research direction for disease prediction and early diagnosis

The application of ML and DL models has demonstrated substantial potential in improving the accuracy and promptness of disease detection; however, these approaches still encounter

Table 18 Quantitative aggregation of results using Mean and sample Variance

Method type	Task type	Method	Dataset	Mean	Variance
Deep Learning Jiao et al. (2018); Wani et al. (2024); Shehita et al. (2025); Horie et al. (2019); Song et al. (2019); Jeyaraj and Samuel Nadar (2019); Khanparia et al. (2021); Hua et al. (2015); Khanparia et al. (2021); Cheng et al. (2016)	Cancer diagnosis (Breast, Lung, Blood, Skin, Oral, etc.)	CNN (ResNet, AlexNet, Inception, VGG, EfficientNet, etc.)	DDSM, MIAS, Kaggle SLC, ISIC archive dataset, C-NMC, etc	0.908	114.28
Deep Learning Ortiz et al. (2016); Bäckström et al. (2018); Feng et al. (2019); Mahmud and Barua (2024); Maji et al. (2025); Odusami et al. (2023); Lee et al. (2019); Venugopalan et al. (2021); Li et al. (2023), Feng et al. (2020)	Alzheimer's disease diagnosis	CNN (VGG16, ResNet, Inception, VGG19, DenseNet169, DenseNet201 etc.), RNN, CNN-SVM	ISIC, ADNI, Kaggle etc	0.864	127.53
Deep Learning Rajendra Acharya et al. (2017); Angel Nancy et al. (2022); Yashudas et al. (2024); Cheng et al. (2016)	Heart disease	CNN, RNN (Bi-LSTM, Bi-GRU etc.)	Sutter PAMF, PhysioBank MIT-BIH arrhythmia database, UCI, NIH Bio-specimen Information Coordination Center and Data Warehouse, Kaggle	0.966	10.80
Deep Learning Alazab et al. (2020); Pandit and Banday (2020); Elsharkawy et al. (2021); Naz et al. (2023); Sharma and Guleria (2023); Hroub et al. (2024); Tawfeek et al. (2025); Krishnaveni et al. (2024)	Lung disease (Nodules, tuberculosis, pneumonia, COVID-19, pulmonary edema, COPD)	CNN (VGG16, ResNet50, Inception-V3, DenseNet121 etc.)	Kaggle, SLC, Cohen's COVID-19 data, CORD-19, University of Louisville and Man soura University, COVID-CT, COVID NET, TCVGH etc	0.958	5.45
Deep Learning Tanim et al. (2025)	Diabetes	DNN	Frankfurt hospital, Pabna diabetes hospital, PIMA Indian diabetes dataset	0.963	5.57
Deep Learning Zilly et al. (2017); Li et al. (2020b), Hybrid MMDL Ranjith and Sakthivaniha (2025)	Eye disease (glaucoma, AMD, PM, DR, cataracts etc.)	CNN (ResNet), GAN, attention mechanisms	DRISHTI-GS, RIM ONE v3, Ichallenge AMD, Ichallenge PM, EyePACS, Kaggle, Fundus FFA etc	0.933	19.75

Table 19 Critical evaluation of Deep Learning Studies in healthcare

Key literature	Validation type	Metrics used	Data balance	Data leakage risks	XAI used
1. Disease Prediction and Early Diagnosis					
Ortiz et al. (2016)	Subject-level k-fold CV on ADNI using AAL-parcellated GM patches, with region-wise unsupervised DBNs combined via ensemble voting and SVM fusion	Accuracy, sensitivity, specificity, ROC-AUC	Moderate class imbalance, notably for converters, with many patches derived from a limited set of subjects	Subject-level CV limits patch leakage, but pre-CV preprocessing and no external test set may still yield dataset-specific optimism	Interpretability is limited to AAL-based importance maps, with no model-agnostic XAI, keeping explanations task-specific
Choi et al. (2017)	Train/test evaluation on a single health-system EHR, comparing RNN-GRU with several baselines, with varying observation windows but no external validation	AUROC is the main metric, with the RNN outperforming baselines at both 12- and 18-month windows, plus limited sensitivity/specificity reporting	Highly imbalanced cohort (12% cases) from a single health system, with class balance kept realistic rather than synthetically adjusted	Patient-level, temporally aligned splits limit leakage, but same-system tuning without external validation may yield optimistic performance	None
Rajendra Acharya et al. (2017)	10-fold CV conducted at the beat level on MIT-BIH, using both original and synthetically balanced data, without strict patient-level separation	Per-class and overall accuracy, sensitivity, specificity, positive predictive value	Original classes highly imbalanced; synthetic augmentation equalized counts, increasing total beats	Beat-level 10-fold CV allows patient overlap and pre-split synthetic beats, potentially inflating metrics	None
Jiao et al. (2018)	Pretrained CNN fine-tuned on mammography; train/val/test splits used, though patient overlap may exist	Accuracy and ROC-AUC reported, with qualitative claims of competitive performance	Class imbalance between benign and malignant masses addressed with augmentation and sampling	Pre-split augmentation may cause lesion overlap; patient- and center-level separation unclear, risking inflated performance	None
Bäckström et al. (2018)	Train/validation/test at scan or subject level with grid search; external datasets not used	Accuracy, AD detection rate, false alarm rate; ROC-based metrics	Scan-level splits inflate N but mix subject scans; subject-level splits reflect more realistic performance	Random scan-level split allows subject overlap, causing leakage; accuracy drops with true subject-level split	None; focus on architecture and hyperparameter impacts; no XAI

Table 19 (continued)

Key literature	Validation type	Metrics used	Data balance	Data leakage risks	XAI used
Feng et al. (2019)	Subject-level 10-fold CV with separate MRI and PET 3D-CNNs, followed by FSBi-LSTM on 3D features and comparison of multiple fusion stages	Accuracy, sensitivity, specificity, F1, balanced accuracy, ROC-AUC	Moderate class imbalance (pMCI and sMCI smaller than NC and AD) with per-subject data, many learned features, and all sourced from ADNI	Subject-level folds prevent classic leakage, but pre-CV preprocessing and tuning on the same dataset, without external testing, leave generalization to other sites unverified	Interpretability is limited to ROI masking-based analyses, with no model-agnostic XAI, keeping explanations custom and task-specific
Feng et al. (2020)	Random 85/15 split for train/validation; 10-fold CV within SVM training on features; no independent external validation	For ternary classification: accuracy up to 95.74%, for binary tasks, accuracy, sensitivity, specificity, AUC (e.g., NC vs AD accuracy 99.1, AUC 0.999)	Multiple scans per subject; moderate class imbalance; single-source imaging with uniform acquisition	Preprocessing done pre-split; scan-level random split likely causes subject-level leakage and inflated metrics	None
Mahmud and Barua (2024)	Train/validation/test split; 80% for train+validation, 20% test; no k-fold CV; no external cohort	Accuracy, precision, recall, F1-score	Data balanced via augmentation from highly skewed classes, potentially causing overfitting	Pre-split augmentation and image-based splitting risk leakage; no clinical external validation	Yes; MRI decision regions interpreted via saliency maps and Grad-CAM across models
Yashudas et al. (2024)	Single setup using CloudSim on Framingham and Statlog; train/test split lacks patient or temporal details; no external cohorts	Accuracy, precision/recall	Moderate-size, imbalanced CVD datasets; detailed class counts unavailable	Preprocessing and tuning done pre-split; no subject-level separation, risking leakage and inflated accuracy	None
Ramu et al. (2025)	Single protocol on a large clinical dataset with Synthetic Minority Over-sampling Technique (SMOTE); train/test split only, no k-fold CV or external validation	Accuracy, recall (sensitivity), precision, specificity, F1	SMOTE balances imbalanced data, increasing sample size but altering real-world prevalence	Pre-split SMOTE and CNN feature learning risk test-set leakage and inflated performance; no temporal or center-level separation reported	Limited; highly interpretable results when combined with clinical indicators

Table 19 (continued)

Key literature	Validation type	Metrics used	Data balance	Data leakage risks	XAI used
Maji et al. (2025)	Train–test split for each network; ablation but no k-fold or external validation	Accuracy, loss; separate train/test accuracy; ablation results	Class balance not fully detailed; potential sampling bias toward specific clinics	Risk of leakage arises from optimizing scaling, model design, or fusion before splitting, with no explicit assurance of patient-level separation	LIME and SHAP used for visualizations of feature and region contributions
2. Medical Imaging Analysis and Radiology					
Elsharkawy et al. (2021)	Multiple CV schemes were used, 10-fold, 4-fold, and 2-fold cross-validation, with the best results obtained from 2-fold CV using neural-network fusion over classical classifiers	Best 2-fold CV achieved high sensitivity, specificity, accuracy, and Dice, with other CV schemes slightly lower	Balanced binary outcome (death vs. recovery) with 200 subjects and single X-ray modality	Subject-level CV minimizes image-level leakage, but tuning on the same 200-case dataset without external validation may inflate performance	No interpretability; method relies on stochastic texture features and shallow neural-network fusion without XAI tools
Hroub et al. (2024)	Hold-out validation on two public X-ray datasets with augmentations; no external clinical data; compared 16 DL models across multiple DA settings	Class imbalance exists and is modified through augmentation (e.g., CROP+ROT+HF)	InceptionV3+CRP+ROT+HF: top accuracy/sens/spec/F1-score	Originally imbalanced and then augmented, with patient counts not reported	Pre-split augmentation can duplicate images across splits, and tuning is limited to public data with no external clinical validation
Tawfeek et al. (2025)	Internal train–validation splitting used, comparing sequential, functional, and transfer learning models with dual Adam/SGD optimization against pretrained networks	acc, sens, spec, prec, F1-score; lower params/compute than baselines	Multi-class imbalance persists; segmentation and augmentation increase sample count but not patient diversity	Pre-split segmentation/augmentation may leak data, and no external cohorts are used beyond public sets	Limited qualitative insights; lacks formal CAM or saliency analysis
Kar et al. (2024)	k-fold CV on public ICH dataset	Acc, sens per subtype (typical)	subtype imbalance persists at the slice level	Slice-level cross-validation permits patient leakage due to the absence of a group-wise split	GradCAM / saliency for bleed localization
Naz et al. (2023)	Hold-out + k-fold on NIH/CheXpert; XAI-integrated	Acc, AUC; XAI fidelity high	multi-label imbalanced	Public datasets exhibit label noise and potential multi-label leakage	LIME / SHAP / Grad-CAM for multi-disease heatmaps

Table 19 (continued)

Key literature	Validation type	Metrics used	Data balance	Data leakage risks	XAI used
Chatto- padhyay and Maitra (2022)	Train/test split upon public dataset	Acc, sens	Small-moderate balanced	Aug pre-split	CNN attention/ GradCAM
3. Multimodal Deep Learning in Healthcare					
Krishnaveni et al. (2024)	Validated only on a single clinical cohort with no external hospital test set	Accuracy, per-stage sensitivity / specificity, macro-F1	Approximately balanced across stages after grouping	Single-center data with no patient-level external split; one-time preprocessing may favor cohort-specific overfitting	Limited interpretability; offers modality-level cues but lacks systematic CAM/SHAP for joint outputs
Gezimati and Singh (2025)	Small-sample dataset validated via k-fold or leave-one-subject-out CV	Accuracy, sensitivity, specificity, ROC-AUC	Very small sample size with frequent class imbalance	High overfitting risk; patch-based training may cause specimen-level leakage without patient-level folds	None
Tsai et al. (2025)	Patient-level train / val / test split on a single health-system EHR, with occasional k-fold sensitivity analysis	AUROC, AUPRC, micro/macro-F1, calibration	Large retrospective cohort with structured EHR and other modalities; disease labels are highly imbalanced	Encounter-level splitting risks leakage; patient-level splits are reported but temporal separation may be lacking	Limited interpretability; shows attention and feature importance, but lacks standardized SHAP/LIME across tasks
Elman (1990)	Temporal train/val/test split with potential internal-external hospital validation	AUROC, AUPRC, Brier score, calibration, decision-curve metrics	Large multi-year hospital cohort with notes, labs, and imaging; long-term outcomes are highly imbalanced	Imperfect temporal cuts or multiple admissions per patient can cause outcome leakage; large model size complicates detection	Prompt- and attention-based explanations with saliency on text/images and occasional SHAP for tabular data
Li et al. (2023)	Internal k-fold or hold-out validation on single or multiple hospital datasets	AUROC, F1, and accuracy reported, with occasional per-modality ablation metrics	Visit-level splits risk leakage; fusion features may be precomputed on the full cohort	Visit-level splits and precomputed fusion features may risk data leakage	Mostly modality-level interpretability; lacks fine-grained XAI like per-feature SHAP

Table 19 (continued)

Key literature	Validation type	Metrics used	Data balance	Data leakage risks	XAI used
Mahalakshmi et al. (2024)	Train/val/test split on hospital cohort, occasionally patient-level k-fold	AUROC, accuracy, sensitivity, specificity	Single- or few-center dataset with mammograms, prescriptions, and labs; moderate size with class imbalance	Multiple exams per patient or pre-split feature aggregation may cause leakage	Limited feature-level interpretation and occasional CAM; no unified XAI framework

various challenges and limitations. As the healthcare domain continues to evolve, future research is anticipated to explore advanced methodologies, with a detailed overview of key directions provided in the following section.

Data preprocessing represents a fundamental phase in data analysis and ML, serving to transform raw data into a suitable format for subsequent analysis. This process is particularly vital for enhancing model performance and ensuring data quality, especially when working with complex datasets such as those derived from medical imaging or genomic studies. Future advancements in data preprocessing are expected to focus on improving standardization protocols (Feng et al. 2020), incorporating sophisticated techniques capable of managing increasingly large and heterogeneous datasets (Bäckström et al. 2018).

- *Standardization:* Standardizing MRI data is vital for ensuring voxel-wise anatomical consistency across images (Feng et al. 2020). This includes spatial normalization using tools like SPM12, along with essential preprocessing steps such as skull stripping and grey/white matter segmentation to improve data quality (Feng et al. 2020).
- *Incorporation of sophisticated methodologies:* Advanced preprocessing methods, including tools like FreeSurfer for MRI scans (Ortiz et al. 2016) and AI-based approaches such as CNNs for data preprocessing (Alazab et al. 2020), have demonstrated potential in enhancing model performance and data preparation efficiency, suggesting promising avenues for further improvement.

The quality and complexity of data are pivotal in shaping the development and effectiveness of DL models. Maintaining high data quality necessitates the resolution of issues such as missing values, outliers, and inconsistencies, all of which can adversely affect model accuracy and reliability. Increasing data complexity introduces further challenges, demanding innovative strategies to sustain optimal model performance. Future directions in this domain include data quality enhancement (Tanim et al. 2025; Shehta et al. 2025), effectively managing data complexity, and fostering methodological innovations (Maji et al. 2025).

- *Data quality enhancement:* Data augmentation techniques play a pivotal role in mitigating the limitations of datasets with insufficient images, which can distort model performance. By artificially expanding the dataset, these methods help achieve balance and improve image quality, ultimately enhancing model accuracy (Mahmud and Barua 2024). Preprocessing methods such as normalization and data cleaning are essential for improving data quality, ensuring that DL models receive accurate and relevant input, thereby boosting their reliability and performance (Tanim et al. 2025). Ethical consideration in data quality requires ensuring unbiased model outcomes. Future research should explore dataset biases with guidance from ethics and healthcare experts to enhance ethical standards (Shehta et al. 2025). The integration of real-time data from wearable health-monitoring devices further enhances data quality by reducing dependence on potentially inaccurate user-entered data, thus increasing system reliability, particularly for users less experienced with tracking or inputting health information (Maji et al. 2025).
- *Managing data complexity:* Enhancing the robustness of DL models requires expanding dataset size and diversity, particularly by incorporating varied demographics and disease stages to ensure broader generalizability (Shehta et al. 2025). Future efforts should prioritize inclusive data collection across age, ethnicity, and geography to

strengthen real-world applicability (Maji et al. 2025). Moreover, optimizing hyperparameters through advanced techniques can significantly improve performance on complex datasets (Shehta et al. 2025). Dimensionality reduction methods, such as PCA, aid in managing high-dimensional data by isolating the most informative features (Tanim et al. 2025). The adoption of advanced model architectures, including deep and Convolutional Neural Networks, enables effective extraction of hierarchical features from complex data (Janghel and Rathore 2021; Ramu et al. 2025). Moreover, ensuring scalability and efficiency in models—through lightweight classifiers like LightGBM and XGBoost—remains a key research direction for handling large-scale datasets without compromising accuracy (Ramu et al. 2025).

- *Incorporation of advanced technologies:* The integration of wearable health-monitoring devices enhances data quality and system reliability by providing real-time, automated health metrics, minimizing dependence on potentially inaccurate user input (Maji et al. 2025). Moreover, adapting models for varied environments through preprocessing techniques—such as histogram equalization and brightness normalization—can improve predictive performance across diverse imaging conditions (Maji et al. 2025).

With the growing reliance on data-driven healthcare, ensuring the availability of high-quality data is vital. This requires overcoming challenges in data collection, storage, and sharing, while upholding ethical and privacy standards. The following sections outline key future research directions concerning data availability in healthcare.

- *Augmenting data and ensuring class balance:* Data augmentation is essential for improving the size and balance of limited datasets, enhancing model performance through techniques such as flipping, rotation, and zooming (Mahmud and Barua 2024). Future research should aim to optimize these methods to better represent minority classes, thereby reducing bias and increasing model robustness (Mahmud and Barua 2024).
- *Multi-source data integration:* Integrating multi-modal data—such as genetic, demographic, and longitudinal information—can enhance diagnostic accuracy and early detection (Mahmud and Barua 2024). Additionally, employing advanced feature extraction techniques, including modern computer vision methods, may improve model performance by capturing complex data patterns and preserving structural details (Feng et al. 2019).
- *Resource-efficient model adaptation:* Optimizing models for resource-constrained environments involves adapting them for low-power devices through techniques like quantization and pruning. Additionally, enabling offline functionality can enhance accessibility in regions with limited internet connectivity. (Maji et al. 2025).

Deep Learning models typically depend on extensive and diverse datasets for optimal performance. However, in fields like medical imaging and diagnostics, acquiring such data is challenging due to privacy constraints and the requirement for expert annotations. Future research is directed toward addressing these limitations related to data scarcity and lack of diversity.

- *Expanding dataset scale and heterogeneity:* Data augmentation enhances dataset size and balance by creating altered versions of existing images, helping improve model

accuracy, especially for underrepresented classes (Ramu et al. 2025; Bäckström et al. 2018; Mahmud and Barua 2024).

- *Pretrained model utilization:* Hyperparameter optimization (Shehta et al. 2025) leverages pretrained models on large datasets and adapts them to smaller (Maji et al. 2025), domain-specific data, improving model performance in data-limited settings. Investigating various pretrained models and their combinations offers potential for enhancing diagnostic accuracy and represents a valuable direction for future research (Mahmud and Barua 2024).
- *Ensemble Learning:* Employing Ensemble methods enhances model accuracy by integrating the strengths of multiple DL models, effectively reducing overfitting in data-limited contexts (Ortiz et al. 2016).

Data imbalance occurs when some classes are significantly underrepresented within a dataset, which can result in biased and unreliable model predictions. Effectively addressing this issue is essential for developing robust and accurate predictive models. Future directions in this domain include the implementation of synthetic data generation approaches (Tanim et al. 2025), adoption of resampling methods (Tanim et al. 2025; Maji et al. 2025), and design of cost-sensitive learning strategies (Bäckström et al. 2018; Ramu et al. 2025).

- *Implementation of synthetic data generation approaches:* The Implementation of synthetic data generation using models like GANs offers a promising approach to addressing class imbalance by augmenting minority classes with realistic, data-driven samples (Tanim et al. 2025).
- *Adoption of resampling methods:* Future research may consider applying resampling techniques, such as oversampling minority classes or undersampling majority ones, to achieve class balance, with methods like SMOTE proving effective in improving model training and performance (Tanim et al. 2025; Maji et al. 2025).
- *Design of cost-sensitive learning strategies:* Exploring cost-sensitive learning methods that assign varying misclassification costs can enhance model focus on minority classes, improving prediction sensitivity and precision in imbalanced datasets (Bäckström et al. 2018; Ramu et al. 2025).

Methodological challenges, including model complexity, interpretability, feature extraction, and the integration of diverse data modalities, must be addressed to enhance the effectiveness and applicability of these models. The following sections highlight key future research directions related to these constraints in healthcare.

- *Improving model interpretability:* Future research should aim to enhance XAI methods to improve the transparency and interpretability of DL models, thereby fostering trust and supporting their adoption in clinical practice (Mahmud and Barua 2024; Shehta et al. 2025).
- *High-level feature engineering techniques:* Incorporating multi-modal data and optimizing feature integration can enhance diagnostic accuracy, highlighting the need for improved dynamic feature weighting in Deep Learning models (Mahmud and Barua 2024; Maji et al. 2025).
- *Unsupervised Feature Learning:* Unsupervised Deep Learning enables representative

feature extraction without reliance on ground truth, benefiting scenarios with unreliable labels (Ortiz et al. 2016).

- *Data fusion techniques:* Integrating imaging, genetic, and clinical data enhances predictive accuracy. Future work should explore end-to-end models combining autoencoders with advanced integration methods beyond simple concatenation, enabling multi-level data fusion to improve disease understanding and outcome prediction (Maji et al. 2025).
- *Strengthening Frameworks for liability and accountability:* In order to facilitate the safe, ethical, and trustful incorporation of AI into Healthcare 5.0, future research should create elaborate liability and accountability schemes specific to AI-facilitating clinical settings. This involves the definition of the sharing of responsibility among the developers, healthcare providers, and institutions especially on high stakes decisions. Transparency and incident attribution can be ensured with help of innovative practice, including robust audit trails, explainable AI models, and standardized reporting mechanisms. The studies should also address technical methods of improving human supervision, reducing automation bias, and incorporation of fail-safe measures into AI processes. To harmonize the liability regulations worldwide, cross-jurisdictional studies are required to cover the differences between the regions in terms of legal frameworks and policies on healthcare. Also, longitudinal research assessing effects of liability on AI adoption, clinical outcomes and patient trust will be used to develop balanced policies that are up to the task of protecting patients without stifling innovation. Such a multi-disciplinary collaboration between legal, technical, and clinical expertise is needed to support the safe and fair implementation of AI in Healthcare 5.0 (Van Leeuwen et al. 2025; Vardas et al. 2025).

DL models in healthcare face challenges like high computational demands, limited real-time capability, scalability, and demographic generalizability. The following sections outline key future research directions to address these issues.

- *Significant resource demands and parameter optimization:* Complex models, like 3D-CNNs, demand high computational resources, limiting deployment in resource-constrained settings (Feng et al. 2019). Lightweight models, such as MobileNet, offer reduced resource requirements but may sacrifice accuracy (Maji et al. 2025). Models like MOD-2D-CNN balance efficiency and performance (Maji et al. 2025). Minimizing the number of trainable parameters improves convergence speed and minimizes memory usage for practical applications (Feng et al. 2019).
- *Real-time model optimization:* Optimizing models for low-power or mobile devices is essential for real-time use, particularly in low-connectivity environments. Techniques like model quantization and pruning reduce computational overhead, improving usability in such scenarios (Maji et al. 2025). Within the low-resource or developing healthcare system, deep learning methods can greatly increase the diagnostic and decision-support functions despite the infrastructural constraints (Williams et al. 2021; Zehra et al. 2023). Lightweight and optimized models may be deployed on a mobile or edge device, allowing prediction of disease and analysis of an image without accessing high-end computational capabilities. Cloud-based and edge computing platforms will be used to evaluate remote data and deploy models to enhance access to diagnostic services in underserved regions. Furthermore, transfer learning enables the application of the existing models to

the medical data of a region, which reduces the scope of the datasets and training data requirements, e.g. brain tumour (glioma) segmentation models based on deep learning have been trained on large datasets and fine-tuned to operate on lower quality or even limited MRI images in Sub-Saharan Africa, allowing diagnostic results where other radiology facilities might be limited (Parida et al. 2024). All of these methods contribute to making deep learning a feasible and scalable solution to enhance the quality and accessibility of healthcare in resource-constrained settings (Hoodbhoy et al. 2019).

- *Cross-platform scalability:* Integrating AI with portable imaging devices offers scalable diagnostic solutions, expanding healthcare access (Aldughayfiq et al. 2023). Ensuring model efficiency across diverse hardware is crucial for widespread adoption (Maji et al. 2025).
- *Comprehensive data acquisition:* Expanding datasets to cover diverse demographics is vital for improving model generalizability and accuracy (Maji et al. 2025). Addressing biases and ensuring representation of underrepresented groups enhances model reliability in real-world applications (Aldughayfiq et al. 2023).

Clinical validation is imperative to ascertain that predictive models demonstrate consistent performance and reliability within real-world healthcare settings. But such models' clinical adoption requires addressing ethical and validation challenges. The following sections outline future research directions to tackle these issues.

- *Ethical and equitable deployment:* Ensuring fairness is vital to mitigate bias and prevent discriminatory outcomes in healthcare (Shehta et al. 2025). Enhancing model transparency through explainable AI is key to clinical trust and ethical use (Tanim et al. 2025). Additionally, adapting models to varied global healthcare environments is necessary to support equitable access and practical deployment (Shehta et al. 2025).
- *Improved protocols for clinical validation:* Clinical validation and integration into healthcare workflows are crucial to ensure the safety, efficacy, and practical utility of such models in real-world medical settings. Meaningful collaboration with healthcare professionals is essential to support the design and execution of clinical studies that assess the safety, efficacy, and practical utility of these models in routine medical practice (Shehta et al. 2025).
- *Ensuring the privacy and integrity of patient data:* Maintaining patient confidentiality and regulatory compliance necessitates the implementation of robust data protection strategies, encompassing secure data management and transparent usage policies to foster patient trust (Shehta et al. 2025).
- *Role of clinical trials and prospective validation:* The need to bring deep learning models to actual Healthcare 5.0 systems implies that such models should be clinically validated. A variety of potential studies and clinical trials have revealed the performance of AI models deployed in the conditions of normal clinical practice. As an example, the IDx-DR system, used to detect diabetic retinopathy and cleared by the FDA, was put in a prospective, multisite clinical trial and demonstrated autonomous diagnostic performance in the primary care setting (Abramoff et al. 2018). On the same note, deep learning stroke detection systems like Viz.ai LVO system have been tested in clinical practice and demonstrated to save time to treatment. The diabetic retinopathy deep learning model developed by Google was also confirmed in the large-scale prospective

implementation trial at Thai hospitals and demonstrated high generalizability but also indicated operative difficulties (Beede et al. 2020). Future studies in the critical care, including the evaluation of deep learning models against early sepsis detection, state that real-time clinical outcomes can vary compared to the retrospective ones owing to data drift and patient heterogeneity. The critical role of clinical evidence, workflow integration, and ongoing observation, as highlighted in these studies, is to make sure that deep learning systems are reliable, safe, and effective in the world of Healthcare 5.0 settings.

- *Involvement of key stakeholders:* Engaging diverse stakeholders—such as patients, clinicians, and ethicists—in model development promotes alignment with practical needs and ethical standards (Shehta et al. 2025).
- *Approaches addressing real-time constraints:* Recent evidence of edge intelligence in the healthcare domain demonstrates that distributed computing to edge or fog devices near sensors can potentially save a lot of latency and bandwidth. Techniques such as model compression, pruning, quantization and lightweight architectures (e.g., MobileNet, EfficientNet-lite, compact CNNs) are being implemented to execute deep models on low-powered hardware at sufficiently diagnostic accuracy and response times suitable to support ongoing monitoring and anomaly detection (Howard et al. 2017; Tan and Le 2019). Research on arrhythmia detection and sepsis prediction also shows that patient safety can be affected by delays or unstable inference pipelines, further explaining the importance of streamlined architecture and well-developed real-time data processing infrastructures (Henry et al. 2015; Hannun et al. 2019). Frameworks based on edge-cloud collaboration and architecture with 5 G/URLLC are also suggested to ensure the trade of accuracy and real-time guarantees to applications such as remote vital-sign prediction and real-time health surveillance (Appavu 2025; Subiramaniyam et al. 2024; Rajagopal and Subramanian 2025).

6.2 Future research direction for medical imaging analysis and radiology

The future of medical image analysis and radiology is set for substantial progress propelled by innovative technologies and techniques. Primary focal points are the incorporation of AI, DL, and radiomics, which are expected to improve diagnostic precision and treatment customization. The subsequent sections delineate the principal research trajectories expected in this domain. Data limitations in medical imaging—such as scarcity, variability, and artifacts—pose significant challenges but can be mitigated through innovative approaches that improve the effectiveness of DL models.

- *Integration of public datasets and clinical partnerships:* The use of open-source datasets offers a practical solution to data scarcity by supporting model training in data-limited domains. Additionally, collaborations with healthcare institutions enable access to diverse, well-annotated datasets, essential for developing robust models (Tawfeek et al. 2025).
- *Data expansion strategies:* Data augmentation methods—such as rotation, scaling, and flipping—enhance the variability of training datasets, thereby improving model performance under limited data conditions (Tawfeek et al. 2025).
- *Preprocessing strategies and attention mechanisms:* Applying preprocessing techniques like normalization and augmentation enhances data quality and model performance by

standardizing input characteristics. Additionally, integrating attention mechanisms into CNNs improves both interpretability and accuracy by enabling the model to prioritize the most relevant image regions, effectively addressing dataset heterogeneity (Kar et al. 2024).

- *Advanced segmentation strategies:* Advanced segmentation techniques like Mask R-CNN improve classification accuracy by isolating relevant features and reducing the influence of artifacts in medical images (Tawfeek et al. 2025).
- *Quality control protocols:* Applying quality control during data acquisition and pre-processing ensures high-quality input data, essential for reliable performance (Kar et al. 2024).

DL has significantly advanced diagnostic accuracy in medical imaging, yet it faces key challenges such as limited interpretability, algorithmic inefficiencies, classification errors, and generalizability issues. Subsequent investigations are focused on mitigating these limitations.

- *AI interpretability frameworks:* Explainable AI frameworks are essential in healthcare to enhance the transparency and trustworthiness of DL models. It needs to be transitioned away in passive visualization to decision-oriented explanations that project model signals into clinical reasoning. Future work should include: (a) creating clinically aligned explanations (e.g., connecting heatmaps or token attributions to diagnostic criteria), (b) integrating uncertainty estimates with explanation (highlight low-confidence cases), and (c) integrate the output of the explanations into the workflow used by clinicians (audit logs, justification notes, counterfactual suggestions). Model explanations that are verified by the domain experts can increase the levels of clinician trust, allow analysis of errors more precisely, and increase the speed of regulatory acceptance-enhancing both the accuracy of early prediction and diagnostic safety (Ribeiro et al. 2016; Amann et al. 2020; Naz et al. 2023).
- *Predictive output transparency:* Ensuring the interpretability of model predictions is essential for fostering trust among healthcare professionals. This requires enhancing the transparency of the AI model's decision-making processes to support informed clinical adoption (Tawfeek et al. 2025).
- *Algorithmic optimization techniques:* Employing dual optimization algorithms, such as Adam and stochastic gradient descent (SGD), can enhance model performance by effectively reducing training parameters and computational overhead (Tawfeek et al. 2025).
- *Model Evaluation and Assessment:* Thorough testing and assessment of models with varied datasets are crucial to reduce incorrect classification results and enhance diagnostic precision (Kar et al. 2024).
- *Integration of multiple imaging modalities:* The integration of multiple imaging modalities, such as CT scans and X-rays, can augment the diagnostic effectiveness of AI models by offering a more holistic representation of medical conditions (Tawfeek et al. 2025).
- *Equitable model performance:* Ensuring fairness and mitigating bias in algorithmic design is fundamental to the ethical implementation of AI in healthcare. This requires training models on diverse and representative datasets to prevent discriminatory or skewed outcomes (Naz et al. 2023).

- *Generalizability assessment:* External validation in real-world clinical contexts is essential to confirm the generalizability of AI models across varied populations and healthcare settings (Kar et al. 2024).

DL models in healthcare encounter challenges in real-world clinical integration, including radiologist workload and complexities in manual interpretation. The subsequent sections highlight key future research directions aimed at addressing these concerns.

- *Automated diagnostic imaging systems:* DL models offer automated medical image screening, effectively reducing radiologists' workload by accurately detecting and classifying chronic diseases such as lung diseases (Tawfeek et al. 2025). This enables radiologists to concentrate on complex cases, thereby minimizing fatigue-related diagnostic errors (Kuchana et al. 2021).
- *Enhanced diagnostic precision:* AI systems, when trained on extensive datasets, can achieve diagnostic performance comparable to or surpassing that of expert radiologists. For example, by delivering objective, quantitative metrics—such as the extent of lung involvement—they promote greater consistency and reliability in clinical decision-making (Kuchana et al. 2021).
- *Advanced medical image interpretation:* Different DL algorithms, particularly CNNs, have significantly advanced medical image analysis by extracting hierarchical features that enhance diagnostic accuracy. They enable automated detection of conditions such as cerebral hemorrhages, thereby reducing manual interpretation and lowering the risk of human error (Kar et al. 2024).
- *Integration into clinical workflows and model refinement:* Workflow integration is a crucial prerequisite of DL in Healthcare 5.0, but it is one of the least developed fields of current research. In practice, a large number of AI systems have a difficult time in clinical environments due to lack of alignment of their output with clinical information flows, their inability to integrate with hospital IT ecosystems, and the fact that they add cognitive and operational overhead to final users. Thus, despite some models demonstrating good performance in the experiment, they fail to serve the purpose of clinical impact (Amann et al. 2020). Future research should put in place technically strict frameworks of integrating deep learning tools into end to end clinical processes. Among the requirements are standard data interfaces, compatibility with EHR, Picture Archiving and Communication System (PACS), Laboratory Information System (LIS), and Fast Healthcare Interoperability Resources (FHIR)-based systems as well as mechanisms that enable constant ingestion of multimodal data provided by IoT devices, wearables, and remote monitoring platforms. The microservice-based architecture, API-based orchestration, and ontology-based knowledge graph can be used to overcome interoperability barriers and provide data exchange among different healthcare systems on a semantically consistent basis (Miotto et al. 2018; Shortliffe et al. 2018). The modeling of the dynamic workflow behavior is another significant direction. AI tools can be programmed to adapt their suggestions depending on the patient acuity, workload of the staff, and changes in resource availability using adaptive workflow engines, multi-agent coordination systems, digital twins, and context-aware decision-support modules. In the real world, reinforcement learning can also help optimize triage, task allocation

and time-sensitive clinical processes. There should also be the issues of scalability and robustness. Federated learning, decentralized integration pipelines, and privacy preserving computation will play a crucial role in the implementation of AI systems in multiple institutions without violation of security and regulatory requirements. Longitudinal evaluation systems, which will include drift detection, real-time performance checks, and feedback mechanisms, are required to make deployed models stable and safe in the course of their life (Pingili et al. 2025). The AI systems should also be incorporated in the clinical workflows in a manner that assists in real-time diagnostics and decision-making. This involves enhancing the interpretability of the model, its reliability, and operational transparency to allow the utilization of outputs at the point of care. On-going updating of models with different datasets and taking into consideration the real-life use feedbacks will be crucial to the preservation of accuracy of the diagnostic in different clinical environments. To make sure that deep learning applications support clinical priorities and in fact decrease cognitive load, cooperation among developers, workflow engineers, and healthcare professionals is paramount to achieve this goal, as well as to make sure that the tools are aligned with clinical priorities (Horie et al. 2019). *Advancing accuracy in clinical diagnosis:* AI systems can be embedded into clinical workflows to support real-time diagnostics, thereby improving care quality. To ensure effective clinical adoption, future research should prioritize enhancing model interpretability and reliability. Continuous training with diverse datasets and close collaboration between developers and healthcare professionals are vital for maintaining diagnostic accuracy and aligning AI tools with clinical needs (Horie et al. 2019; Kar et al. 2024).

- *Minimizing diagnostic inaccuracies:* DL enhances CADx by minimizing errors linked to manual interpretation. Models like the SDAE outperform traditional methods by autonomously extracting and leveraging meaningful features from medical images (Cheng et al. 2016).
- *Optimizing clinical efficiency:* AI-powered diagnostic tools can improve hospital efficiency by decreasing clinicians' workload and enabling greater focus on patient care, especially in rural settings with limited medical personnel (Tawfeek et al. 2025).
- *Physician acceptance and trust:* The acceptance of deep learning in Healthcare 5.0 depends on the acceptance of physicians. The clinicians tend to be reluctant to use AI systems when the reasoning of a model is not clear and where the outputs fail to respond to the clinical judgment. Trust can be enhanced by ensuring systems have understandable explanations, uncertainty estimates, and actionable insights as opposed to opaque predictions (Caruana et al. 2015). Human-centered design needs to be considered in the future, with the clinicians involved in the model development, consideration of the explanation form, and cognitive workflow requirements. Further research is also required to quantify the interaction of clinicians with AI tools in practice, such as overriding tendencies and responsibility or automation bias issues (Crigger et al. 2022). Enhancement of these trust issues will be critical to any meaningful use of deep learning in daily clinical practice.
- *Real world deployment:* A major translational gap in deep learning research for Healthcare 5.0 is the relative scarcity of real-world, large-scale deployments that extend beyond controlled academic prototypes. However this trend is changing very fast. As an illustration, Advocate Health has implemented more than 40 AI solutions, which are ready to production, such as Microsoft Dragon Copilot in ambient documentation and

imaging skills like Aidoc and Rad AI, within the scope of its clinical enterprise. Kaiser Permanente implemented Abridge's generative documentation system in more than 40 hospitals and 600+ medical offices, the largest generative AI deployment in clinical operations to date, demonstrating that scalable, field-tested tools can dramatically reduce documentation burden while integrating with EHR ecosystems. Predictive sepsis and respiratory failure models, digital twin models, and closed-loop ventilator management have been tested and implemented in various ICUs around the world using frameworks such as the Learn-Predict-Monitor-Detect-Correct (LPMDC) architecture, which have demonstrated significant reductions in mortality, ICU length-of-stay and cognitively load on clinicians. Also, existing medical imaging triage AI systems like Aidoc are already used in the US and European hospitals on a regular basis to rank radiology workflows and identify critical results to be reviewed faster (Boussi Rahmouni et al. 2025). Nevertheless, the uptake of deep learning systems is projected to be very low (estimated at 22% of health organizations around the world) due to the complexity of integration, compliance mandated by the government, and workforce adaptation issues. Therefore, the future development is to create more robust deployment models, scalable interoperability models, and large-scale longitudinal studies that can help close the ongoing research-to-practice gap based on theoretical experimentation and sustainable and pragmatic deployment in various and complicated healthcare settings (Younesi et al. 2025).

6.3 Future research direction for multimodal deep learning in healthcare

The field of MMDL in healthcare will be a topic of future research to resolve the existing limitations and widen its functions to produce exact, steady, and clinical combination diagnostic schemes. The most important areas involve model optimization on large and diverse datasets, adding new types of data, easing model explainability, and potentially exploiting novel training approaches such as federated learning to alleviate data privacy and sharing issues.

6.3.1 Data limitations solutions

- *Expansion and diversification of datasets:* An initial issue to address is the relatively small sample size of many medical datasets, e.g., the ADNI dataset to study AD, and limited purely publicly accessible THz imaging datasets to study cancer. The future work will be focused on testing models on increased, ultra-rich, diverse datasets to enhance generalizability and resiliency (Venugopalan et al. 2021; Gezimati and Singh 2025).
- *Approaches to missing data:* The potential problem with missing data can be observed in clinical data, especially data related to a certain test, such as the gene mutation status. It is planned that more advanced imputation techniques will be investigated in the future to deal with such an unavoidable issue (Li et al. 2023).
- *Normalization:* Normalization of clinical data entails rescaling numerical variables to a defined range and transforming categorical variables into binary representations, commonly through approaches such as one-hot encoding (Venugopalan et al. 2021).
- *Alternative training datasets:* Future work could find and utilize subsets of already well-

known and complementary modalities (such as IR imaging for THz imaging) as an alternative training dataset to address the problem of scarcity of training data with newer and more unproven imaging schemes (Gezimati and Singh 2025).

- *Optimization on varying datasets:* Future work will be to optimize different models, such as DenseNet, over larger and more diverse data to increase their power to predict (Kumar et al. 2025).

6.3.2 Model architecture and performance improvement

- *Improved fusion techniques:* The early and late fusion techniques are advanced techniques and oversimplistic. The future research will be aimed at more sophisticated fusion strategies, including models utilizing attention capabilities and graph neural networks, which describe complex relationships between the types of data in a better way (Ranjith and Sakthivanitha 2025).
- *Fusion of new modalities:* Multimodal data can be integrated to get an improved result. E.g., an investigation can be undertaken into the integration of additional data sources, such as cognitive assessments and genetic markers, to develop more robust and effective predictive models (Kumar et al. 2025).
- *End-to-end training:* A future direction is to support end-to-end training of multimodal models: integrating feature extraction steps (such as auto-encoders) and integration and classification, instead of training them separately (Venugopalan et al. 2021).

6.3.3 Clinical integration and explainability

- *Increased explainability:* One of the largest impediments to clinical uptake is the *black box* nature of most DL models. Future work will focus on the enhancement of explainability, with methods such as Grad-CAM and SHAP to make it more amenable to clinician trust and practices (Ranjith and Sakthivanitha 2025).
- *Advancing clinical DL through emerging methods:* Emerging methods like foundation models, large multimodal AI models, and federated learning offer tangible ways to address major drawbacks of the existing clinical deep learning. Large-scale medical and general-domain trained foundation models can offer powerful generalization, can support low-resource tasks, and lower the need for task-specific labeled datasets (Moor et al. 2023). Large multimodal models combine imaging, text, genomics and signals into a unified architecture to allow clinical reasoning in a more holistic manner and to decrease errors from single-modality blind spot (Wan et al. 2023). The federated learning will play a critical role in overcoming the issue of data security and privacy. This can enable training models on decentralized data across multiple institutions, without sharing raw patient data, and so provide access to larger, more diverse datasets to train high-precision, multimodal diagnostic systems (Kaissis et al. 2020; Li et al. 2020c; Ranjith and Sakthivanitha 2025). All these new approaches offer tangible and practical improvements that can enhance generalizability, decrease bias, reinforce multimodal integration, and render clinical realization.
- *Increased scope:* Models can be set to the behavior of unusual or complex cases and can

be scaled to be able to diagnose more diseases, and hence that increases their clinical value (Ranjith and Sakthivanitha 2025).

- *Regulatory and compliance requirements* Companies that regulate the field have started to codify the specifications of AI-powered medical tools. The U.S. Food and Drug Administration (FDA) (Benjamins et al. 2020) has published frameworks including the Software as a Medical Device (SaMD) guidelines and the regulatory approach proposal of AI/ML-based adaptive algorithms which focus on transparency, real-world performance monitoring, and risk control. Likewise, the European Medicines Agency (EMA) offers a guideline in its AI reflection paper, which identifies data quality, model validation, explainability and patient safety requirements (Kelly et al. 2019; Wiens et al. 2019). These frameworks facilitate the standardization of the development and implementation of DL applications to make emerging Healthcare 5.0 systems to address the clinical, ethical, and safety expectations.

7 Conclusions

The integration of DL into smart healthcare systems has opened new frontiers in disease prediction, diagnostics, medical image analysis, and radiology. This survey systematically addresses the research questions outlined at the beginning. In relation to RQ1, our review has demonstrated that DL serves as a cornerstone for advancing Healthcare 5.0 by enabling automation, personalization, and precision, thereby surpassing the capabilities of traditional ML approaches, as discussed in the introduction. For RQ2, we have presented a detailed examination of the most widely utilized DL algorithms, emphasizing their architectures, strengths, and clinical suitability, which are elaborated in the background section. With respect to RQ3, we have synthesized findings from approximately 20 studies in each of the three primary application domains, including disease prediction and early diagnosis, medical imaging and radiology, and MMDL. These studies are analyzed regarding their methodologies, outcomes, and clinical implications. Our survey further connects diseases, predictive models, accuracy levels, and dataset characteristics, demonstrating how factors such as dataset size, modality, and diversity affect model performance and generalizability, as reported in the DL in healthcare 5.0 section. Addressing RQ4, we have identified major limitations, including a lack of interpretability, scarcity of annotated data, heterogeneity of data sources, high computational requirements, and ethical/privacy concerns, as outlined in the challenges and limitations section. In response to RQ5, we have highlighted promising research directions such as the development of XAI frameworks, integration of multimodal data, privacy-preserving learning techniques, and lightweight architectures for real-time clinical applications, as presented in the future research directions section. Responding to RQ6, we found that the majority of studies are based on the standard validation methods and performance measurement procedures, yet they do not consider the external validation, data leakage, bias evaluation, and reproducibility, and standardized assessment protocols and open reporting practices are necessary. Lastly, to answer RQ7, we highlighted the role of interpretability, fairness, and regulatory compliance with the XAI approaches, bias-conscious assessment, and compliance with regulatory standards being essential to fostering clinical trust and enabling the safe use of deep learning systems.

This study systematically reviews deep learning methodologies in healthcare, highlighting how innovations such as generative models and attention-based architectures are being applied to tackle complex medical challenges. Our bibliometric analysis further highlighted the intellectual landscape of this interdisciplinary field, underscoring emerging research hotspots and collaborative trends. Despite the remarkable progress, significant hurdles persist, including data scarcity, ethical concerns, lack of interpretability, and the difficulty in deploying models in real-world clinical settings. Addressing these challenges requires a balanced focus on algorithmic innovation, regulatory frameworks, and cross-disciplinary collaboration. The review provides a highly empirical and objectively distinguished viewpoint through the comprehensive comparative analysis of DL studies on Healthcare 4.0 and 5.0 and combining reported findings. It further points out a lack of rigor in evaluation and pinpoints interpretability, fairness, and regulatory compliance as the needs of reliable and deployable Healthcare 5.0 systems. As the healthcare sector increasingly embraces AI-driven solutions, DL stands at the core of this transformation, holding great promise for building resilient, accessible, and personalized healthcare systems for the future.

Appendix

Background of Foundational Deep Learning Architectures

In this section, we give an overview of the most important structures that have dominated contemporary AI: CNNs, RNNs, LSTM and GRUs.

Convolutional neural networks (CNNs)

CNNs are commonly used for feature extraction from pixel images and classification tasks. It was first introduced by LeCun et al. in their paper on document recognition in 1998 (LeCun et al. 2002). It lowered the reliance on manual extraction of features by sliding the convolution filters over input data (Basha et al. 2020) and captured the spatial structures. The pooling layers condense the feature maps (Zhao et al. 2024) and improve computational efficiency. A fully connected layer takes the flattened feature maps as the input and produces the output (Alzubaidi et al. 2021). Equation 5 represents the forward pass of a CNN.

$$y_{k,i,j} = \sigma \left(b_k + \sum_{c=1}^{C_{in}} \sum_{m=0}^{K_h-1} \sum_{n=0}^{K_w-1} W_{k,c,m,n} X_{c,i+m,j+n} \right) \quad (5)$$

where:

$y_{k,i,j}$ = Output for channel k at position (i, j) ,
 σ = Activation function,
 b_k = Bias for channel k ,
 C_{in} = Total input channels,
 K_h, K_w = Width and height of Kernel K ,
 $W_{k,c,m,n}$ = For input channel C and output channel K , the weights at (m, n) position in the filter,
 $X_{c,i+m,j+n}$ = Pixel value for channel c , after moving m rows and n columns from the start.

Recurrent neural networks (RNNs)

RNN has three layers: input, hidden, and output layers, in which the outer layers are feed forward but the hidden layers are recurrent, in which the output of the present state is fed to the next, which is crucial for understanding sequences (Bianchi et al. 2017). The hidden state contains the short-term memory. RNNs are slow and fail to handle long sequences and suffer from vanishing and exploding gradient problems. The hidden state computation is represented in equation 6. RNNs are especially well applied to clinical problems that deal with sequential or time-dependent or variable data like patient history, physiology, and clinical text. Their memory, dependency modeling, and future health outcome forecasting capabilities render them essential in time-sensitive medical activities such as modeling the course of diseases, real-time monitoring, and predictive analytics. For example, they are used in alzheimer's detection (Nguyen et al. 2018), detecting medical events (Jagannatha and Yu 2016), etc.

$$h_t = \phi(W_{xh}x_t + W_{hh}h_{t-1} + b_h) \quad (6)$$

where:

h_t = Hidden state or internal memory of RNN,
 ϕ = Activation function,
 x_t = Input vector for present state,
 W_{xh} = Weight matrix from current state input to hidden state,
 W_{hh} = Weight matrix from previous hidden state to present hidden state,
 b_h = Bias for the hidden layer.

Long short-term memory (LSTM) and gated recurrent units (GRUs)

LSTM introduces another cell state, which is the long-term memory state, along with the short-term memory present in traditional RNNs. It consists of three gates discussed below and represented in equation 12 (Filippo et al. 2017).

The equations represent the LSTM operations, which include forget gate, input gate, candidate cell state, cell state update, output gate, and hidden state update computations.

1. Forget gate: to decide what to discard from the previous state.
2. Input gate: decides the amount of new information to be added to the state.
3. Output gate: decides what to show as output.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (7)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (8)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (9)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (10)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (11)$$

$$h_t = o_t \odot \tanh(c_t) \quad (12)$$

Where:

x_t = Input vector at time t ,

h_t = Short-term memory – hidden state at t ,

c_t = Long-term memory – cell state at t ,

$\tilde{c}_t$ = Candidate cell state,

f_t, i_t, o_t = Activation for forget gate, input gate, and output gate,

W_f, W_i, W_c, W_o = Weight matrices for respective gates,

b_f, b_i, b_c, b_o = Bias vectors for respective gates,

σ = Sigmoid activation function,

$\odot$ = Hadamard multiplication [237],

$[h_{t-1}, x_t]$ = Current input and previous hidden state concatenation.

GRU was proposed by Cho et al. in 2014 (Cho et al. 2014). It has two gates (Filippo et al. 2017):

1. Update gate: How much information must be retained from the past state?
2. Reset gate: Decides what is to be forgotten during state calculation. The calculations are represented in equation 16

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z) \quad (13)$$

$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r) \quad (14)$$

$$\tilde{h}_t = \tanh(W_h[r_t \odot h_{t-1}, x_t] + b_h) \quad (15)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (16)$$

Where:

- x_t = Input vector at time t ,
 h_t = Hidden state at t ,
 z_t = Update gate,
 r_t = Reset gate,
 $\tilde{h}_t$ = Candidate hidden state,
 W_z, W_r, W_h = Weight matrices for the respective gates and candidate state,
 b_z, b_r, b_h = Bias vectors for the gates and candidate state,
 σ = Sigmoid activation function,
 $\odot$ = Hadamard multiplication,
 $[h_{t-1}, x_t]$ = Current input and previous hidden state concatenation.

Autoencoders

We train the autoencoder to minimize differences between the input and the reconstructed output. Although autoencoders are unsupervised and do not require labeled data, they use input data as the target to reconstruct. Thus have a self-supervised training. Errors are calculated by taking the differences between the input and the output, usually calculating the mean squared error, to deal with positive values only. During training, the latent space becomes organized by learning the representation, and results converge, mostly forming clusters based on similarity. The encoder and decoder equations have been represented in equation 17 and 18. Due to their data compression ability, autoencoders are used in dimensionality reduction. They are used to reduce noise from data and also to detect anomalies.

$$h_j = \sigma \left(b_j^{(h)} + \sum_{i=1}^{n_x} W_{ji}^{(h)} x_i \right) \quad (17)$$

$$\hat{x}_i = \sigma \left(b_i^{(o)} + \sum_{j=1}^{n_h} W_{ij}^{(o)} h_j \right) \quad (18)$$

Where:

- x_i = The i -th input feature
 h_j = The j -th latent feature
 $\hat{x}_i$ = Reconstructed i -th feature
 $W^{(h)}, W^{(o)}$ = Encoder and decoder weight matrices
 $b^{(h)}, b^{(o)}$ = Biases
 σ = Activation function
 n_x = Number of input features
 n_h = Number of neurons in the hidden layer

In healthcare-related fields, autoencoders are used learning representations from unlabeled data, as medical data is extensively available but labeled data is limited (Xu et al. 2020). They have also been used in detecting anomalies in biomedical data (Nawaz et al. 2024),

noise removal (Gondara 2016), imputation, multimodal integration, and unsupervised feature learning. Their ability to handle scarce and heterogeneous data makes them highly valuable for disease detection, medical imaging, EHR modeling, and privacy-preserving healthcare solutions.

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Declarations

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Consent to participate We consent to participate in this Study.

Consent for publication We authorize the use of my data for research publication purposes.

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